

Ministry of Health
National Center for Tuberculosis
and Leprosy Control

Kingdom of Cambodia
National Religion King



Technical Guidelines on Tuberculosis Control

2nd Edition

National Tuberculosis
Control Program
September 2016



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FOREWORD

At present tuberculosis is still a major public health problem in Cambodia and worldwide. The Ministry of Health of the kingdom of Cambodia gives high priority to TB control, one of the main communicable diseases in the country, with the purpose of contributing to improving health as well as reducing poverty for Cambodian people.

This Technical Guidelines for TB Control was prepared with a view to provide overall information on TB control, which includes TB control network, strategies and technical principles for service provision, case detection and treatment, TB health education and prevention, encompassing Childhood TB, MDR-TB and TB-HIV/AIDS.

This document is a result of process of revising the guidelines published in 2003 and based on other related new documents with close collaboration with NTP partners. The revision process takes into consideration major changes of health system as well as evolution of TB control at both national and international levels.

This National guide was prepared for TB health workers and health managers at all levels, and for those whose works are related to TB control within and outside the government sector as well as for other people who are interested in TB control activities.

I hope this document is a basis of key information for all concerned parties for their respective works for contributing to the target attainment of the national Strategic Plan for TB Control for the period 2016-2020 in line with the overall Health Strategic Plan.

Phnom Penh, 13 September, 2016

For Minister of Health *Eng Huot*

Secretary of State



Prof. Eng Huot

ACKNOWLEDGEMENT

The development of this Technical Guidelines on Tuberculosis Control is an achievement of the close collaboration between the National Tuberculosis Control Program (NTP) in Cambodia and its health partners, especially the Japan International Cooperation Agency (JICA), the World Health Organization (WHO) and the Global Fund, which provided both technical and financial support to develop this document.

The National Tuberculosis Control Program (NTP) would like to deeply thank to the technical working group which includes:

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Dr. Khun Kim Eam	Dr. Koeut Pichenda	Dr. Saint Saly
Dr. Tan Kundara	Dr. In Sokhanya	Dr. Nou Chanly
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Dr. Peng Veasna	Dr. Kien Soriya	Dr. Keo Sokonth
Dr. Kim Samoeun	Dr. Ly Sothin	Dr. Leng Chhenglay
Dr. Team Bak Khim	Dr. Khann Sokhan	Dr. Ly Mena

And other NTP and partner staff who contributed to the successful development of this national technical guidelines for TB control.

The NTP would also like to express our sincere thanks to Dr. Osuga Katsunori, WHO technical advisor, and Dr. Kosuke Okada, Japan Anti-Tuberculosis Association, who provided technical support for the development of this important document.

Phnom Penh, 13 September, 2016
Director of National Center for TB and Leprosy Control

Dr. Mao Tan Eang

1. INTRODUCTION

1.1. Background

Although the World Health Organization (WHO) has declared tuberculosis (TB) as emergency in 1993 together with the efforts in mobilizing resources for TB control activities from all sources, tuberculosis remains a major problem to public health worldwide including Cambodia. According to the WHO's new classification, Cambodia is still one among the 30 countries with high burden (HBC) of TB in the world.

Overall objectives of the national TB control program are to reduce morbidity and mortality due to TB and to prevent the development of drug resistant TB in line with the End TB Strategy 2016-2035 as well as Sustainable Development Goals (SDGs) 2016-2030.

Any National TB Control Program that detects bacteriologically confirmed pulmonary TB at least 70% and successful treatment of more than 85% will result in:

- Rapid reduction of TB prevalence and transmission;
- Gradually reduce the incidence of TB;
- Low level of drug resistant development.

In order for NTP to better implement its program activities especially to achieve the new targets set, the important thing is to have clear policies, strategic plan, and technical guidelines together with sufficient resources, as well as properly implemented by concerned partners/stakeholders.

This document aims to provide technical guidance including TB burden, TB control network, new ways of TB screening, diagnosis and treatment; as well as TB prevention and education focusing on TB control activities in the Kingdom of Cambodia.

1.2. Global TB Situation

Globally, in 1994, 9.6 million new TB cases occurred, of which only 6 million were detected and reported to the WHO. TB killed 1.5 million people, of whom 400,000 were HIV-positive. TB now ranks alongside HIV/AIDS as a leading cause of death among communicable diseases. The death due to HIV/AIDS was estimated about 1.2 million in 2014. HIV positive among TB patients were 12%.

2015 was the last year of the Millennium Development Goals (MDGs) and it was a year of transitions from the MDGs to a new era of Sustainable Development Goals (SDGs), and from the Stop TB Strategy to the End TB Strategy. Between 1990 and 2015, MDGs had three main targets:

1. Reduction of TB prevalence by 50%;
2. Reduction of TB mortality by 50%; and
3. Reverse the incidence of TB.

Globally, from 1990 to 2015, TB mortality rate had fallen 47% while prevalence had fallen 42% (compared to target of MDGs: 50% reduction each). Incidence rate has fallen since 2000 (on average 1.5% reduction per year). In particular, among the WHO's six regions, only three regions and nine out of the 22 high burden TB countries have achieved the three main targets of MDGs. Those three regions are the Americas, the South-East Asia Region and the Western Pacific Regions; and the remaining three regions did not achieve the targets (Eastern Mediterranean, European and African Regions).

Three main targets at the global level are:

- Reduction of TB mortality: 90% by the end of 2030 and 95% by the end of 2035;
- Reduction of TB incidence: 80% by the end of 2030 and 90% by the end of 2035;

- No catastrophic costs for patients due to TB from 2020.

Based on the End TB Strategy 2016-2035, the world has set Global Plan to End TB 2016-2020 with three main targets: “90%”, “90%”, and “90%”. That is to detect at least 90% of TB cases, to detect at least 90% of the key populations, and to achieve at least 90% of successful treatment. International Community against TB is urging member countries to prepare proper plan in line with the global plan and also mobilize resources in order to successfully implement the program.

1.3. TB Situation in Cambodia

In the recent years, TB control activities in Cambodia were remarkably achieved. In overall, total number of TB cases of more than 530 thousand (531, 316) and more than 250 thousand (251, 543) bacteriologically confirmed TB cases were detected and treated in the last 15 years. This means that tens of thousands lives of TB patients were saved. That is why the prevalence, incidence and mortality of TB in Cambodia have been rapidly and steadily reduced compared to other countries in more than a decade (i.e. TB incidence rate in Cambodia decreased 3.5% per year compared to average global reduction of 1.5%).

Mortality rate decreased from 157/100,000 populations in 1990 to 58/100,000 populations in 2014 which account for 63% compared to the target 50% by the end of 2015. In more than a decade, mortality rate decreased on average 5% per year. Prevalence rate of all forms of TB decreased from 1,670/100,000 populations in 1990 to 668/100,000 populations in 2014 which account for 60% compared to the target 50% by the end of 2015. This reduction of prevalence was 6% per year on average. Incidence rate has fallen on average 3.5% per year since 2001.

The WHO Global TB Report 2015 has shown that Cambodia has already achieved MDG targets of reduction in TB prevalence, incidence and mortality rates. This is a tremendous and historical success in TB control in the kingdom of Cambodia.

1.4. NTP Structure and Network

The National TB Program (NTP) covers the entire public health care delivery system, which includes the National Centre for Tuberculosis and Leprosy (CENAT) at the national level, all 25 provincial and municipal health departments, 91 operational districts and 1,255 health facilities (102 referral hospital and 1,153 health centers) in 2015. Each level has designated TB staff and is responsible for specific functions of the TB program as below:

Central Level:

Leadership and managerial responsibility for the NTP lies with CENAT operating around 200 staff members, of which 30 are working full-time for the NTP headquarters, or Program Central Unit. In this Unit, there are the focal points responsible for developing policies and plans, training, supervision, monitoring and evaluation of the NTP activities, drug procurement for the whole country and coordinating with other partners supporting the NTP. The NTP is financed directly by the Ministry of Health, and directly or indirectly by external partners. CENAT also houses a referral chest hospital with 130 beds and the National TB Reference Laboratory (NTRL), (see Chapter 7).

Provincial and Municipal Levels:

Every province/municipality has a provincial/municipal TB Medical Supervisor, in charge of the TB control activities in the province; and a provincial/municipal TB Laboratory Supervisor, appointed by the provincial/municipal Health Director. They are responsible for all TB services in the

province/municipality, especially planning, training, coordination and regular supervision of the operational districts (ODs), TB microscopy centers and health centers. In some provinces, there are MDR (multi-drug resistant) -TB Treatment Units which are directly involved in the clinical care for MDR-TB patients.

Operational District

Every OD has an OD TB Supervisor whose main responsibilities are to maintain the OD TB registry, plan, train, coordinate and supervise health centers every month, and to interact with the clinical teams caring for patients with TB. OD has a referral hospital which consists of a TB unit with beds, TB staff, and TB microscopy; and some ODs have Xpert machine for diagnosis. In all, there are 215 microscopic centers (MCs) in the country in 2015.

Health Center Level

In the health centers, cross-trained general health workers implement the DOTS Strategy. Usually, at least two workers in each health center have received three-day training on DOTS, including smear-making. They are designated as TB health workers at health center level. Over 10% of all health centers (mostly former district hospitals) have TB laboratories. These health centers are staffed with 3 or 4 workers including at least one responsible for the microscopy work.

TB Laboratory

The National TB Reference Laboratory (NTRL) located in the CENAT premises works for the NTP, under the authority of the Director. It has 21 staff members and is capable of performing isolation and culture for TB from specimens, together with drug susceptibility testing (DST) for first-line drugs. NTRL is responsible for planning, training and supervising of TB laboratory services; and also responsible for the External Quality Assurance (EQA) of sputum microscopy services, which are provided through a network of 215 microscopic centers covering the entire country in 2015. In addition to the NTRL, the microscopic centers in Battambang and Kampong Cham also perform TB cultures.

TB Network in Community

The NTP has mobilized support from a number of development partners for implementation of services at the most peripheral level. Through Community DOTS (C-DOTS) initiatives, a network of community volunteers usually members of Village Health Support Groups (VHSG), have been engaged to refer presumptive TB patients to public health facilities and serve as treatment supporters (DOT watchers) for patients in their own communities. These activities are coordinated by many national and international NGOs.

Community-based TB activities are often run, being integrated with other health interventions that are technically and financially supported by different partners, such as maternal and child health and malaria. These are conducted in line with the draft Community Participation Policy for Health¹, through VHSG and Health Center Management Committees (HCMC). For TB, two VHSG members per village are trained to refer presumptive TB, collect and transport sputum specimens, conduct community sensitization, support patients on treatment, and collect drugs from the health center. Transport cost for collection of drugs from the health center for their patients is financially supported. The routine monthly or bi-monthly meetings with health center workers and VHSGs at health centers sometimes function for refreshing the basic knowledge and skills of the VHSGs about TB screening, diagnosis and treatment.

¹Community Participation Policy for Health, Kingdom of Cambodia Ministry of Health, 2008.

1.5. Basic Facts about TB

1.5.1. What is TB?

TB is an infectious and not hereditary disease. It is caused mostly by *Mycobacterium tuberculosis* called BK (and rarely caused by *Mycobacterium bovis* and *Mycobacterium africanum*). In general, TB germ penetrates into the body through respiratory tracts. It mostly develops disease in lungs called pulmonary TB and it also occurs in other organs outside lungs such as pleural, lymph node, bone etc. called extra-pulmonary TB (see section 4).

1.5.2. How does TB transmit?

Infection occurs when someone inhales the infected droplets produced by TB patient especially when bacteriologically confirmed pulmonary TB patient is coughing, sneezing, laughing or talking. The infection mostly happens at close environment where the droplet nuclei containing TB bacilli can remain suspended in the air for long periods.

1.5.3. Risk of TB Infection

An individual's risk of infection depends on the extent of exposure to droplet nuclei containing TB bacilli and his or her susceptibility to infection. Two factors determine an individual's risk of exposure includes 1) the concentration of droplet nuclei containing TB bacilli in contaminated air; and 2) the length of time he or she breathes that contaminated air.

1.5.4. How does TB Develop?

TB develops in human body in two stages. First stage occurs when someone exposes to TB bacilli from pulmonary TB patient and one gets infected (infection). Second stage is happening when infected person develops TB symptoms or signs (disease).

Once infected with TB bacilli, a person can stay infected for the whole life. The vast majority (90%) of people without HIV infection who are infected with TB bacilli do not develop TB. In these, asymptomatic but infected individuals, the only evidence of infection may be a positive tuberculin skin test. However, infected person can develop TB at any time. The chance of developing disease is greatest shortly after infection and steadily lessens as time goes by. Various physical or emotional stresses may trigger progression of infection to disease. The most important trigger is weakening of immune resistance especially by HIV infection.

1.5.5. How does Untreated TB Disease Develop?

The progression of TB disease without treatment depends on the type and severity of disease. If no treatment, the progression towards death is high for pulmonary disease especially for bacteriologically positive pulmonary TB and severe cases such as meningitis and pericarditis cases. For bacteriologically positive pulmonary TB, without treatment, by the end of five years, 50% of pulmonary TB patients will be dead; 25% will remain ill with chronic infectious TB; and 25% will be healthy, self-cured with or without consequences.

2. CASE DEFINITION

- Tuberculosis (TB) is an infectious disease caused in most of cases by micro-organism called "*Mycobacterium Tuberculosis*". Its characteristics are common lesions in tissue, especially in the lungs.
- Presumptive TB is person who has signs or symptoms that may think about TB, especially having chronic cough of more than 2 to 3 weeks.
- Case detection: all activities to identify TB cases among adult and children at outpatient department and inpatient department, in communities and other setting such as prisons.

2.1. Definition of Presumptive TB

Person who has signs or symptoms suggestive of TB, especially chronic cough for 2 weeks or longer, fever, drenching night sweat and weight loss.

2.2. Case Definition

2.2.1. Classification based on Anatomical Site of Disease

2.2.1.1. Pulmonary Tuberculosis (PTB): referred to any bacteriologically confirmed or clinically diagnosed case of TB involving the lung parenchyma or the tracheobronchial tree. Miliary TB is classified as PTB because there are lesions in the lungs. Tuberculous intra-thoracic lymphadenopathy (mediastinal and/or hilar) or tuberculous pleural effusion, without radiographic abnormalities in the lungs, constitutes a case of extra-pulmonary TB. A patient with both pulmonary and extra-pulmonary TB should be classified as a case of PTB.

2.2.1.2. Extra-Pulmonary Tuberculosis (EPTB): referred to any bacteriologically confirmed or clinically diagnosed case of TB involving organs other than the lungs, e.g. pleura, lymph nodes, abdomen, genitourinary tract, skin, joints and bones, meninges etc.

2.2.2. Classification based on History of TB Treatment

Classifications based on history of TB treatment are slightly different from those previously published. They focus only on history of previous treatment and are independent of bacteriological confirmation or site of disease. Note also that the registration groups for drug-resistant TB (DR-TB) are slightly different.

2.2.2.1. New Patients: those who have never been treated for TB or have taken anti-TB drugs for less than one month.

2.2.2.2. Previously Treated Patients: those who have received 1 month or more of anti-TB drugs in the past and right now they are diagnosed as active TB.

- **Relapse Patients** have previously been treated for TB, were declared *cured* or *treatment completed* at the end of their most recent course of treatment, and are now diagnosed with a recurrent episode of TB (either a true relapse or a new episode of TB caused by reinfection).

- **Treatment after Failure Patients** are those who have previously been treated for TB and whose *treatment failed* at the end of their most recent course of treatment (i.e. positive result of sputum follow up at the 5th month of treatment or in between the 5th month and end of treatment).

- **Treatment after Loss to Follow-up Patients** are those have previously been treated for TB for at least one month and have not taken anti-TB drugs for at least two consecutive months (in their most recent course of treatment) and now are diagnosed as active TB.

- **Other Previously Treated Patients** are those who have previously been treated for TB but whose outcome of their most recent course of treatment is unknown or undocumented and now are diagnosed as active TB.

2.2.2.3. Unknown TB Treatment History Patients: those who are not fit to any of the categories listed above.

2.2.3. Classification based on HIVStatus

- **HIV-positive TB patient:** referred to any bacteriologically confirmed or clinically diagnosed case of TB who has a positive result from HIV testing conducted at the time of TB diagnosis or during TB treatment or other documented evidence of enrolment in OI/ART clinic, such as enrolment in the pre-ART register or in the ART register once ART has been started.

- **HIV-negative TB patient:** referred to any bacteriologically confirmed or clinically diagnosed case of TB who has a negative result from HIV testing conducted at the time of TB diagnosis or during TB treatment. Any HIV-negative TB patient subsequently found to be HIV-positive should be reclassified accordingly.

- **Unknown HIV status TB patient:** referred to any bacteriologically confirmed or clinically diagnosed case of TB who has no result of HIV testing and no other documented evidence of enrolment in HIV care. If the patient's HIV status is subsequently determined, he or she should be reclassified accordingly.

2.2.4. Classification based on Drug Resistance

Cases are classified in categories based on drug susceptibility testing (DST) of clinical isolates confirmed to be *M. tuberculosis*:

- **Mono-resistance:** resistance to one first-line anti-TB drug only.
- **Poly-resistance:** resistance to more than one first-line anti-TB drug, other than both Isoniazid and Rifampicin.
- **Multidrug Resistance (MDR):** resistance to at least both Isoniazid and Rifampicin.
- **Extensive drug resistance (XDR):** resistance to any Fluoroquinolone, and at least one of three second-line injectable drugs (Capreomycin, Kanamycin and Amikacin), in addition to multidrug resistance.
- **Rifampicin resistance (RR):** resistance to rifampicin detected using phenotypic or genotypic methods, with or without resistance to other anti-TB drugs. It includes any resistance to rifampicin, in the form of mono-resistance, poly-resistance, MDR or XDR.

2.2.5. Treatment Outcome Definitions

The revised treatment outcome definitions make a clear distinction between two types of patients:

- Patients treated for drug-susceptible TB
- Patients treated for drug-resistant TB using second-line treatment (defined as chemotherapy for drug-resistant tuberculosis)

The two groups are mutually exclusive. Any patient found to have drug-resistant TB and placed on second-line treatment is removed from the drug-susceptible TB outcome cohort. This means that management of the standard TB register and of the second-line TB treatment register needs to be coordinated to ensure proper accounting of the outcomes of treatment.

2.2.6. Treatment Outcomes for TB Patients (excluding patients treated for RR-TB or MDR-TB)

All bacteriologically confirmed and clinically diagnosed TB cases should be assigned an outcome from this list **except** those with rifampicin resistant TB (RR-TB) or multi-drug resistant TB (MDR-TB), who are placed on a second-line drug regimen.

2.2.6.1. Cured: a pulmonary TB patient with bacteriologically confirmed TB at the beginning of treatment who was smear- or culture-negative in the last month of treatment and on at least one previous occasion (including C2 or C3).

2.2.6.2. Treatment Completed: a TB patient who completed treatment without evidence of failure BUT with no record to show that sputum smear or culture results in the last month of treatment and on at least one previous occasion were negative, either because tests were not done or because results are unavailable.

2.2.6.3. Treatment Failed: a TB patient whose sputum smear or culture is positive at month 5 or later during treatment.

2.2.6.4. Died: a TB patient who died for any reason before starting or during the course of treatment.

2.2.6.5. Loss to follow-up: a TB patient who did not start treatment or whose treatment was interrupted for 2 consecutive months or more.

2.2.6.6. Not evaluated: a TB patient for whom no treatment outcome is assigned. This includes cases “transferred out” to another treatment unit as well as cases whose treatment outcome is unknown to the reporting unit.

2.2.6.7. Treatment success: The sum of *cured* and *treatment completed*.

2.2.7. Treatment Outcomes for RR-TB/MDR-TB/XDR-TB Patients treated with Second-Line Drugs

- **Cured:** Treatment completed as recommended by the national policy without evidence of failure AND three or more consecutive cultures taken at least 30 days apart are negative after the intensive phase.
- **Treatment completed:** Treatment completed as recommended by the national policy without evidence of failure BUT no record that three or more consecutive cultures taken at least 30 days apart are negative after the intensive phase.
- **Treatment failed:** Treatment terminated or need for permanent regimen change of at least two anti-TB drugs because of:
 - o lack of conversion by the end of the intensive phase, *or*
 - o bacteriological reversion in the continuation phase after conversion to negative, *or*
 - o evidence of additional acquired resistance to Fluoroquinolones or second-line injectable drugs, *or*
 - o adverse drug reactions (ADRs).

- **Died:** a patient who died for any reason during the course of treatment.
- **Loss to follow-up:** a patient whose treatment was interrupted for 2 consecutive months or more.
- **Not evaluated:** a patient for whom no treatment outcome is assigned (this includes cases “transferred out” to another treatment unit and whose treatment outcome is unknown).
- **Treatment success:** The sum of *cured* and *treatment completed*.

Remarks: Patients found to have an RR-TB or MDR-TB strain at any point in time should be started on an adequate second-line drug regimen (CAT4). These cases are **excluded** from the main TB cohort when calculating treatment outcomes¹ and included only in the second-line TB treatment cohort analysis or CAT4. If treatment with a second-line drug regimen is not possible, the patient is kept in the main TB cohort and assigned a treatment outcome of drug susceptible cases.

3. STRATEGY AND PATHWAY TO TB CASE DETECTION

Generally, the new strategy for TB case detection in Cambodia is to promptly diagnose for all presumptive TB regardless of age and sex everywhere in line with the End TB strategy, especially to ensure “universal health coverage” and “leave no one behind.”

In Cambodia, by late 2013, the main symptom to identify TB cases was focusing only on cough. Since 2014, TB symptom screening for the suspect is focused on multiple symptoms which includes cough, fever, night sweat, weight loss lasts for 2 weeks and swollen lymph nodes and this method is quite different from the previous one which is focused only on cough.

Chest X-ray plays more important roles than ever before, right now chest X-ray plays two roles, as a tool for TB screening or identifying presumptive TB, and a tool for diagnosis. For instance, X-ray is used as a tool to identify presumptive TB in enhanced TB case finding at prison or for diabetes patients at clinic, and it is also used as a complementary diagnostic tool for bacteriologically negative TB patients.

There are two pathways to detect TB cases: active and passive. Passive case finding is to find TB cases in routine health facilities (health center, hospital or health post) where general patients come to OPD or IPD for seeking health care. In this case, it is also including presumptive TB referred by village health supporting group (VHSG) or other sources. Passive case finding is a backbone of TB control program, especially in the routine activities.

To complement to passive case finding, active case finding or enhanced case finding requires central team from national level moving to the specific sites such as prisons or communities in target zones where TB cases are prevalent and the team is equipped with X-ray machine, Gene Xpert machine or microscopes. For some health centers, health workers go to villages or pagoda for finding TB cases; it is called semi-active case finding and is also a part of active case finding. Active case finding is a main priority for TB program activities to complement passive case finding at least in the period of 2016-2020.

4. DIAGNOSIS OF TUBERCULOSIS

4.1. Introduction

For early and proper diagnosis of tuberculosis (TB) based on the Cambodia National TB Control Program (NTP)'s guidelines, physicians who are working in TB control at the national hospitals, provincial or district referral hospitals and health workers who are working at the health centers or health posts, play important roles in screening and diagnosis of pulmonary and extra-pulmonary TB. In general, hospital has main diagnostic tools in TB screening and diagnosis such as microscopy, GeneXpert machine and Chest X-ray.

Most TB cases are diagnosed by passive case finding through routine activities at health centers and hospitals. Health center has main three roles in TB screening and diagnosis:

1. TB Screening and presumptive TB identification.
2. Diagnosis of pulmonary smear-positive TB.
3. Register TB cases and transferred-in TB patients for treatment.
4. Provision of health education on TB disease.

Health center has a role in screening for all forms of TB. For diagnosis of TB, health center has only role for pulmonary smear-positive TB especially collecting sputum from presumptive TB and referring to the microscopic center. For childhood TB, smear-negative pulmonary TB and extra-pulmonary TB, health center staff usually refers presumptive TB case after identification to referral hospital for diagnosis of TB.

4.2. Diagnostic Tools

4.2.1. Smear Microscopy

Because of its feasibility and sustainability, NTP acknowledges that smear microscopy still fulfill an important role in TB control as the basic principle for case detection and treatment monitoring at least from 2016 to 2020. NTP has been implementing 215 microscopic centers (MC), including 29 MCs equipped with LED microscopes (Light-Emitting Diode) or fluorescence microscopy using Auramin staining.

4.2.2. GeneXpert MTB/RIF

GeneXpert MTB/RIF, a new TB diagnostic tool, was officially recommended by the World Health Organization (WHO) since 2010. This test provides simple, highly effective and fast result within 2 hours. Moreover, it does not only detect *M. tuberculosis*, but also detect Rifampicin resistance. For this reason, the Cambodia NTP uses this new test to detect MTB among the target groups, especially 1) presumptive MDR-TB patients, 2) people living with HIV/AIDS, 3) high risk population including diabetes, closed contact with smear-positive TB, the elderly aged from 55 years old and 4) for intervention of active case finding (ACF). NTP considers expanding the use of Xpert test to other target populations based on the WHO's new recommendation.

4.2.3. TB Culture and DST

At the end of 1999, in collaboration with JICA, NTP implemented solid culture at the National Center for Tuberculosis and Leprosy Control (CENAT) as the National TB Reference Laboratory (NTRL). Then, liquid culture was implemented at CENAT and Battambang laboratory centers in 2011, and at Kampong Cham laboratory center in 2014.

Drug Susceptibility Testing (DST) for first-line anti-TB drugs using liquid culture was implemented at CENAT in 2011 and at Kampong Cham hospital in 2013. The performance of DST for

the second-line anti-TB drugs using liquid culture at CENAT has been evaluated by the supra-national reference laboratory (SNRL) of the Research Institute of Tuberculosis (RIT) of Japan since 2014.

Culture and DST is part of the NTP's strategy for both detecting MDR-TB patients and research purposes.

4.2.4. Chest X-Ray

Chest X-ray Radiography (CXR) is an important component for detecting TB patients. About half of pulmonary smear-positive TB cases detected in the National TB Prevalence Survey in 2011 didn't present any typical TB symptoms such as cough more than 2 weeks or hemoptysis, which means that symptom screening has a limitation in case detection of TB. NTP recognized that CXR plays two important roles in TB control, “**Screening**” and “**Diagnosing**”. For example, CXR can be used to provide further information on whether or not a patient has active TB, after their smear microscopic examination result was negative or no improvement by broad spectrum antibiotics. Currently, CXR is nationwide available in all national hospitals and almost all in provincial and operational districts hospitals.

4.2.5. Tuberculin Skin Test

According to the World Health Organization, the Tuberculin Skin Test (TST) can be used to determine that a person has TB infection or not, but not for diagnosing active TB disease. Moreover, in case of immune-compromised host (for example, HIV infected persons), TST may show a negative result even in TB infected persons. Hence, NTP applies the TST to detect TB infection only in children under 5 years old who are living with pulmonary smear-positive TB or having one of four symptoms suggestive of TB (cough, fever, weight loss or night sweat). Positive result of the TST and CXR suggestive of TB may contribute to detecting clinical TB cases in children (see section 12). TST is useful in TB diagnosis in children of whom *M. tuberculosis* could not be detected by laboratory.

4.3. Diagnosis of TB

4.3.1. Main Points for TB Diagnosis

- All patients with or without sputum, with unexplained cough (unknown reason) lasting for two or more weeks, should be evaluated for TB disease.
- All patients clinically suggestive of pulmonary TB (adult and children are capable of producing sputum), should submit three sputum specimens to microscopy (or GeneXpert MTB/RIF).
- Of all patients suggestive of Extra-pulmonary TB (adult and children), appropriate specimens from the involvement site should be obtained for microscopy, if possible, the specimens should be submitted to GeneXpert MTB/RIF, culture or histological examination.
- For all patients with CXR shadow suspected of TB, sputum should be collected & tested for AFB
- Diagnosis of pulmonary smear-negative or clinically diagnosed TB (PTB BK-) should be based on the three principles: 1) at least three sputum specimen examinations with smear-negative results, 2) CXR suggestive of TB, and 3) no improvement by the symptomatic treatment with anti-biotic.

4.3.2. Main Symptoms of TB

- **Common TB symptoms:** fever/febrile at the evening, night sweat, weight loss, poor appetite and fatigue.
- **Specific TB symptoms:**
 - Pulmonary presumptive TB: Cough with or without sputum, hemoptysis and chest pain lasting two weeks or more (except hemoptysis, which is regardless of time).
 - Extra-pulmonary presumptive TB: It depends on the organ affected:

- Lymph node TB: Swelling, sometimes fistula
- Bone and joint TB: Swelling with pain
- TB Meningitis: Headache, fever, stiff neck (rigidity) and convulsion.
(please see the detailed in section 4.5).

4.4. Pulmonary Tuberculosis

4.4.1. Introduction

Pulmonary TB (PTB) is diagnosed by confirming that a patient has active TB disease in lung parenchyma, bronchi or bronchiolar. There are two types of pulmonary TB namely “pulmonary BK(+) TB or bacteriologically confirmed TB” and “pulmonary BK(-) TB or clinically diagnosed TB”. Miliary tuberculosis is classified as pulmonary TB because there are lesions in the lungs. A patient with both pulmonary and extra-pulmonary TB should be classified as a case of pulmonary TB.

4.4.2. Diagnosis of Pulmonary TB

Taking accurateness and standardization into consideration, two diagnostic algorithms of TB were developed: “the first algorithm is for presumptive PTB among general population” and “the second algorithm is for presumptive PTB among high risk population (such as people living with HIV/AIDS, elderly people aged 55 years old and above, people in contact with pulmonary smear-positive TB, people with diabetes etc.).”

4.4.2.1. Diagnosis of Pulmonary TB among General Population

(See diagnostic algorithm-1)

- In case of patient **with cough** more than 2 weeks, with or without other symptoms such as fever, weight loss and night sweats, we should do as follows:
 - Fill laboratory request form
 - Fill in presumptive TB register (TB Suspect Book)
 - Collect three sputum specimens (three different times)
 - Make smearing in three different slides (one specimen, one slide)
 - Refer three smear-slides to TB microscopic center (MC) for mycobacterium TB examination.
 - Register for pulmonary smear-positive TB treatment (Baccille de Koch positive TB/TB BK+), if at least one slide has positive-result.
 - Patient is recommended for large spectrum antibiotics (AB) symptomatic treatment for 10 days, if all slides are negative for TB. He/she is taken CXR, if not improved with antibiotic treatment. Register for TB clinic treatment (Smear-negative TB or Bacille de Koch Negative TB/TB BK-), if CXR shadow suggestive of active TB. But when CXR has no abnormal shadow, patient may have disease other than TB or patient need to be followed up.
- In case of patient **without cough** or cough less than 2 weeks, but with other chronic symptoms such as fever, weight loss or night sweats, we should do as follows:
 - Register at the Out-Patient Department (OPD) or another site as appropriate
 - Large-spectrum antibiotics symptomatic treatment for 7 to 10 days
 - Patient should be taken CRX, if not improved with AB symptomatic treatment
 - If CXR shadow suggestive of active TB, three sputum specimens will be collected by two options:
 - 1) Register for smear-positive TB treatment, if at least one smear-slide is positive or
 - 2) Decision making by physician, if all three slides are negative results (Register for TB clinic treatment or follow-up for other diseases).

4.4.2.2. Diagnosis of Pulmonary TB among High Risk Population

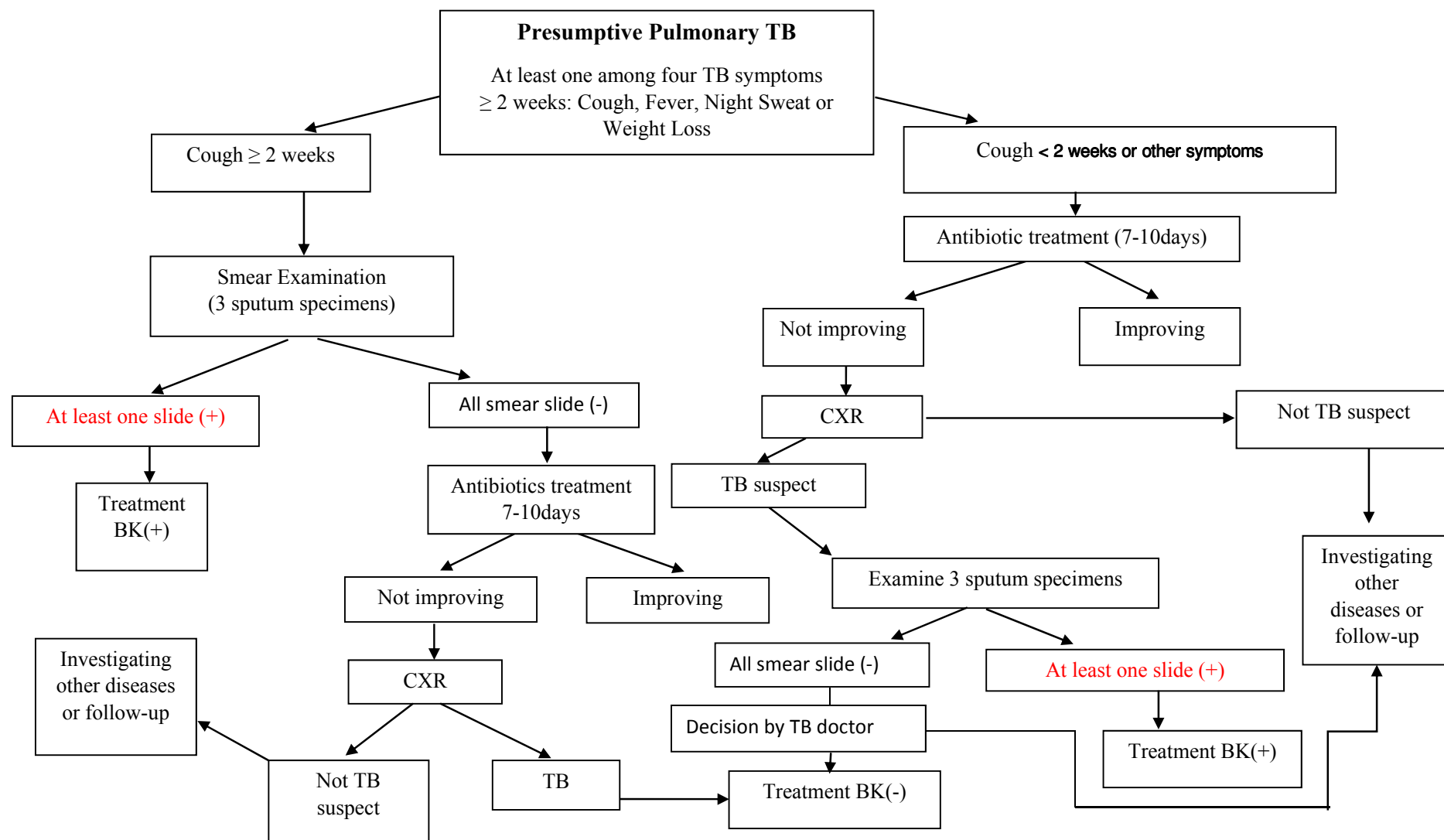
(See diagnostic algorithm-2)

In Cambodia, the high risk population include people living with HIV/AIDS, elderly people aged 55 years old and above, people in contact with pulmonary smear-positive TB or bacteriologically confirmed TB (TB contact), people with diabetes and inmate people (prisoners).

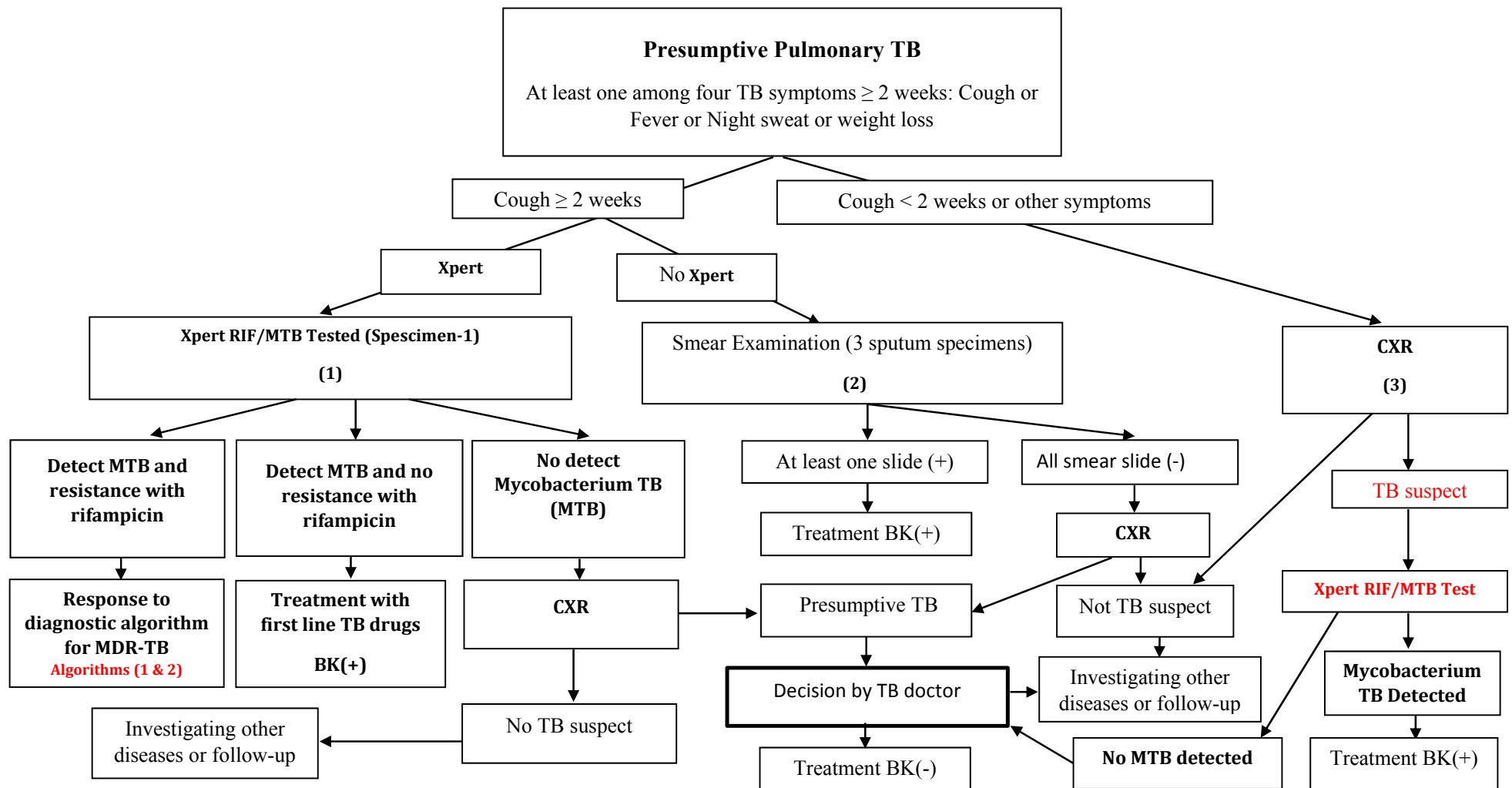
People with at least one among four TB symptoms below are suspected of TB:

- In case of high risk people with cough, we should do as follows:
 - ✓ If GeneXpert test is available:
 - Fill in laboratory request form
 - Collect sputum one specimen and refer it to GeneXpert site for testing
 - If mycobacterium TB without resistance to rifampicin is detected, it is bacteriologically confirmed TB (or TB with BK+) which is registered for TB treatment with the first line anti-TB drugs.
 - If mycobacterium TB with resistance to rifampicin is detected, it is bacteriologically confirmed TB (or TB with BK+) and resistance to rifampicin which is registered for MDR-TB treatment with the second line anti-TB drugs (see diagnostic algorithm-2 TB/HIV).
 - If mycobacterium TB is not detected, it is referred for CXR for further examinations for TB.
 - ✓ If GeneXpert test is not available:
 - Fill in laboratory request form
 - Collect three sputum specimens (three different times)
 - Make smearing in three different slides (one specimen, one slide)
 - Refer three smear-slides to TB microscopic center (MC) for mycobacterium TB examination.
 - Register for pulmonary smear-positive TB treatment (Baccille de Koch positive TB/TB BK+), if at least one slide has positive-result.
 - Refer patient for CXR taking, if all three smear-slides are negative for TB. If CXR shadow suggestive of active TB, physician clinically makes a decision whether he/she should be treated as TB (clinically diagnosed TB or Bacille de Koch Negative TB/TB BK-). But if CXR has no abnormal shadow, patient may have disease other than TB or patient need to be followed up.
- In case of high risk people without cough, and with other symptoms suggestive of TB, we should do as follows:
 - Fill in referral slip for CXR taking
 - If CXR shadow suggestive of active TB, we should do as follows:
 - Collect three sputum specimens (three different times) in two days for OPD patient and 3 consecutive mornings for hospitalized patient
 - Refer first sputum specimen to the GeneXpert site for testing
 - If mycobacterium TB without resistance to rifampicin is detected, it is bacteriologically-confirmed TB (or TB with BK+) which need to be registered for TB treatment with the first line anti-TB drugs.
 - If mycobacterium TB with resistance to rifampicin is detected, it is bacteriologically-confirmed TB (or TB with BK+) and resistance to rifampicin which need to be registered for MDR-TB treatment with the second line anti-TB drugs (see diagnostic algorithm-2 TB/HIV).

Algorithm-1
Diagnostic Algorithm for Presumptive Pulmonary TB among General Population



Algorithm-2
Diagnostic Algorithm for Presumptive Pulmonary TB among High Risk Population



(*)High Risk Population: Elderly people aged 55 years old or more, people in contact with pulmonary smear-positive TB (see diagnostic algorithm for children if child is under 15 years old), people living with HIV/AIDS, people with diabetes and inmate in prison (prisoner).

4.5. Extra-Pulmonary TB

4.5.1. Introduction

Extra-pulmonary TB affected organs out of the lung parenchyma, such as lymph node (especially in the neck or under the arms), pleural (usually one-sided pleural effusion), bone and joint TB, kidney, intestine and meningitis TB etc. Extra-pulmonary TB accounts for about 25% of all TB cases. All patients with suggestive of Extra-pulmonary TB (adult and children), appropriate specimens from the suspected sites of involvement should be obtained for microscopic examination; and if the facility has more capable diagnostic tools, specimens should be sent to GeneXpert MTB/RIF, culture or histological examination.

4.5.2. Diagnosis of Extra-Pulmonary TB

The same as the pulmonary TB, extra-pulmonary TB has two types which includes BK(+) TB or bacteriologically confirmed TB [or extra-pulmonary TB BK(+)] and extra-pulmonary BK(-) negative or clinically diagnosed [or Extra-pulmonary TB BK(-)]. In order to practice easily and properly, physician/clinician should read the diagnostic algorithm-3 for presumptive EP-TB patient in general population.

4.5.3. Types of Extra-Pulmonary TB

A). TB Meningitis

General symptoms: General symptoms and signs of TB meningitis are:

- Cough
- Fever
- Weight loss
- Night sweat
- Evidence for TB elsewhere

Local Symptoms: Symptoms to be suspected of TB meningitis are:

- Headache
- Somnolence
- Neck rigidity/stiffness
- Central nerve palsy

Diagnosis:

When TB meningitis is suspected, it is important to do a lumbar puncture and perform biochemical and bacteriological examination of cerebrospinal fluid (CSF). We can confirm meningitis TB if patient presenting symptoms mentioned above, and the results of cerebrospinal fluid examination:

- CSF opening pressure is elevated
- Increasing white blood cell, especially lymphocyte
- Cerebrospinal fluid is clear with high protein and low glucose
- AFB smear rarely positive

B). TB Lymphadenopathy

TB lymphadenopathy frequently occurs in cervical region, but sometime it occurs in axillary region or inguinal region. Generally, it develops slowly and painless. If this TB lymphadenopathy occurs, a primary lesion can be associated with fistula to other lymph nodes.

General symptoms: General symptoms and signs of TB lymphadenopathy are the followings:

- Cough
- Fever
- Weight loss
- Night sweat
- Evidence for TB elsewhere

Local Symptoms: Symptoms to be suspected of TB lymphadenopathy are:

- The lymph node is enlarged with 2 cm or more in size, asymmetrical or localized,
- Painless, firm, fluctuant and sometimes associated with fistula

Diagnosis:

- It presents symptoms above
- No improvement in size of lymph node on a course of antibiotics
- Sputum examination for AFB: If sputum is positive on bacteriology, register as bacteriologically confirmed pulmonary TB or pulmonary TB BK+
- Aspiration with needle for AFB and cytological examinations: If AFB is positive, register for bacteriologically positive extra-pulmonary TB or Extra-pulmonary TB BK+
- Tuberculin skin test (TST) in children: positive
- Abdominal ultrasound: enlarged lymph nodes found

C). Bone and Joint TB

General symptoms: General symptoms and signs of bone and joint TB are:

- Cough
- Fever
- Weight loss
- Night sweat
- Evidence for TB elsewhere

Local Symptoms: Symptoms to be suspected of bone and joint TB are:

- Chronic pain in bone/joint
- Sign of Gibbus deformity (spine), or joint swelling

Diagnosis:

- Bone and joint X-ray
- Tissue biopsy
- Tuberculin Skin Test in children
- Bone destruction and narrowing of joint suggestive of TB
- AFB is positive from the site specimens
- Cytology: Granulomatous reaction

D). TB Pleurisy

General Symptoms: General symptoms and signs of TB pleurisy are:

- Cough
- Fever
- Weight loss
- Night sweat
- Evidence for TB elsewhere

Local Symptoms: Symptoms to be suspected of TB Pleurisy are:

- Unilateral pleural effusion
- Evidence for TB elsewhere

Diagnosis:

- Chest X-ray: Finding site and level of pleurisy
- Sputum bacteriological examination if coughing
- Aspirate and inspect fluid: Generally, AFB is not detected. If AFB detected, register for bacteriologically positive extra-pulmonary TB or Extra-pulmonary TB BK+
- Lymphocyte and protein is elevated (exudate: protein > 30g/l)
- **Note: Leave the aspirated liquid in the vertical container. If it is precipitated, it is exudate.**

Algorithm-3

Diagnostic Algorithm for Presumptive Extra-Pulmonary TB among General Population

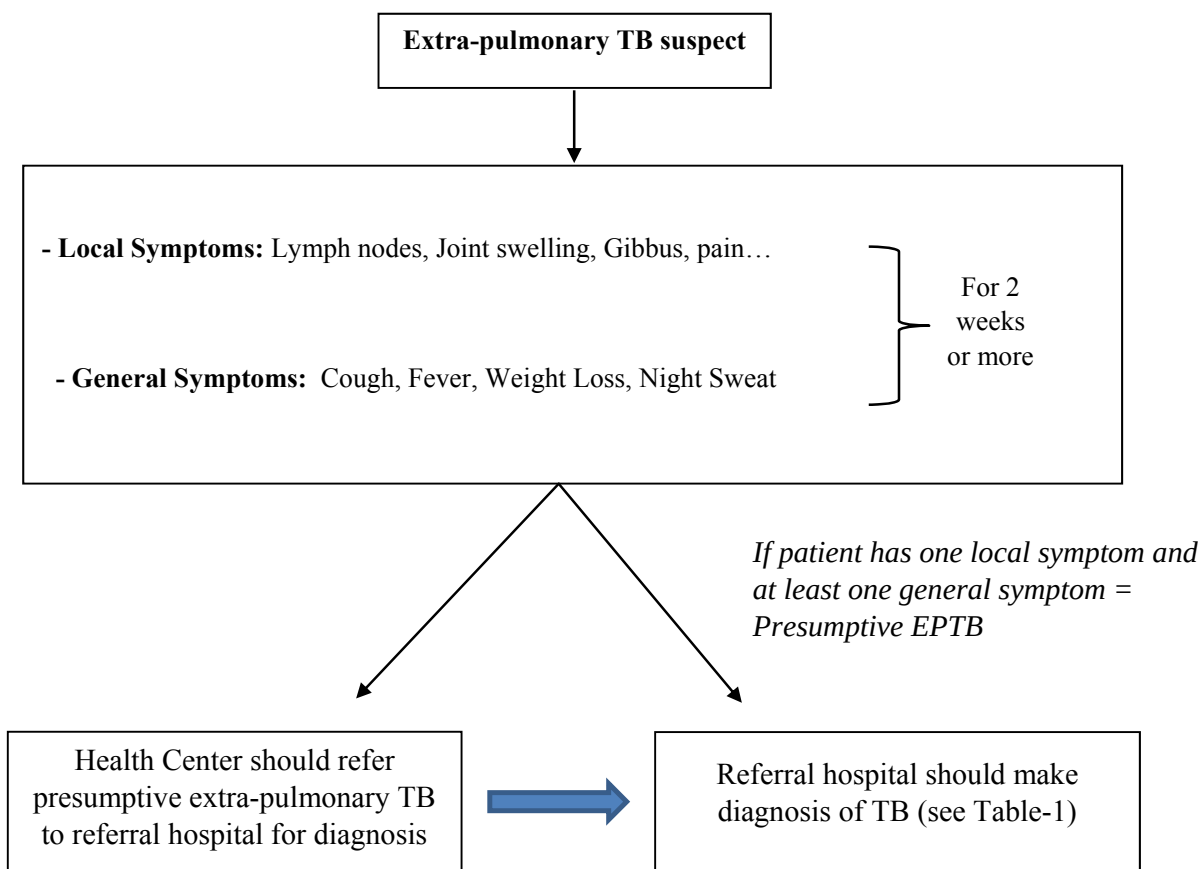


Table 1: Summary of Symptoms/Signs and Diagnosis of Extra-Pulmonary TB among General Population

	TB Meningitis	TB Lymphadenopathy	Bone and Joint TB	TB Pleurisy
Symptom/Sign	• Genral symptoms • Headache • Somnolence • Neck rigidity/stiffness • Cranial nerve palsy	• Genral symptoms • The lymph node is swelling with 2 cm or more in size, symmetrical or localized • Painless, firm, fluctuant • Associated with/without chronic drainage fistula	• Genral symptoms • Local symptoms such as pain, red skin and joint swelling • Chronic pain in bone/joint with sign of gibbus (spine), psoas abscess, radicular nerve compression	• Genral symptoms • Unilateral effusion • Abdominal ascites • Pericarditis • Evidence for TB elsewhere
Diagnosis	• Lumbar puncture, biochemical and bacteriological examination of cerebrospinal fluid (it may be dangerous if intracranial pressure is high). • Should be treated with empirical treatment and do lumbar puncture.	• Aspirate lymph node with noodle for AFB and cytology examinations; tuberculin skin test in children • Abdominal ultrasound for enlarged lymph nodes	• Plain X-Ray • Tissue biopsy	• Physical examination • Chest X-ray • Ultrasound • Aspirate and inspect pleural fluid and tissue biopsy

Results	• CSF opening pressure is high • Cerebrospinal fluid is clear with moderate increase of white blood cells especially lymphocyte • CSF is high in protein • CSF is low in glucose • AFB smear rarely positive	• Caseation • AFB smear may be positive	• AFB smear may be positive • Histology: Granulomatous reaction	• CXR shadow shows pleural effusion or pericarditis • Ultrasound can find ascites and abdominal lymph nodes • Tissue biopsy: AFB and/or Granulomas. Aspirated fluid: • Generally, AFB negative • Culture usually need 4-8 weeks • White blood counts change • Exudate: Protein content > 30 g/l (if it is precipitated when leaving the aspirated liquid in the vertical container, it is exudate)
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CSF= Cerebrospinal Fluid, AFB= Acid Fast Bacilli, CBC= Complete Blood (Cell) Count, CXR= Chest X-Ray, LN= Lymph Node

5. TREATMENT OF TUBERCULOSIS

5.1. Aims and Principles of Tuberculosis Treatment

Aims of TB Treatment

- to cure the patient and restore quality of life and productivity;
- to prevent death from active TB or its late effects;
- to prevent relapse of TB;
- to reduce transmission of TB to others; and
- to prevent the development and transmission of drug resistance.

Principles of TB Treatment

- to treat patient immediately after diagnosis and should ***not exceeding 3 days***
- to ask patients about history of TB treatment including MDR contact history before start TB treatment
- to educate patients for strengthening patient's adherence
- before start the continuation phase, weigh TB patient to adjust the number of pills is needed

5.2. First-Line anti-TB Drugs

TB treatment consists of two phases: initial phase (or intensive phase) and continuation phase.

Optimal Dosages for Essential anti-Tuberculosis Drugs in Adults

First-Line Drugs	Daily dose in mg per kg of body weight (range)
Isoniazid (H)	5 mg (4-6 mg)
Rifampicin (R)	10 mg (8-12 mg)
Pyrazinamide (Z)	25 mg (20-30 mg)
Ethambutol (E)	15 mg (15-20 mg)
Streptomycin (S)	15 mg (12-18 mg)

5.3. Symptom-based Approach to Managing Side Effects of anti-TB Drugs

Side Effect	Responsible Drug(s)	Management
Minor Symptom		
Anorexia, nausea, abdominal pain	Rifampicin, Pyrazinamide	Give drugs with small meals or just before bedtime
Joint pain	Pyrazinamide	Aspirin or non-steroidal anti-inflammatory drug, or paracetamol
Peripheral neuropathy (Burning, numbness or tingling sensation in the hands or feet)	Isoniazid	Pyridoxine (Vit B6) 100 mg daily
Orange/red urine	Rifampicin	Reassurance (patients should be told when starting treatment, that this may happen and is normal)
Major Symptom		
Skin rash with or without itching	Streptomycin, rifampicin, pyrazinamide, isoniazid	Stop all anti-TB drugs
Deafness (no wax on otoscopy), dizziness, mydriase	Streptomycin	Replace Streptomycin by Ethambutol
Jaundice (other causes excluded), hepatitis	Rifampicin, Pyrazinamide, Isoniazid,	Stop all anti-TB drugs
Confusion (suspect drug-induced acute liver failure if there is jaundice)	Most anti-TB drugs	Stop all anti-TB drugs and check liver function
Visual impairment (other causes excluded)	Ethambutol	Stop Ethambutol
Shock, purpura, acute renal failure	Rifampicin	Stop Rifampicin

5.4. TB Treatment Regimen

5.4.1. Category 1 (CAT-1): in this category, there are two types, CAT-1a and CAT-1b.

Category-1a: 2RHZE/4RH

This treatment regimen is used for new TB case and unknown history TB case:

- Pulmonary TB
- Extra-pulmonary TB (except TB meningitis, joint/bone TB)
- TB/HIV

Intensive Phase (initial phase): for 2 months

In this phase, TB patient has to take 4 anti-TB drugs every morning for 2 months under direct observation by the healthcare worker or trained community DOT watchers (it is called DOT approach). The 4 drugs are Rifampicin, Isoniazid, Pyrazinamide and Ethambutol (combined tablet, 4FDC)

At the end of the second month, if sputum smear examination is still positive then one month more of the initial phase has to be undertaken. In this case, the continuation phase still has 4 months and the total duration of treatment is 7 months.

Continuation Phase: for 4 months

In this phase, TB patient has to take 2 anti-TB drugs every morning for 4 months under direct observation by the healthcare worker or trained community DOT watchers. The 2 drugs are Rifampicin and Isoniazid (combined tablet, 2FDC).

Category-1a: 2RHZE/4RH (Adult)

Body Weight in Kg	Intensive Phase: 2months, daily	Continuation Phase: 4months, daily
	RHZE (150mg+75mg+400mg+275mg)	RH (150mg+75mg)
30-39	2 tablets	2 tablets
40-54	3 tablets	3 tablets
55-70	4 tablets	4 tablets
>70	5 tablets	5 tablets
Dosage/kg/day	R= 10mg H= 5 mg Z= 25 mg E= 15 mg	

Category-1b: 2RHZE/10RH

This treatment regimen is used for new cases of:

- TB meningitis ; and
- Osteoarticular/spinal TB (joint/bone TB)

This regimen lasts 12 months

Intensive Phase (initial phase): for 2 months

In this phase, TB patient has to take 4 anti-TB drugs every morning for 2 months under direct observation by the healthcare worker or trained community DOT watchers. The 4 drugs are Rifampicin, Isoniazid, Pyrazinamide and Ethambutol (combined tablet, 4FDC).

Continuation Phase: for 10 months

In this phase, TB patient have to take 2 anti-TB drugs every morning for 10 months under direct observation by the healthcare worker or trained community DOT watchers. The 2 drugs are Rifampicin and Isoniazid (combined tablet, 2FDC).

Category-1b: 2RHZE/10RH (Adult)

Body Weight in Kg	Intensive Phase: 2months, daily	Continuation Phase: 10months, daily
	RHZE (150mg+75mg+400mg+275mg)	RH (150mg+75mg)
30-39	2 tablets	2 tablets
40-54	3 tablets	3 tablets
55-70	4 tablets	4 tablets
>70	5 tablets	5 tablets
Dosage/kg/day	R= 10 mg H= 5 mg Z= 25 mg E= 15 mg	

5.4.2. Category 2 (CAT-2):

Category 2: 2RHZES/1RHZE/5RHE

This category of treatment is used for treating:

- Relapse cases
- Treatment after failure
- Treatment after loss to follow-up
- Other previously treated patients

Intensive Phase (initial phase): for 3 months ⁽¹⁾ divided into 2 parts:

- ✓ The first part of initial phase lasts for 2 months using 5 anti-TB drugs. Within this period, the TB patients have to take tablets and injection every morning under direct observation by health worker or trained person/trained community DOT watcher. The 5 drugs are Rifampicin, Isoniazid, Pyrazinamide and Ethambutol (combined tablet) and Streptomycin.
- ✓ The second part of initial phase lasts for 1 month using 4 drugs. Within this period, TB patients take anti-TB drugs every morning under direct observation by health staff or trained person/trained community DOT watcher. The 4 drugs are Rifampicin, Isoniazid, Pyrazinamide and Ethambutol (combined tablet).

(1) At the end of the 3rd month, if sputum smear examination is still positive, then one month more of the initial phase has to be prolonged (the second part). In this case, the continuation phase has only 4 months.

Continuation Phase: for 5 months

In this phase, TB patient has to take 3 anti-TB drugs every morning for 5 months under direct observation by the healthcare worker or trained community DOT watchers. The 3 drugs are Rifampicin and Isoniazid (combined tablet, 2FDC), and Ethambutol.

Remarks: for patient over 60 years, Streptomycin should be used in a dose of 10mg/kg only and Streptomycin is prohibited to use for pregnant women.

Category 2: 2RHZES/1RHZE/5RHE (Adult)

Body Weight in Kg	Intensive Phase: take and inject daily			Continuation phase: take daily	
	1 st and 2 nd month		3 rd month	4 th to 8 th month	
	RHZE	S	RHZE	RH	E
30-39	2 tablets	500 mg	2 tablets	2 tablets	1.5 tablets
40-54	3 tablets	750 mg	3 tablets	3 tablets	2 tablets
55-70	4 tablets	1000 mg	4 tablets	4 tablets	3 tablets
>70	5 tablets	1000 mg	5 tablets	5 tablets	3.5 tablets
Dosage/kg/day	R= 10 mg H= 5 mg Z= 25 mg E= 15 mg S= 15 mg				

RHZE (150+75+400+275 mg); RH (150 mg+75 mg); E 400 mg; S 1,000 mg

Remarks: *Streptomycin is allowed to use by trained healthcare worker at health center*

5.4.3. Pyridoxine (Vitamin B6):

- Vitamin B6 is given to all registered TB patients in order to prevent peripheral neuropathy
- Provide 10 mg daily of Vitamin B6 (1 tab of Vit B6) with anti-TB drugs during TB treatment

5.4.4. Prednisolone

The use of corticosteroids is recommended in extra-pulmonary TB, particularly for TB meningitis and pericarditis. High dose steroid treatment for 2-4 weeks and the taper off gradually over several weeks depending on clinical progress as described below:

- 60mg of Prednisolone (1-2mg/kg) daily for weeks 1-4
- 30mg of Prednisolone (0.5-1mg/kg) daily for weeks 5-8
- 20mg of Prednisolone daily for week 9
- 10 mg of Prednisolone daily for week 10
- 5 mg of Prednisolone daily for week 11 and then stop!

5.5. DOT Approach and Procedure

DOT is one of the 5 components of DOTS strategy. DOT means that patient has to take anti-TB drugs with correct dosages, pills and at every morning under direct observation by healthcare worker or trained community DOT watchers.

Hospitalization DOT: Treatment is done for hospitalized patient at health facilities (hospitals or health center) and TB patient has to take anti-TB drugs under direct observation by healthcare worker.

Ambulatory DOT: Treatment is given to TB patient who comes to health facility at every morning and takes anti-TB drug under direct observation by healthcare worker.

Home delivery DOT: Treatment is provided to TB patient at patient's house under direct observation by healthcare worker or trained person. This home delivery DOT is divided into:

- Home delivery DOT under direct observation by healthcare worker; and

- Home delivery DOT under direct observation by community volunteer, called “DOT in Community”

5.6. Places for TB Treatment

Diagnosed TB patients have right to decide the places where more convenience for their treatment. TB patient can choose TB treatment unit where diagnosis is done or she/he is referred to another health facility for TB treatment, for instance his or her homeland.

5.7. Identifying when TB Disease is No Longer Transmit during Treatment

In some condition, TB patients need the referral letter to use in order to allow them to return to work (e.g. garment factory...). In this case, there is a need for sputum follow up examination result. For bacteriologically confirmed TB cases, sputum examination can be done at the end of 1st month of TB treatment. In principle, patient can return to work until sputum examination result is found negative.

For clinically diagnosed TB patients, they can return to work after starting appropriate TB treatment for one month and no severe symptom/clinical signs occurred.

5.8. Treatment Regimens in Special Conditions

5.8.1. Pregnancy and Breastfeeding

Women in reproductive age should be asked about current or planned pregnancy before starting TB treatment. A pregnant woman should be advised that successful treatment of TB with the standard regimen is important for successful outcome of pregnancy. ***With the exception of Streptomycin, the first line anti-TB drugs are safe for use in pregnancy: streptomycin is ototoxic to the fetus and should not be used during pregnancy.***

A breastfeeding woman who has TB should receive a full course of TB treatment. Timely and properly applied chemotherapy is the best way to prevent transmission of tubercle bacilli to the baby. Mother and baby should stay together and the baby should continue to breastfeed. Pyridoxine supplementation is recommended for all pregnant or breastfeeding women taking Isoniazid.

5.8.2. Liver Disorders

This section covers TB treatment in patients with pre-existing liver disease. Patients with the following conditions can receive the usual TB regimens provided that there is no clinical evidence of chronic liver disease: hepatitis virus carriage, a past history of acute hepatitis, current excessive alcohol consumption.

In patients with unstable or advanced liver disease, liver function tests should be done at the start of treatment, if possible. If the serum Alanine Aminotransferase level is more than 3 times of normal level before the initiation of treatment, the following regimens should be considered.

Possible regimens include:

- Two hepatotoxic drugs (rather than the three in the standard regimen):
 - 9 months of Isoniazid and Rifampicin, plus Ethambutol (until or unless Isoniazid susceptibility is documented) 9RHE
 - 2 months of Isoniazid, Rifampicin, Streptomycin and Ethambutol, followed by 6 months of Isoniazid and Rifampicin (2RHSE/6RH)

- 6–9 months of Rifampicin, Pyrazinamide and Ethambutol (6-9RZE)

- One Hepatotoxic Drug:

- 2 months of Isoniazid, Ethambutol and Streptomycin, followed by 10 months of Isoniazid and Ethambutol (2HSE/10HE)

- No Hepatotoxic Drug:

- 18–24 months of Streptomycin, Ethambutol and a Fluoroquinolone (18-24 S.E.Flu)

Expert consultation is advisable in treating patients with advanced or unstable liver disease. Clinical monitoring (and liver function tests, if possible) of all patients with pre-existing liver disease should be performed during treatment.

5.8.3. Renal Failure and Severe Renal Disorder

The recommended initial TB treatment regimen for patients with renal failure or severe renal disorder is 2 months of Isoniazid, Rifampicin, Pyrazinamide and Ethambutol, followed by 4 months of Isoniazid and Rifampicin (2RHZE/4RH). Isoniazid and Rifampicin are eliminated by biliary excretion, so no change in dosing is necessary. There is significant renal excretion of Ethambutol and metabolites of Pyrazinamide, and doses should therefore be adjusted. Three times per week administration of these two drugs at the following doses is recommended: Pyrazinamide (25 mg/kg), and Ethambutol (15 mg/kg). While receiving Isoniazid, patients with severe renal disorder or failure should also be given Pyridoxine in order to prevent peripheral neuropathy.

Because of an increased risk of nephrotoxicity and ototoxicity, Streptomycin should be avoided in patients with renal failure. If Streptomycin must be used, the dosage is 15 mg/kg, two or three times per week, to a maximum of 1 gram per dose, and serum levels of the drug should be monitored.

5.9. Management of Cases Who Interrupted Treatment

Patients who fail to follow-up as scheduled (by healthcare workers or DOT watcher) should be immediately traced through telephone call or home/workplace visit. Assess the cause of interruption and take action on the agreed solutions.

Interrupted treatment of patient is classified as below:

- 1- If patient interrupted treatment for less than 1 month, continue the treatment and prolong it to compensate for the missing doses.
- 2- If patient interrupted treatment for more than 1 month but less than 2 months, perform sputum smear microscopic examination and result will be as below:
 - a- If the sputum smear result is negative, continue the treatment and prolong to compensate for the missing doses.
 - b- If the sputum smear result is positive, check how long the patient has been treated:
 1. If treatment received is less than 5 months, continue treatment and collect and send sputum to check for drug-resistant TB and if drug-resistant TB is found please follow the instructions of drug-resistant TB.
 2. If treatment received is five 5 months or more, close the treatment card, classify the patient's treatment outcome as "Failure". Collect and send sputum to check for drug-resistant TB and follow the instructions of drug-resistant TB.

3. If the patient interrupted treatment for 2 months or more, classify as having a treatment outcome of “Lost to Follow-up” and close the previous registration. Collect and send sputum to check for drug-resistant TB and follow the instructions of drug-resistant TB.

Length of interruption	Do sputum smear examination if > 1 month interruption	Length of treatment provided	Disposition
Less than 1 month	Continue treatment and prolong to compensate		
More than 1 month	S (-)	Continue treatment and prolong to compensate	
	S (+)	Less than 5 months	Continue treatment and prolong to compensate
		More than 5 months	Classify as “treatment failure”
More than 2 months	Classify as “Loss to Follow up”		

6. MONITORING TB TREATMENT AND REFERRAL SYSTEM

6.1. Principle of Monitoring

For monitoring pulmonary TB patients with bacteriologically confirmed TB or clinically diagnosed TB, the 2 main objectives of the monitoring during TB treatment are:

1. To monitor and record the response to treatment: sputum follow up (pulmonary TB patient with bacteriologically confirmed TB), patient weight and clinical signs and symptoms of patient during TB treatment.
2. To manage other problems and especially detect and manage drug-induced toxicity (or side effect of drugs)

6.2. Monitoring Patients

6.2.1. Bacteriologically Confirmed TB Case

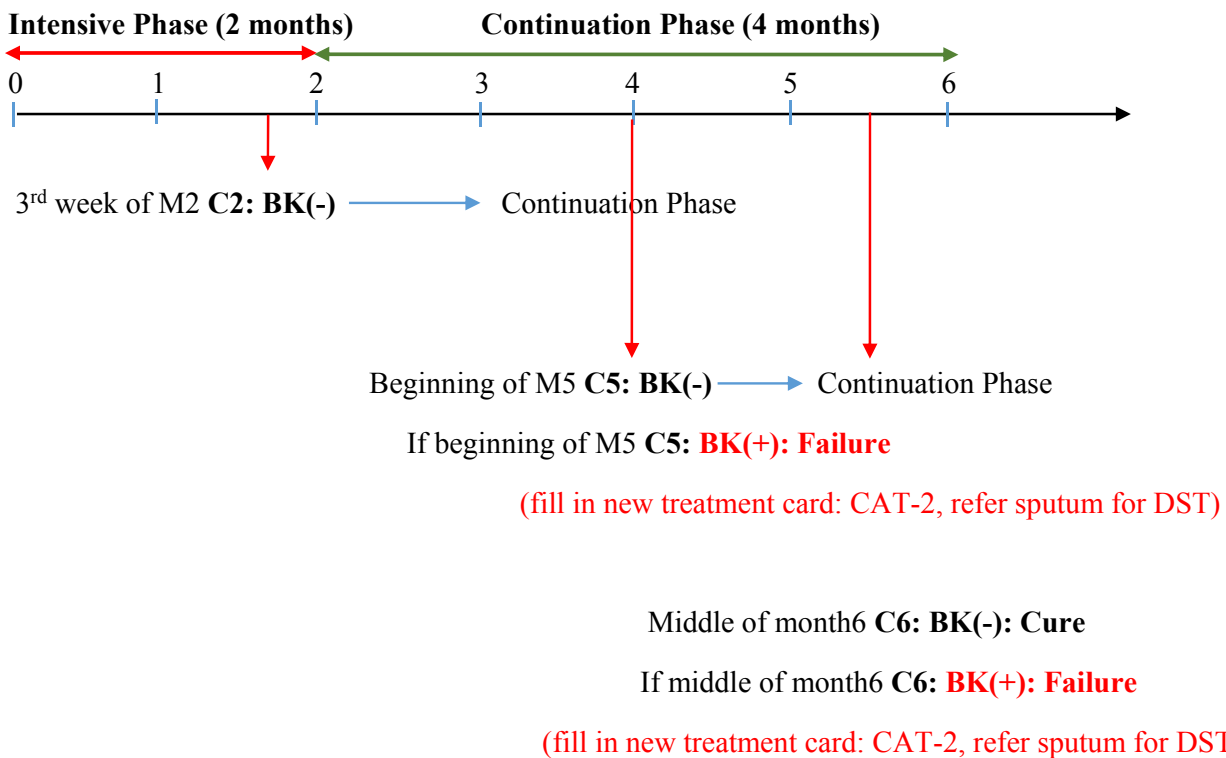
For bacteriologically confirmed TB cases, to see whether the response to TB treatment is good or not, we need to monitor the results of sputum smear microscopic examination as follows:

CAT1 (bacteriologically confirmed case) sputum follow-up at 2nd month C2: BK-)

- **Sputum follow-up at 2nd month (result negative** please see the figure below): sputum follow-up at 2nd month is very important in determining whether patients need further intensive treatment, can enter the continuation phase, or need to do other investigations. This sputum check is made at the end of the 2nd month for patients who stay in the hospital; and at the end of the 3rd week of the 2nd month of treatment for patients treated at the health center. If result of the sputum follow-up is negative, then go to continuation phase and check sputum at 5th month. In case of result of sputum follow-up at 2nd month is positive, please see the detail in next page.

- **Sputum follow-up at 5th month:** If the result of this sputum follow-up is negative, go on for further continuation phase and then check sputum in the middle of the 6th month. But if the result of control sputum is positive, then classify this patient as failure case. Collect and send patient's sputum for drug susceptibility test (DST), and if drug-resistant TB is found please follow the instructions of drug-resistant TB. If it is not as MDR case, fill new treatment card with CAT-2 treatment regimen.

- **Sputum follow-up at the middle of 6th month:** After sputum smear result at 2nd month or at 5th month is negative and the result of this control sputum at this time is also negative, the patient is classified as cured. But if the result of sputum follow-up at 6th month found positive, the patient is considered as failure case. Then collect and send patient's sputum for drug susceptibility test and, if drug-resistant TB is detected, please follow the instructions of drug-resistant TB. If it is not case with drug resistance, please fill in new treatment card with CAT-2 treatment regimen.



Sputum Follow-up at 2nd Month (C2: **BK(+)**)

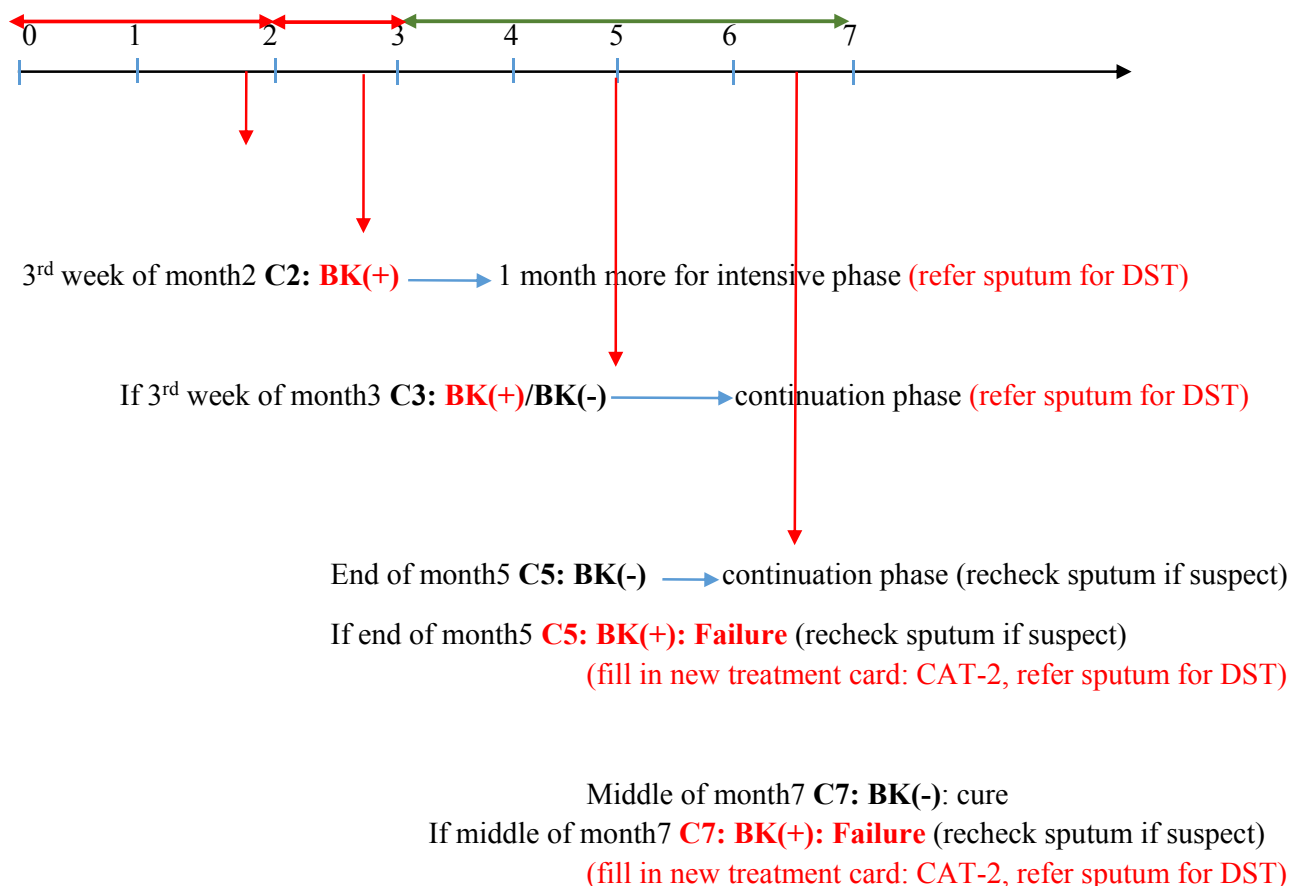
- **Sputum follow-up at 2nd month (result positive)** please see the figure below): In this situation, patients should continue intensive treatment one month more and the patient's sputum should be sent for DST for drug-resistant TB. If the result tests negative for drug-resistant TB, continue the treatment and perform smear sputum examination in the 3rd week of the 3rd month of the treatment. If drug-resistant TB is detected, please follow the instructions of drug-resistant TB.

- **Sputum follow-up at the 3rd week of 3rd month:** In this case, if the result of sputum follow-up tests negative, the treatment goes into the continuation phase. There is a need to perform sputum follow-up at late 5th month of the treatment. And if the result of sputum exam is positive, please collect and send patient's sputum for DST for drug resistance TB and, if TB drug resistance is detected, please follow the instructions of drug-resistant TB.

- **Sputum follow-up at 5th month:** If the result of sputum exam is negative, continue further continuation phase of treatment and perform the final sputum check in the middle of the 7th month of treatment. But if the result of sputum at 5th month is positive, the patient is classified as the failure case. In case, please collect and send sputum for DST for drug resistant TB, and if drug-resistant TB is detected, please follow the instructions of drug-resistant TB. If it is not a case with drug resistance, please fill out new treatment card with CAT-2 regimen treatment.

- **Sputum follow-up at the middle of 7th month:** If the result of sputum exam is negative, the patient is classified cured. But if the result of sputum is positive, the patients is classified a failure case. In case, please collect and send patient's sputum for DST and, if drug resistance is detected, please follow the instructions. If it is not a case with drug resistance, please fill in a new treatment card with CAT-2 treatment regimen.

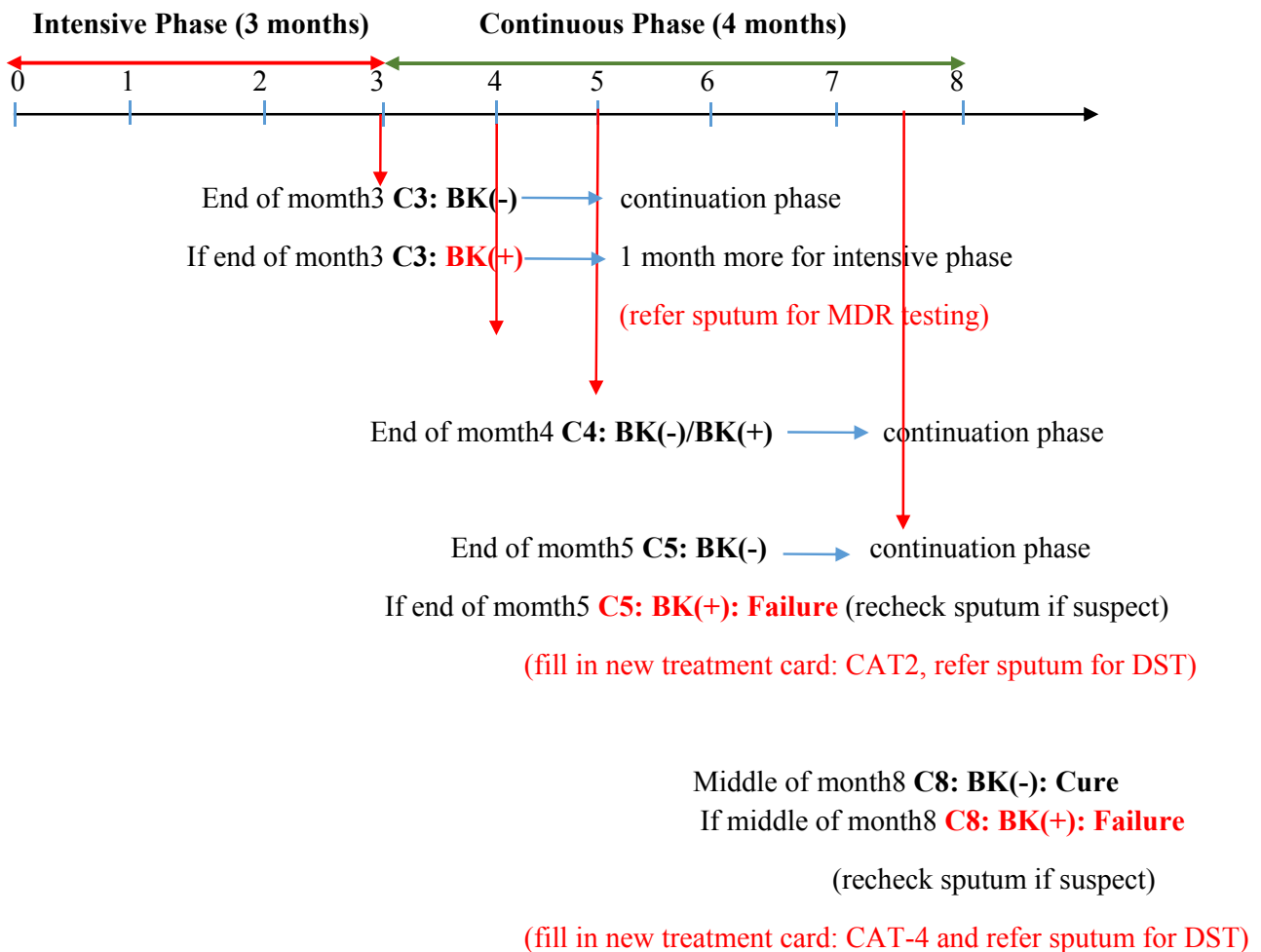
Intensive Phase (3 months) Continuous Phase (4 months)



Remark: if intensive phase is extended one month more, the total duration of treatment will be 7 months.

CAT-2 (bacteriologically confirmed case)

- **Sputum follow-up at 3rd month** (please see the figure below): after 3 months of intensive phase, if follow-up sputum at the 3rd month is negative, please go into the continuation phase. If the 3rd month sputum exam is positive, the intensive phase has to be extended for additional one month; and then collect and send patient's sputum for DST. The next sputum follow-up will be done at the 3rd week of 4th month of treatment.
- **Sputum follow-up at 3rd week of 4th month**: In this case, irrespective of the result of sputum exam, the patient should come into continuation phase. Meanwhile, please collect and send patient's sputum for DST. The schedule of the next sputum follow-up will be at 5th month of treatment.
- **Sputum follow-up at 5th month**: if the result of this sputum exam is positive, the patients is classified a failure case. Then collect and send patient's sputum for DST for drug resistance. If the result of the sputum is negative, the next sputum check will be in the middle of the 8th month.
- **Sputum follow-up at the middle of 8th month**: if result of this sputum exam is negative, the patient is classified as cured. If the result of sputum is positive and the patient is classified as failure case, please collect and send patient's sputum for DST and manage the case based on the result of DST.



6.2.2. Clinically Diagnosed TB Case

For these patients, the monitoring of their treatment is based on clinical monitoring, especially weighing patient and other clinical symptoms and signs. If the disease becomes worse, the health staff should closely monitor the results of the sputum at the end of 2nd month of treatment, check if the patient is not strictly following the principles of treatment, or bacteriologically positive patients was misclassified initially as clinically diagnosed TB. They should monitor drug intake and side effects in order to cope with the issues timely.

6.2.3. Extra-Pulmonary TB Case

The same to clinically diagnosed pulmonary TB cases, follow-up treatment is based on clinical observations, monitoring patient weight, regular drug intake, and side effects of medicines ((please refer to the procedure of treatment monitoring based on treatment regimen and types of diseases)).

6.3. TB Service Referral System

In Cambodia, TB service is available nationwide. TB service includes public sector, private sector and other community providers. In principle, TB service is to identify presumptive TB (TB suspect), diagnose TB disease and treatment, including TB disease with or without TB-drugs resistance.

Main reasons of referring TB suspect and TB disease are as followings:

1. TB Diagnosis:

This includes the cases referred from the community to Health Centre (HC) or Referral Hospital (RH), and from the private sector or HC to RH (for example, presumptive TB in children).

2. TB Registration and Treatment:

This includes the cases referred from the diagnostic center after confirming TB disease to treatment center for registration of TB treatment (for example, refer TB cases from RH to HC for treatment).

3. TB Continued-treatment:

This refers to the case referred from one treatment center to another treatment center for continued-treatment (for example, refer TB drugs-resistance to community for continued-treatment, and refer TB in prison to HC etc.).

4. TB-drugs Resistance Diagnosis:

In this case, it refers from HC or RH to Xpert Machine center etc.

5. TB-Drugs Side-effect Solution:

This refers to the case referred from community to HC, or HC to RH, or RH to another RH with higher capacity in solving the side effect of the medicines.

6. TB Screening:

In this case, systematic collaboration between referrer and receiver (services related) is needed for all referred cases with referral slips, record/registrations and timely feedback results.

7. TUBERCULOSIS LABORATORY

7.1. Tuberculosis Laboratory Networks

TB bacteriology is an important part of the National TB Program as well as of the DOTS strategy. Both diagnosis and monitoring of treatment for TB mainly rely on bacteriological examination.

The National TB Reference Laboratory (NTRL) is a part of and belongs to NTP; and is located in the CENAT premises. There are 21 staff members working for the NTRL where besides sputum smear microscopic and Xpert® MTB/RIF tests, TB isolation by solid and liquid culture, identification test, and drug susceptibility testing (DST) for first and second-line drugs can be performed. Hence, the NTRL is responsible for planning, training and supervision of TB Laboratory services. It is also responsible for the External Quality Assurance (EQA) of sputum smear microscopy services, which are provided through a network of 215 microscopic centers covering the entire country. As well as NTRL, two TB laboratories in Battambang and Kampong Cham are capable of performing Xpert® MTB/RIF test and TB culture. Moreover, the Kampong Cham TB Laboratory is capable of performing of first-line drug susceptibility testing as well. Connecting to the international network, NTRL is working with the Research Institute of Tuberculosis of Japan, which is a Supranational Reference Laboratory (SNRL).

A new diagnostic tool, called Xpert® MTB / RIF, has been put into operation in Cambodia since 2011 after an authorization by WHO in late 2010, and currently 58 sets are working, out of which, 49 sets are used in routine activities. The National Strategic Plan 2014-2020 of NTP recognizes a need to strengthen and expand the machine to all referral hospital throughout the country at the latest by the end of 2018. Therefore, NTP is planning to purchase 20 sets every year to continuously provide for the referral hospitals with a higher priority.

7.2. Roles and Responsibilities of TB Laboratories

The roles and responsibilities of TB laboratory vary with the levels: reference level and operational level. NTRL undertakes more important and leading roles and responsibilities than operational laboratories.

A. Roles and Responsibilities of National TB Reference Laboratory (NTRL):

- Develop guidelines and plans on TB laboratory activities
- Diagnose TB and treatment follow-up by sputum smear microscopy
- Diagnose TB by Xpert® MTB/RIF test
- TB culture
- Perform TB drug resistant testing
- Perform quality assurance
- Manage and maintain TB laboratory equipment
- Record, report, analyze and utilize TB laboratory information
- Train for TB laboratory staff
- Conduct researches

B. Roles and Responsibilities of Operational TB Laboratories:

- Diagnose TB and treatment follow-up by sputum smear microscopic examination

- Diagnose TB by Xpert® MTB/RIF test
- Perform quality assurance
- Record, report, analyze and utilize TB laboratory information
- Manage and maintain TB laboratory equipment
- Participate in TB laboratory staff training
- Participate in researches

7.3. Role of Diagnostic Tools and Treatment Follow-up by TB Bacteriology

NTP has stressed the importance of WHO's guidance on choosing methods and tools to diagnose and monitor treatment of TB on a practical basis as follows:

7.3.1. Roles of Microscopy in TB Diagnosis

It is widely recognized that smear microscopy has been used for TB diagnosis for a long time in spite of its lower sensitivities. However, NTP still continues to use smear microscopy for TB diagnosis and treatment monitoring since there are not any devices that are affordable and acceptable to replace it completely in practice. In addition to conventional light microscope with Ziehl-Neelsen staining method (Fuchin solution, Acid alcohol 3%, Methylene Blue 0.3%), In 2010, NTP introduced Fluorescent microscope with Light-Emitting Diode (LED) with Auramine-O staining method (Auramine-O 0.1%, Acid alcohol 0.5%, Potassium permanganate 0.5%). Currently, the NTP is running 215 microscopic centers including 29 LED microscopes with Auramine-O staining method. NTP recognizes the basic role of microscope for TB diagnosis and treatment monitoring and still uses smear microscopy as a major diagnostic tool like in other countries.

7.3.2. Roles of GeneXpert (Xpert® MTB/RIF test)

WHO officially endorsed Xpert® MTB/RIF test in 2010. This test is simple, highly effective with short turnaround time of less than two hours. This new tests can detect not only TB but also Rifampicin-resistant TB. Based on its specific characteristics, NTP decided to roll out the machines to use for case detection among presumptive MDR-TB, people living with HIV/AIDS, and other high risk groups (the elderly from 55 years old, close contact with smear positive pulmonary TB, and diabetes mellitus patients) in addition to active case finding (ACF) activities for TB.

7.3.3. Roles of TB Culture and Drug Susceptibility Testing (DST)

In late 1999, NTP introduced TB culture with solid medium in NTRL under the technical assistance from JICA. After that, NTP strengthen and expand the laboratory capacity step by step; culture exams with liquid medium (MGIT) for more rapid isolation of MTB started at CENAT and Battambang TB laboratories in 2011, and then at Kampong Cham TB laboratory in 2014.

The first-line drug susceptibility testing (DST) using liquid medium (MGIT) was introduced first at CENAT, and later in Kampong Cham TB laboratory in 2014. The second-line DST using liquid medium (MGIT) was evaluated by the supranational TB reference laboratory, Research Institute of Tuberculosis (RIT) in Japan and was put into service in 2014.

Culture and DST are tools that NTP guided and introduced to detect MDR-TB for clinical use and conduct various researches such as national TB prevalence survey and national drug resistance survey.

8. MANAGEMENT OF SUPPLY FOR DRUGS AND REAGENTS

8.1. Introduction

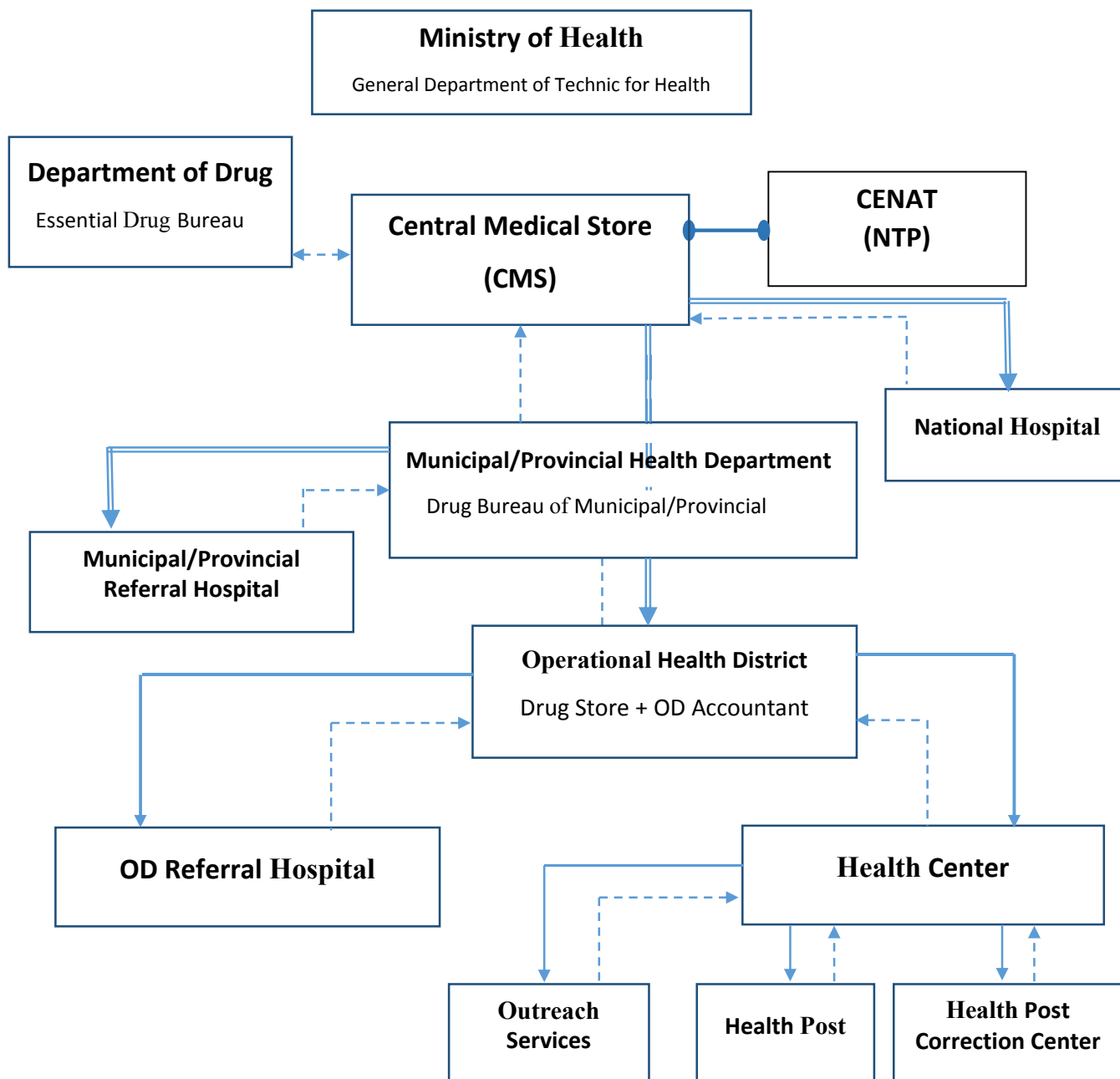
Management of supply for anti-TB drugs and reagents for diagnostic and TB laboratory are key elements of NTP. An uninterrupted supply of TB drugs, reagents, equipments diagnostic supplies is essential for the sustained provision of quality TB diagnostic and treatment services through DOTS in all service delivery facilities nationwide. It promotes patient care, fosters the credibility with public health services, and increases the patients' trust and participation in the program. Ensuring uninterrupted of TB drugs, reagents, equipment, diagnostic supplies leads to better treatment success and reduced TB deaths. This chapter aims to provide the principles for TB networks to ensure that uninterrupted and sustained supply of quality-assured anti-TB drugs, laboratory reagents and equipment diagnostic supply are fundamental to TB control. For this purpose in the chapter, an effective drug, laboratory reagents and equipment diagnostic supply and management system are essential for OD supervisor and TB staff. A reliable system of procurement and distribution of them to all relevant health facilities should be in place.

8.2. Policies

- NTP will monitor the consumption of current anti-TB drugs, TB diagnostics/laboratory supplies, estimate future requirement and provide information about anticipated requirement and estimated budget.
- MoH will ensure that there is uninterrupted supply of the 1st and 2nd line anti-TB drugs, TB diagnostics/lab reagents and materials and consumables for TB network.
- NTP will take appropriate and immediate actions to correct shortages of drugs and diagnostic/laboratory supplies and to prevent similar situations in the future.
- MoH will ensure that good quality drugs and diagnostic/laboratory materials and consumables are supplied to the Central Medical Store (CMS), properly stored and timely distributed.
- Municipal/provincial RH pharmacies will be responsible for proper storage and timely distribution to the TB units providing DOTS; and also be responsible for maintaining the buffer and security stocks and monitoring expiry dates of drugs and laboratory supplies.
- OD pharmacies will be responsible for proper storage and timely distribution to the health centers providing DOTS; and also be responsible for maintaining the buffer and security stocks and monitoring expiry dates of drugs and laboratory supplies.
- NTP will ensure that TB drugs are used according to the national protocol and should adhere to the rational use of drugs.

8.3 Management Procedures

Management and dispensing of anti-TB drugs, reagents, equipment, diagnostic materials and TB laboratory supplies is included in the supply system of drugs, medical equipments and health products of Ministry of Health as presenting in below algorithm.



Remarks:

- ==> The supplies of drugs, medical equipment and health product from MoH to OD and Municipal/Provincial RH with coordination by M/PHD's bureau of drug.
- > The supplies of drugs, medical equipment and health product from OD to subordinated health facilities.
- - -> System of stock in and out reports, medical equipment and health product.
- < - - -> Information system between CMS and EDB (Essential Drug Bureau).
- — ● Information system between CENAT and CMS.

8.4. Management of Medicines, Reagents, Equipment and TB Diagnostic Supplies

8.4.1. Calculation of Medicines, Reagents, Equipment and TB Diagnostic Supplies

-Estimate the number of presumptive TB patients expected to be tested and treated for the order period. The number of patients treated in the previous quarter can be used to guide the estimation. Alternatively, number of patients treated in the same quarter last year can be used to guide the estimation for required quarter/month. Consider special activities (intensified case finding, semi-active case finding, health promotion activities, etc.) that may result in more patients diagnosed and requiring treatment.

-Calculate the quantity of buffer stock equivalent to one quarter for Operational District (OD) and one month for Health Center (HC). For reagents and equipments diagnostic supplies, the calculation is the same as the calculation of drugs.

-Fill-up order request form and submit to Provincial Health Department (PHD)

8.4.2. Receiving Medicines, Reagents, Equipment and TB Diagnostic Supplies

The person who is in charge of drug management should perform the following procedures when receiving medicines, reagents, equipment and diagnostic supplies as below:

- Obtain the supply receipt forms accompanying the delivery
- Check the followings: quantities for each item listed, medicine labels for name, strength, and dosage form, presence of damages, and the expiry dates
- Record the discrepancies noted and send feedback to the distributing unit
- Record the quantity of good items and quantity that are missing or damaged on the receipt form
- Sign the receipt form. If possible, ask a fellow health worker to verify the quantities received and to sign the forms
- Keep a copy and file the delivery receipt form for your records
- Encode data into Operational District Drug Information Database (ODDID)

8.4.3. Storage of Medicines, Reagents, Equipment and Diagnostic Supplies

- Maintain clean storeroom with regular cleaning, prohibit food consumption where stocks are kept. Perform regular inspections to check for potential thievery. Check hygiene, drug store and safety system.
- Organize medicine's name so that the label and expiry date can be easily read
- Promote air circulation in the storage room, high ceilings with vents; if feasible, install air conditioner, an exhaust fan, or a window or air vents
- Keep medicine containers closed to avoid exposure to humid air
- Store medicines and supplies under recommended storage temperatures
- Store medicines only on shelves or pallets, never on the floor
- Keep medicine "First Expire, First Out (FEFO)"
- Return excess medicines for redistribution

8.4.4. Maintaining Records for Medicines, Equipment and TB diagnostic Supplies

The health facility should maintain proper records for medicines, reagents and diagnostic equipment to facilitate monitoring of available stocks and consumption.

- Maintain and update medicines, reagents, equipments and diagnostic supply stock records.
- Perform physical inventory medicines, reagents, equipment and diagnostic supplies
- Data entry and update data regularly into ODDID system
- Limitation of the authorized persons for the access to store room

8.4.5. Need for Health Center

Distribution will be made based on the need request for one month in addition to the buffer stock for one month.

8.5. Rational Uses OF Medicines

Arrangement and uninterrupted uptake of drugs for TB patients is an important role of TB staff in health facilities. Treatment should follow the TB standard treatment regimens and DOTS implementation as stated in the technical guidelines of NTP and treatment process needs to be recorded in TB treatment cards and TB register book. Regarding the use of reagents, equipment and diagnostic supplies, please refer to sections for TB case detection and treatment.

9. INFORMATION SYSTEM

9.1. General Principle

Health information system aims to provide necessary information for health system management including information for planning, monitoring and evaluation of the implementation of health activities. Health information system includes mechanisms and forms for collecting, analyzing, and using information for:

- Health managers and planners at all levels to prepare plans for programmatic, financial, monitoring and evaluation, and health program coordination.
- Other health workers such as researchers, educators, and other personnel for their own works.
- Policy makers, socio-economic planners, as well as general public entities in both government and non-government.

The current health information system in Ministry of Health has been updated from paper-based to electronic recording and reporting system, which helps to facilitate health information users and speed up the reporting flow from health facility level to central levels. On the other hand, health information system of TB control activities is a part of general health information system and is one of the five important components of DOTS. TB health information system emphasizes on three points as follows:

- Collecting, recording and reporting information through the routine system;
- Collecting, recording and reporting information from other sources; and
- Analyzing and using information

Current TB information is recorded and reported in paper-based format from health facilities. According to the revised definitions and reporting forms updated in 2014, the current TB information system planned to be shifted from paper-based to electronic recording and reporting system for individual TB patient in order to help the users. This health information update will improve data collection, reduce the delay of sending reports from health facilities to central level, and help the decision making on time. The electronic recording and reporting system will be introduced from 2018 onwards.

9.2. Data Collecting, Recording and Reporting through Routine System

The following recording and reporting forms are the documents currently used by NTP for drug-susceptible TB and drug-resistant TB in both adults and children (all ages). These documents are used at health facilities with DOTS services. These documents include:

- TB Laboratory Request Form (Lab TB 01)
- Drug-resistant TB Laboratory Request Form (Lab TB 02)
- Presumptive TB Book (Lab TB 03)
- Laboratory Register for Smear and Xpert (Lab TB 04)
- Laboratory Register for Xpert, Culture and DST (Lab TB 05)
- TB Treatment Card (White card for service provider)
- TB Patient Card (Red card for TB patient)
- DOT Observer Card (Yellow card for DOT Watcher)
- TB Patient Book
- TB Register Book

- Referral Slip
- Presumptive MDR-TB Book
- MDR-TB Treatment Card
- MDR-TB ID Card
- MDR-TB DOT Observer Card
- MDR-TB Patient Register
- TB Monthly Report
- TB Quarterly Report

9.2.1. TB Laboratory Request Form (Lab TB 01)

This form is used to record information on all requests for sputum/specimen examination including TB diagnosis and treatment follow-up for every TB patient. This request form is used by health staff at OPD/IPD (health center, health post and referral hospital).

9.2.2. Drug-resistant TB Laboratory Request Form (Lab TB 02)

This form is used to record information on all requests for sputum/specimen examination including TB diagnosis and treatment follow-up for high risk groups of MDR-TB. This request form is used by health staff responsible for MDR-TB at referral hospital.

9.2.3. Presumptive TB Book (Lab TB 03)

(Sputum collection, slide sending and presumptive extra-pulmonary TB)

This book is used to record all information related to presumptive TB patients and/or TB patients with both pulmonary and extra-pulmonary TB at OPD, admitted patients in the hospital, and anyone who are screened in community when they have or have no specimen for smear microscopic examination and/or GeneXpert test to diagnose or follow-up the treatment efficacy. This book is used at OPD (health center or health post) and TB ward of the referral hospital.

9.2.4. Laboratory Register for Smear and Xpert (Lab TB 04)

This register is used to record all information related to presumptive TB patients and/or TB patients with both pulmonary and extra-pulmonary whose sputum were collected and requested by physician for TB diagnosis and/or treatment follow-up. This laboratory register is used at TB laboratory.

9.2.5. Laboratory Register for Xpert, Culture and DST (Lab TB 05)

This register is used to record all information related to presumptive MDR-TB patients whose specimen were collected and requested for culture and drug susceptibility tests. This laboratory register is used at culture center.

9.2.6. Treatment Cards

9.2.6.1. TB Treatment Card (White card)

This card is used as a document to record general information on TB patients as well as other information related TB diagnosis, treatment and treatment follow-up for each patient from the beginning to the end of TB treatment. This TB treatment card is used and kept by TB health workers (health post, health center, referral hospital) to record all TB patients in both adults and children who are treated at TB unit or in the community.

9.2.6.2. TB Patient Card (Red card)

The same as TB treatment card, this card is use to record information related to TB treatment of a patient from the beginning to the completion. This patient or red card is used by TB patient him/herself and kept at his/her own home. TB health workers need to explain TB patient about how to keep this treatment card well during the treatment period and after the completion of treatment.

9.2.6.3. DOT Observer Card (Yellow card)

The same as TB patient card, this card is use to record information on a patient from the beginning to the end of TB treatment. This card is kept at DOT-provider or DOT-watcher. TB health workers also need to tell DOT watchers to bring yellow card or red card every time when they come to collect anti-TB drugs from TB unit.

9.2.7. TB Patient Book

This book is used to record important information about TB diagnosis and treatment for every patient registered at any health facility (health center or referral hospital). The TB register book is used and kept by health workers responsible for TB (health post, health center or referral hospital) to record information on TB patients in both adults and children who are treated at TB unit or in the community. The TB patient book has 52 columns which require TB staff to fill in all information on TB patient without missing any information in a timely manner. Please use the information in the treatment card of TB patient to complete both on the left and right sides of TB patient book.

9.2.8. TB Register Book

This register book is for OD TB supervisor to record important information on TB diagnosis and treatment for every TB patient treated in the OD by extracting information from all TB patient books within the OD. TB register book has 53 columns which require TB staff to fill in all information on TB patient without missing any information in a timely manner. Please use the information in the TB patient book to complete both on the left and right sides of TB register book.

9.2.9. Referral Slip

This slip is used for referring TB patients from one treatment center to another.

9.2.10. Presumptive MDR-TB Book

This presumptive MDR-TB book is used to record general information related to presumptive MDR-TB patients at referral hospital.

9.2.11. MDR-TB Treatment Card

MDR-TB treatment card is used to record general information on MDR-TB patients as well as other information related MDR-TB diagnosis, treatment and treatment follow-up from the beginning to the end of TB treatment.

9.2.12. MDR-TB ID Card

The same as MDR-TB treatment card, this MDR-TB ID card is used to record general information on each MDR-TB patient from the beginning to the end of MDR-TB treatment. This MDR-TB treatment card is used and kept by MDR-TB patients themselves. TB health workers need to explain patient about

how to keep this MDR-TB treatment card well during the treatment period and after the completion of treatment.

9.2.13. MDR-TB DOT Observer Card

The same as MDR-TB treatment card, the MDR-TB observer card is used to record information on each MDR-TB patient from the beginning to the end of MDR-TB treatment. This card is used and kept by MDR-TB observer.

9.2.14. MDR-TB Patient Register

This MDR-TB patient register is used to record information on MDR-TB patients related to MDR-TB diagnosis and treatment who were admitted at MDR-TB treatment sites (provincial referral hospital). This MDR-TB patient register is used and kept by MDR-TB health workers of the provincial referral hospital.

9.2.15. Monthly TB Report

Monthly TB report needs to be filled by health staff responsible for TB at each facility in collaboration with other concerned people such as OPD staff, store keeper of retailed pharmacy or laboratory staff, etc. This monthly TB report needs to be made every month and sent to OD TB supervisor in a regular basis by the 10th day of the next month to the reporting month. The monthly report will provide information as below (see Annex...):

- Monthly report on case finding
- Case notification
- TB/HIV collaborative activities
- Report on screening and referral activity of drug-resistant TB suspects
- Drug inventory and requirement
- Laboratory activities
- Other activities
- Comments and suggestions

9.2.16. Quarterly TB Report

Quarterly TB report needs to be filled by OD TB supervisor in collaboration with OD store keeper/provincial referral hospital store keeper and laboratory staff at each laboratory. This quarterly TB report needs to be started in filling information in the first week of the next month to the relevant quarter. This quarterly TB report needs to be made every quarter and sent to NTP in a regular basis by the 15th day of the first month next to the relevant quarter. The quarterly report will provide information as below (see the Annex...):

- Number of presumptive TB cases examined and source of referral
- Case notification
- Sputum conversion at the end of intensive phase
- Treatment outcomes

- TB/HIV collaborative activities
- Drug inventory and requirement
- Inventory of laboratory materials and other equipment
- Laboratory activities
- Inventory and IEC materials

9.3. Information from Other Sources

In order to fulfill the need for comprehensive information on TB control activities, NTP requires other information, which is not routinely captured, by specific data collection system. This includes information on TB drug and health product management; finance; and human resource as well as other information acquired by studies or researches such as prevalence survey, drug-resistant TB survey, study on knowledge, attitude and practices in health care seeking; and economic related surveys etc.

9.4. Information Analysis and Use

The same as general information, TB health information is designed to be used at all levels of health system both nationally and internationally. In order to make it useable, the data need to be prepared as measurable and comparable indicators; e.g. a case notification rate of new TB cases is an important indicator of TB control activity.

9.4.1. Definition of Indicator

Indicator is a tool to quantify the phenomenon (event) to be assessed. The indicator selection is based on what to measure. In general, indicators will lead to:

- Assess situation at the beginning of program or work
- Measure the changes that may happen in a specific period

Indicator selection is crucial. Indicators must be responded to the questions raised and should be flexible and specific to the phenomenon to be measured; while the computation will vary depending on the types of the indicator.

9.4.2. Indicator Analysis

At this stage, the collected information is organized in a comparable and timely manner so that its change and trend can be assessed:

- With targets
- With reference figure or standard

Or can be compared with other groups such as:

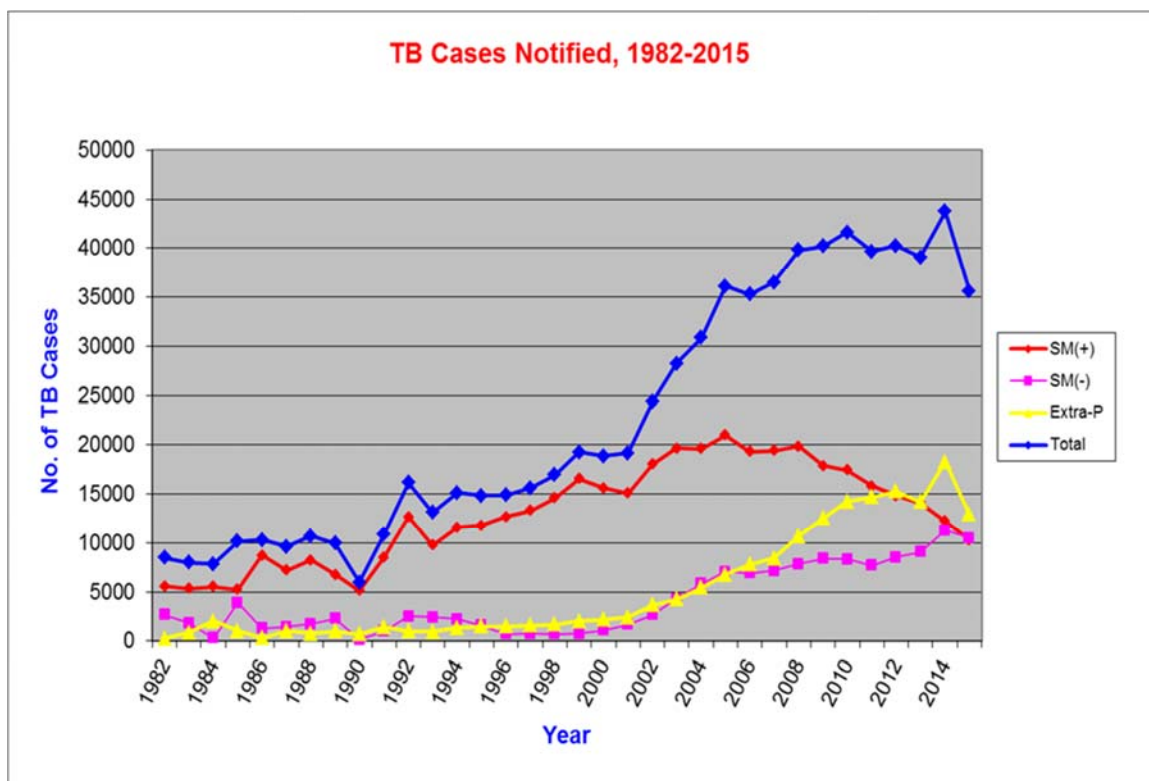
- New cases by types of disease or by age group
- Populations in different locations of the relevant areas, etc.

These comparisons can find: Differences, Similarities and Changes.

9.4.3. Pictorial Representation of Indicators

Once indicators have been computed, they need to be organized and displayed in a way that is easy to understand. This will help to see the figures clearly and is easy to compare. The indicators are most frequently presented in table, graph, histogram, diagram, bar chart, pie chart and maps.

E.g. Graph of TB case notification from 1982 to 2014 in Cambodia.



9.4.4. Indicator Interpretation

It is not an easy stage that similarities, differences, and variations found need to be explained. It is necessary to explore the reasons and factors that relate to the event and that contribute to the changes of the figures. However, at any circumstances, the figures need to be verified in order to avoid miscalculation or improper pointing of findings.

For example, the above comparable graph shows that case notification of bacteriologically confirmed TB cases has declined in the recent years. The question is “what factors that lead to this achievement?” Answer could be:

- Presumptive TB patients accessed to TB service on time
- Smear microscopic diagnosis was limited
- Living condition of people in the country was improved

Solution could be:

- Active TB case findings especially among high risk groups should be strengthened.
- Using GeneXpert test as a diagnostic tool for TB should be promoted.

10. MULTI-DRUG RESISTANT TUBERCULOSIS

10.1. Introduction

This section aims to provide the information on the prevalence of Multi-drug resistant TB (MDR-TB), the technical definition of drug resistant TB, MDR-TB suspect screening and referral mechanism, and MDR-TB treatment and follow up in Cambodia.

Results from the second drug resistance surveillance (DRS) 2006-2007 indicate that the problem of drug resistant TB in Cambodia is relatively low, although an increase was noted in the prevalence of MDR-TB since the first survey in 2001. The combined MDR rate increased from 0% to 1.4% among new cases, and from 3.1% to 11% in previously treated cases. Although its causes are microbial, clinical and programmatic, DR-TB is essentially a man-made phenomenon.

The programmatic management of drug resistant TB (PMDT) has been integrated into the DOTS strategy of the NTP, ensuring that MDR-TB prevention and control is not a separate and parallel undertaking but rather is integral to the NTP. Drug resistant TB cases are classified in categories based on drug susceptibility testing (DST) of clinical isolates confirmed to be *M. tuberculosis*:

- Mono-resistance: resistance to one first-line anti-TB drug only
- Poly-drug resistance: resistance to more than one first-line anti-TB drug (other than both Isoniazid and Rifampicin)
- Multi-drug resistance: resistance to at least both Isoniazid and Rifampicin
- Extensive drug resistance: resistance to any fluoroquinolone and to at least one of three second-line injectable drugs (Capreomycin, Kanamycin and Amikacin), in addition to multi-drug resistance
- Rifampicin resistance: resistance to Rifampicin detected using phenotypic or genotypic methods, with or without resistance to other anti-TB drugs. It includes any resistance to Rifampicin, whether mono-resistance, multidrug resistance, poly-drug resistance or extensive drug resistance.

10.2. Case Finding Strategies for Drug-resistant TB

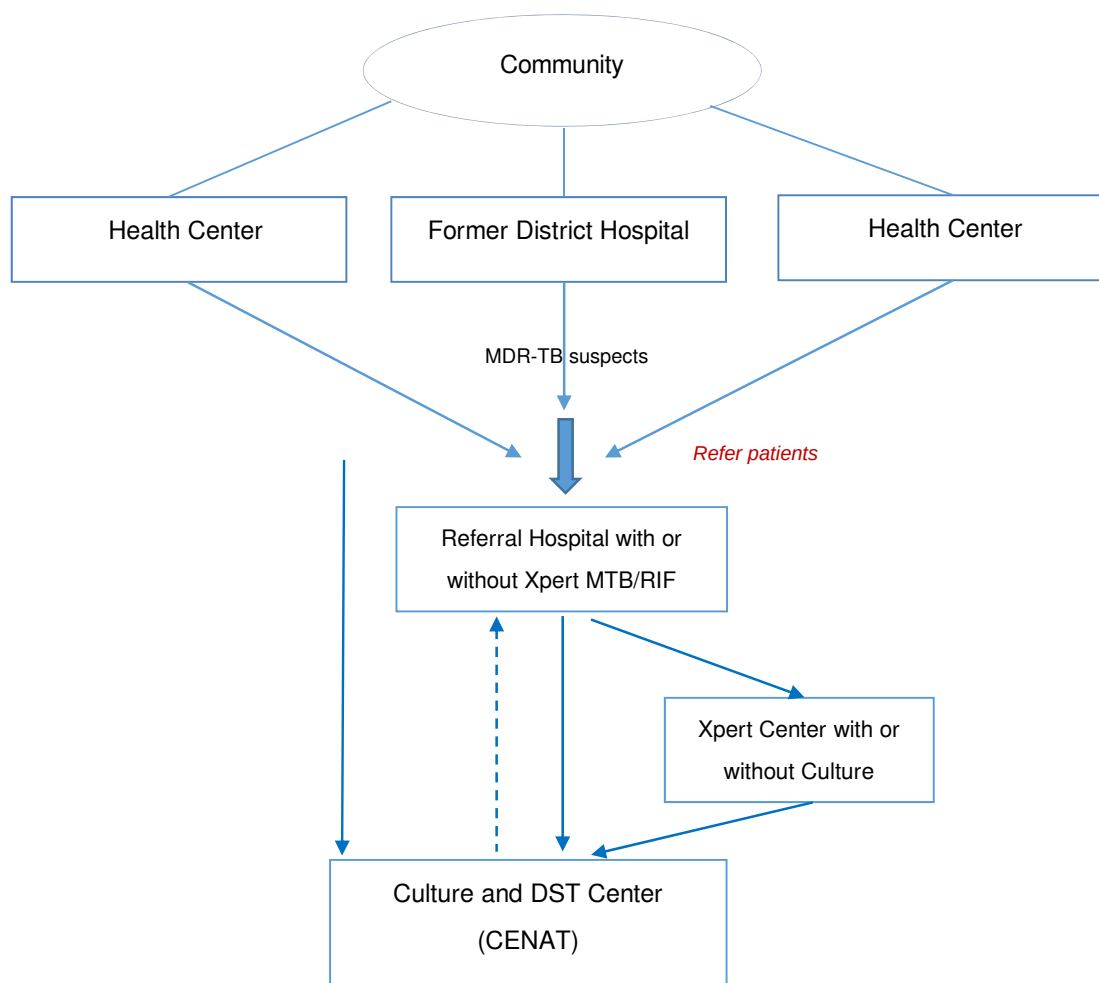
DR-TB is confirmed through drug susceptibility testing (DST) which shows that the infecting isolates of *M. tuberculosis* grow in vitro in the presence of anti-TB drug and through molecular technique like line probe assay (LPA), or Xpert® MTB/RIF assay using the GeneXpert® platform for the rapid diagnosis of R-resistance.

NTP currently prioritizes the high risk group for testing MDR-TB (DST or Xpert MTB/RIF). Those high risk groups are called MDR-TB suspects which consist of the following:

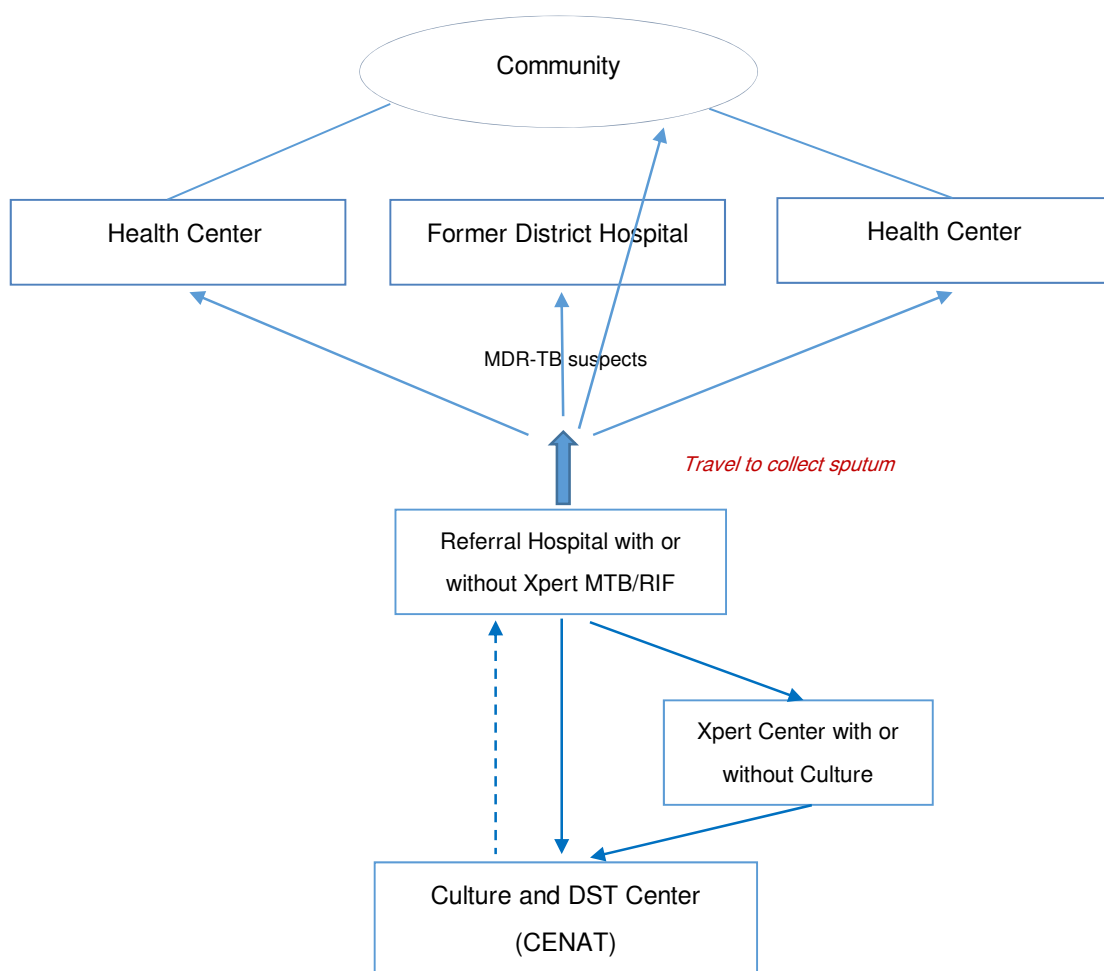
- Pulmonary TB previously treated TB cases
- Non-converters at month 2/3 of treatment with the first line anti-TB drugs
- Symptomatic contacts of DR-TB cases
- Smear positive TB infected by HIV (TB/HIV cases)

MDR-TB suspects are identified either by the health care workers (HCWs) at the periphery, i.e. by HC or referral hospital staff or by OD/provincial TB Supervisors through patient's information such as TB treatment history or risk factors. After identification of a presumptive MDR-TB case, one of the three options for sputum collection may be done in order to collect and transport sputum samples for Xpert or Culture/DST (please refer to MDR-TB Suspect Screening SOP for detail information).

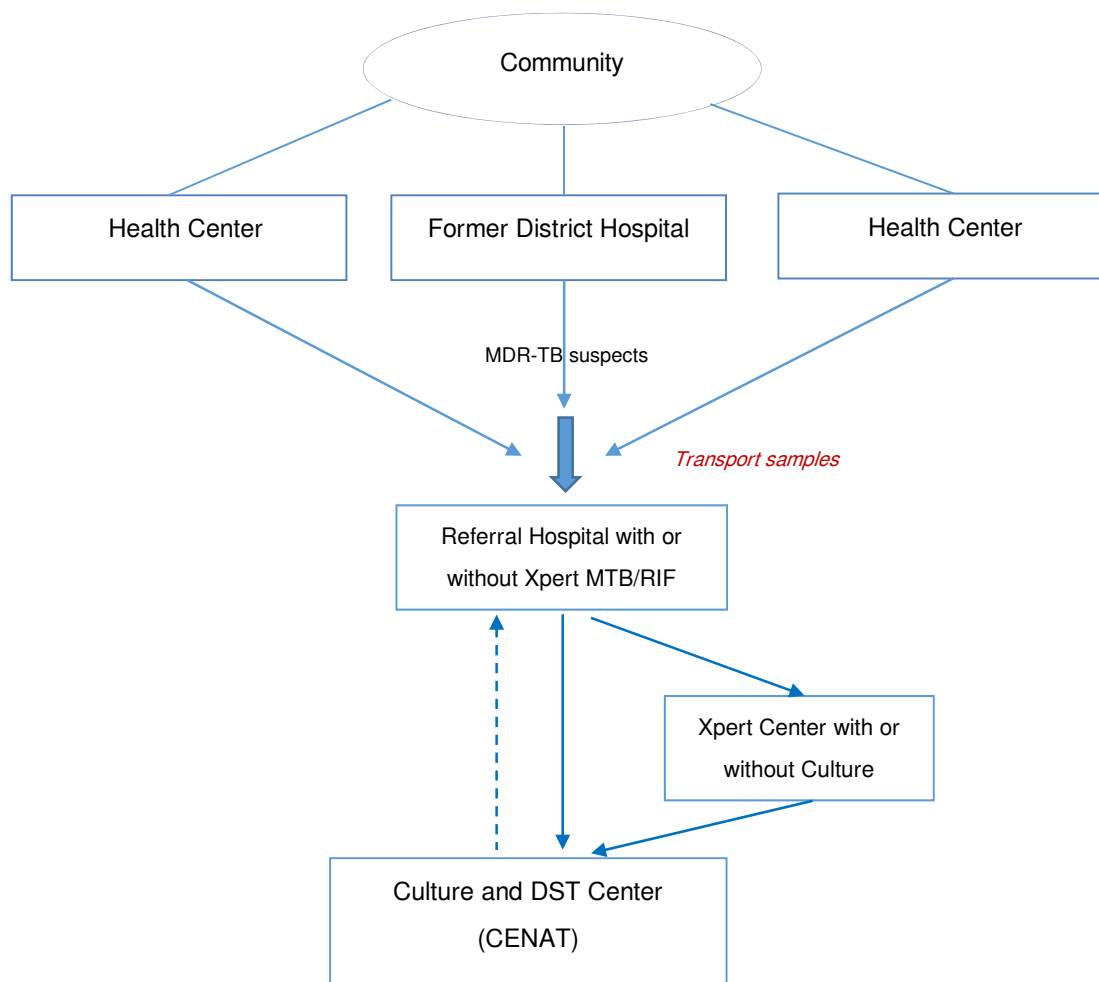
Option I: is an option which MDR-TB suspect patients were identified by local healthcare workers (HC or Former District Hospital). The local HCWs refer the patients to the nearest referral hospital to collect and transport sputum samples to the designated diagnostic laboratory.



Option II: is an option which healthcare worker at referral hospital or TB supervisors (OD/PHD) travel to health center and community/patient's home to collect and transport sputum samples to the designated diagnostic laboratory.

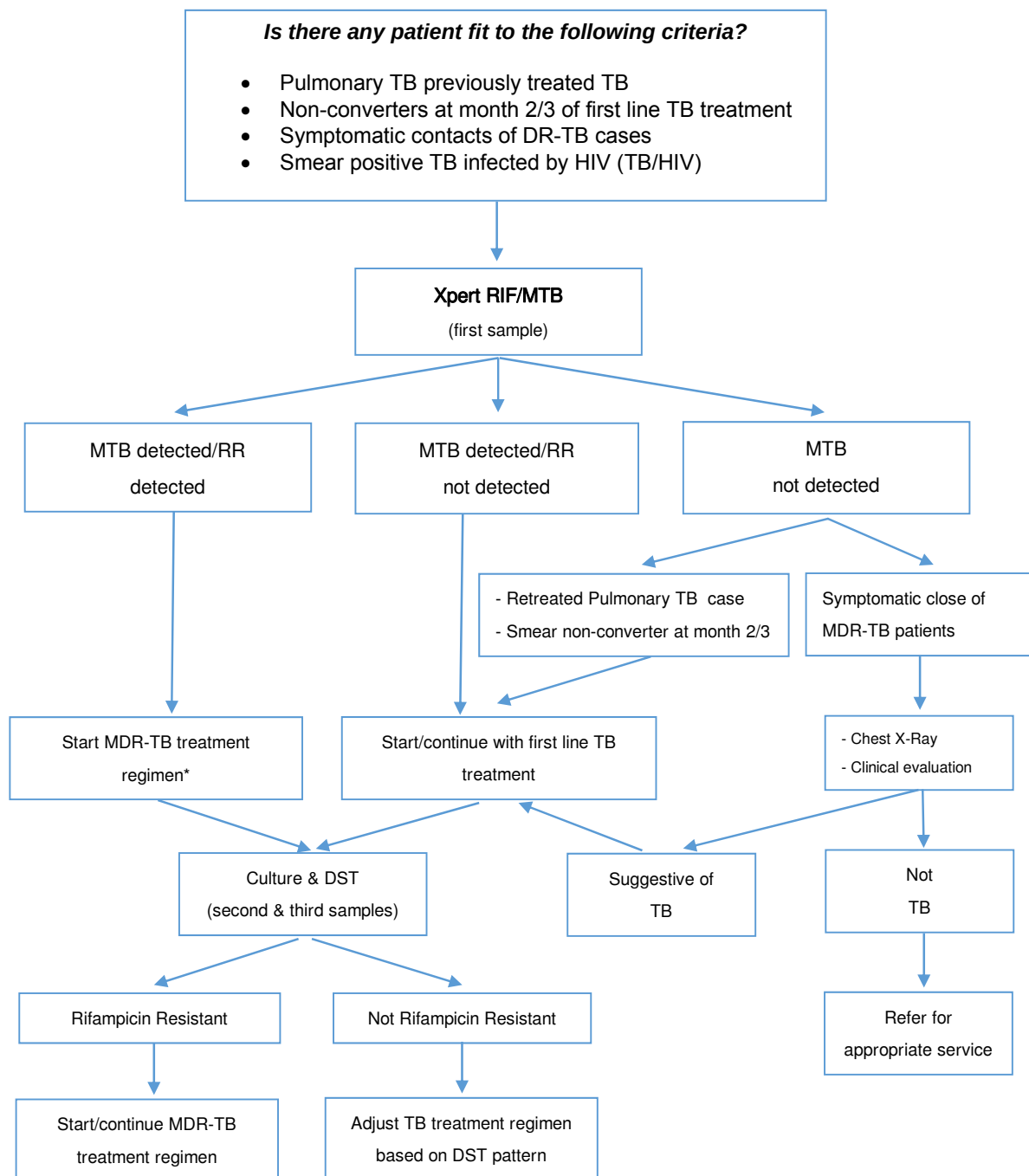


Option III: is an option which local healthcare worker at health center or former district hospital collects and transports sputum samples to the designated diagnostic laboratory.



All MDR-TB suspects are recommended to test by Xpert MTB/RIF and drug sensitivity testing with other first line anti-TB drugs after culture positive for MTB. In case Rifampicin resistant detected (RR-TB) by Xpert MTB/RIF, the patient is considered for an appropriated treatment with second line TB (see the following algorithm).

Algorithm for Diagnosing MDR-TB among Population High Risk for MDR-TB



* Note: detailed information on MDR-TB treatment regimen, please refer to MDR-TB treatment algorithm

10.3. MDR-TB Treatment Strategy

10.3.1. MDR-TB Treatment Regimen Design

This section describes the approach to manage patients with MDR-TB, including MDR-TB treatment regimen design, treatment duration, DOT approach, treatment follow-up, drug side effect monitoring and management, and patient-centered strategies to enabling patients overcome the difficulties and long term treatment complex such as counseling and community involvement. According to WHO guidelines on Programmatic management for drug resistant TB published in 2014, there are three main treatment strategies in MDR-TB:

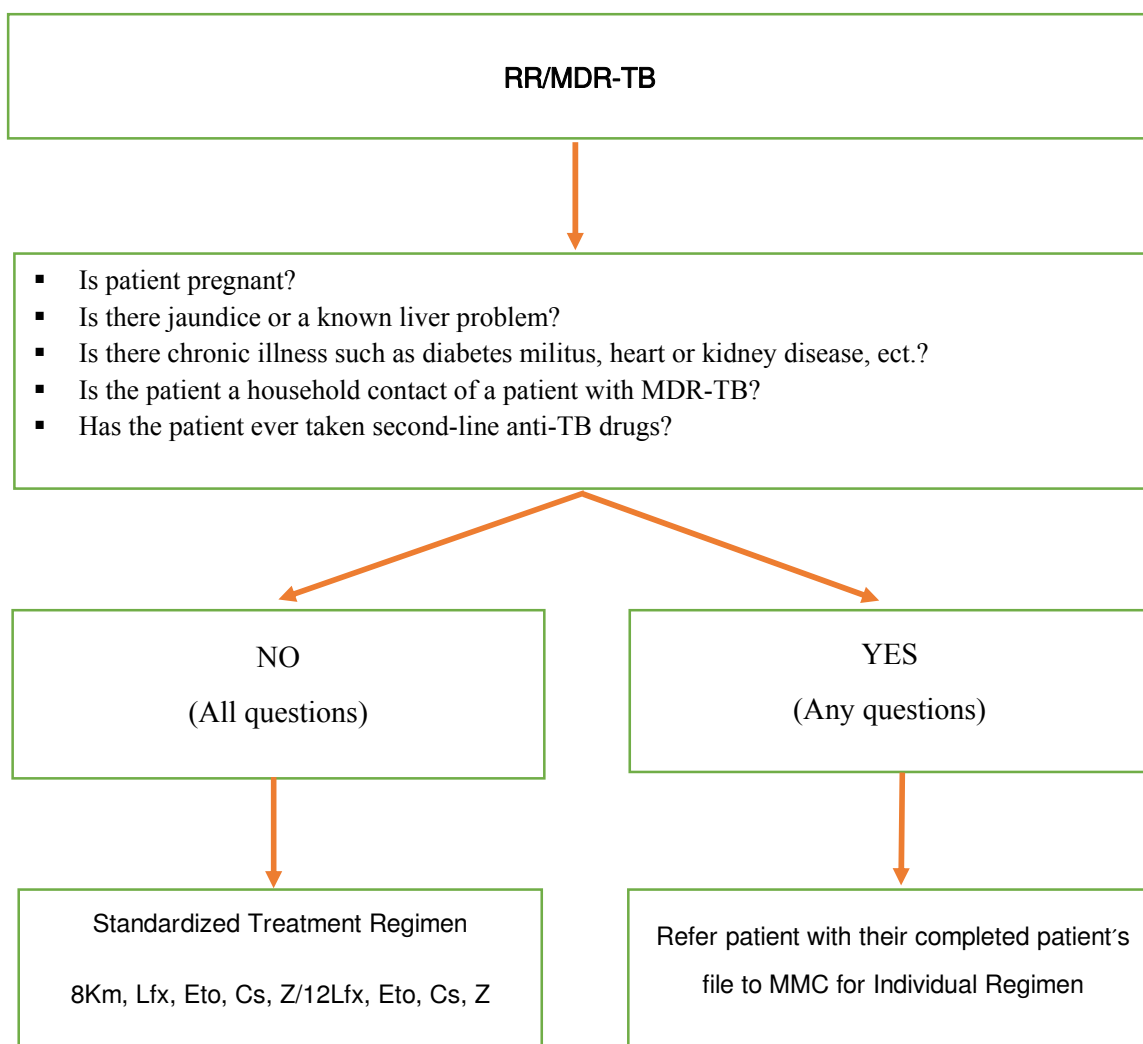
- Standardized treatment: the designed regimen is done based on the resistant pattern of drug resistant survey (DRS) data. Standardized treatment is used for all RR/MDR-TB patients who never treated with second line TB drugs and second line DST result is not available, especially for Fluoroquinolone and injectable agents.
- Empirical treatment: Each regimen is individually designed based on the patient's previous history of treatment with consideration of DRS data from the representative patient population. The regimen is adjusted once DST results of the individual patient become available.
- Individualized treatment: This strategy uses regimens that designed based on the patient's previous history of treatment and individual second line DST results.

Cambodia has been updated its PMDT guidelines based on WHO guidelines on Programmatic management for drug resistant TB published in 2014. In line with Cambodian NTP PMDT guidelines 2011, NTP has been designed a standardized treatment regimen for RR/MDR-TB based the MDR-TB prevalence of national drug resistant survey and the assumption of low prevalence of SLD resistant, especially Kanamycin (Km) and Levofloxacin (Lfx).

8 Km⁶, Lfx⁷, Eto⁷, Cs⁷, Z⁷/12 Lfx⁷, Eto⁷, Cs⁷, Z⁷

Note: Intensive phase composes of at least 5 molecules for 8 months. Km is provided 6 days/week and other molecules are provided daily; such as Levofloxacin (Lfx), Ethionamide (Eto), Cycloserine (Cs) and Pyrazinamide (Z)¹ Continuation phase composes of 4 molecules (Lfx, Eto, Cs, Z) for 12 months without injectable agent.

Figure 1: RR/MDR-TB Treatment Regimen Selection



All MDR-TB treatment regimens should be designed based on the written guidelines. MDR-TB regimen composes of Pyrazinamids (Z), even it was found resistant or not, and at least 4 second line anti-TB drugs with high efficacy consisting of one injectable agent, one Fluoroquinolone, and two from bacteriostatic group (Table 1). Appropriate empirical and individual regimens should be designed based on the condition:

- **Group 1:** Include Z in the treatment regimen. Z is important for its action in acidic environment that exists in chronically damaged lungs.
- **Group 2:** Select one second line injectable drug. There is no superiority in effect among the SLD injectable agents, but use Kanamycine (Km) as the first option because of low cost. Amykacine (Am) may be used if necessary because the two agents are almost 100% cross-resistance. If Km cannot be used for whatever reasons (resistance or intolerable adverse reactions) then Capreomycine (Cm) can be used. Do not use streptomycin, because of its high resistance in many settings.
- **Group 3:** Select one Fluoroquinolone. A later-generation Fluoroquinolone rather than an earlier-generation Fluoroquinolone must be used as the former may have some efficacy against Ofx-resistant strains. For the time being, Lfx is the Fluoroquinolone of choice for MDR-TB regimen.

- **Group 4:** Among the oral bacteriostatic drugs belonging to this group, the association with cure was strongest in Eto > Cs. PAS performed the worst, showing no significant effectiveness in the main analysis. Its use would thus be recommended only if an additional drug is needed to achieve a 5-drug regimen or if Eto and Cs cannot be used or are unlikely to be effective.
- **Group 5:** This group is not recommended to be among the first drugs of choice for the standard regimen. Although all of them have demonstrated some activity at least in vitro or in animal models, the quality of the evidence of their efficacy and safety in humans for the treatment of drug-resistant TB varies. However, they remain as options in cases where adequate regimens are impossible to design with medications from Group 1 to Group 4. If a situation requires the use of Group 5 drugs, often experts will recommend using two to three drugs from the group given the limited knowledge of efficacy.

Table 1: Group of Medicines for Treating Drug-resistant TB based on their Body Weight

Group of Medicines (Name Abbreviation and Form)	Daily Dose	30-35 kg	36-45 kg	46-55 kg	56-70 kg	> 70 Kg
Group 1- First Line TB Drugs						
Isoniazid (H), tablet	4-6 mg/kg once daily	150 mg	200 mg	300 mg	300 mg	300 mg
Rifampicin (R), tablet	8-12 mg/kg once daily	300 mg	450 mg	450 mg	600 mg	600 mg
Ethambutol (E), tablet	15-25 mg/kg once daily	600 mg	800 mg	1,000 mg	1,200 mg	1,200 mg
Pyrazinamide (Z), tablet	20-30 mg/kg once daily	800 mg	1,000 mg	1,200 mg	1,600 mg	2,000 mg

Group 2- 2 nd Line Injectable Agent	Daily Dose	30-33kg	34-40 kg	41-45 kg	46-50 kg	51-70 kg	> 70 kg
Kanamycin(Km), 4ml Ampoule	15-20 mg/kg once daily	500 mg	625 mg	750 mg	875 mg	1,000 mg	1,000 mg
Amikacin (Am), Vial	15-20 mg/kg once daily	500 mg	625 mg	750 mg	875 mg	1,000 mg	1,000 mg
Capreomycin (Cm), Vial	15-20 mg/kg once daily	500 mg	600 mg	750 mg	800 mg	1,000 mg	1,000 mg
Streptomycin (S), Vial	12-18 mg/kg once daily	500 mg	600 mg	700 mg	800 mg	900 mg	1,000 mg
Group 3- Fluoroquinolones	Daily Dose	30-35 kg	36-45 kg	46-55 kg	56-70 kg	> 70 Kg	
Levofloxacin (Lfx), tablet	750-1,000 mg once daily	750 mg	750 mg	1,000 mg	1,000 mg	1,000 mg	
Moxifloxacin (Mfx), tablet	400 mg once daily	400 mg	400 mg	400 mg	400 mg	400 mg	
Gatifloxacin (Gfx), tablet	400 mg once daily	400 mg	400 mg	400 mg	400 mg	400 mg	
Group 4- 2 nd Line anti-TB Drugs, Bacteriostatic							
Ethionamide (Eto), tablet	500-750 mg/day in 2 divided doses	500 mg	500 mg	750 mg	750 mg	1,000 mg	
Prothionamide (Pto), tablet	500-750 mg/day in 2 divided doses	500 mg	500 mg	750 mg	750 mg	1,000 mg	
Cycloserine (Cs)	500-750 mg/day in 2 divided doses	500 mg	500 mg	500 mg	750 mg	750 mg	
Terizidone (Trd)							
P-aminosalicylic acid (PAS)	8 g/day in 2 divided doses	8 g	8 g	8 g	8 g	8 - 12 g	
Para-aminosalicylate sodium (PAS Na)							
Group 5- Anti-TB Drugs with limited data on efficacy and/or long term safety in the treatment of drug resistant TB (This group includes new anti-TB agents)							
Bedaquiline (Bdq)	400 mg once daily for 2 weeks then 200 mg 3 times per week						
Delamanid (Dlm)	100 mg twice daily (total daily dose = 200 mg)						
Linezolid (Lzd)	600 mg once daily	600 mg	600 mg	600 mg	600 mg	600 mg	
Clofazimine (Cfz)	200–300 mg daily (first 2 months) then reduce to 100 mg daily (alternative dosing 100 mg daily)						
Amoxicillin/ clavulanate (Amx/Clv)	80 mg/kg/day in 2 divided doses	2,600 mg	2,600 mg	2,600 mg	2,600 mg	2,600 mg	
Imipenem/cilastatinf (Ipm/Cln)	1,000 imipenem/1,000 mg cilastatin twice daily						
Meropenem (Mpm)	1,000 mg three times daily (alternative dosing is 2,000 mg twice daily)						
High-dose isoniazid (H _{high-dose})	16-20 mg/kg once daily	600-1,000 mg	1,000-1,500 mg	1,500 mg	1,500 mg	1,500 mg	
Thioacetazone (Thz)							
Clarithromycin (Clr)							

Case Studies of MDR-TB Treatment:

Case Study 1: Mr Sok, 58 years old and he is a farmer. He is found to be RR-TB detected by Xpert MTB/RIF. He is referred by OD TB supervisor to Siem Reap hospital for SLD TB treatment. He had received TB treatment for 6 months in previous 2 years. He is not founds another disease than TB and his weight is 45Kg.

Answer: Mr Sok has never been treated with second line anti-TB drugs. He has no other health problems such as HIV infected, diabetes, renal problem or hepatitis ect. Mr Sok is decided by local clinican to be treated with Standardized Treatment Regimen for MDR-TB and dosage of each molecule are as following:

Name (form and dose)	Intensive phase (8 months)	Continuation phase (12-16 ms)
Kanamycin (Km), 1g	750 mg once daily (6days/week)	
Levofloxacin (Lfx), 250 mg	750 mg once daily (7days/week)	750 mg once daily (7days/week)
Ethionamid (Eto), 250 mg	250 mg/day Twice daily (7days/week)	250 mg/day Twice daily (7days/week)
Cycloserine (Cs), 250 mg	250 mg/day Twice daily (7days/week)	250 mg/day Twice daily (7days/week)
Pyrazinamid (Z), 400 mg	1,000 mg once daily (7days/week)	1,000 mg once daily (7days/week)

Case Study 2: Mr S.P, 36 years old and 42Kg, was found RR-TB detected by Xpert MTB/RIF. He had been diagnosed MDR-TB which resistant to Rifampicin, Isoniazid, Ethambutol, and Streptomycin. He had been treated for MDR-TB with second line anti-TB drugs such as Km, Lfx, Pto, Cs, and Z in previous one year. He dropped his MDR-TB treatment after 3 months of treatment due to severe drugs side effect and lack of support from his family members and neighbors because of he was found HIV infected. He was not found with another health problem.

Answer: Physician decides to retreat him (Mr S.P) again. At the same time, physcian takes sputum for referring to culture center for the second line DST, especially with Fluoroquinolone and Injectable agents. The standardized treatment regimen for MDR-TB is not appropriate for Mr S.P, because he was previously treated TB with the second line anti-TB drugs, HIV infected, and his second line DST result not available yet. Physician treats him with individual treatment for MDR-TB and dose as following:

Name (form and dose)	Intensive phase (8 months)	Continuation phase (12-16 ms)
Kanamycin (Km), 1g	750 mg once daily (6days/week)	
Levofloxacin (Lfx), 250 mg	750 mg once daily (7days/week)	750 mg once daily (7days/week)
Ethionamid (Eto), 250 mg	250 mg/day Twice daily (7days/week)	250 mg/day Twice daily (7days/week)
Cycloserine (Cs), 250 mg	250 mg/day Twice daily (7days/week)	250 mg/day Twice daily (7days/week)
Pyrazinamid (Z), 400 mg	1,000 mg once daily (7days/week)	1,000 mg once daily (7days/week)

The above regimen can be adjusted if the second line DST shown resistant to any second line anti-TB drugs in the current regimen.

10.3.2. Phases of Treatment

The duration of MDR-TB treatment is 20-24 months dividing in two phases: Intensive phase and continuation phase.

Intensive Phase (IP): The duration of the intensive phase is at least 8 months of injection period depending on its culture conversion.

Continuation Phase (CP): After stopping the injectable agent, all oral drugs are continued during the continuation phase. Usually, adverse effects of drugs are better tolerated during this phase. The total duration of treatment is at least 20 months without previously treated DR-TB with SLD.

CENAT uses a mix method in managing drug resistant TB patients including hospitalizing and providing care at community. The main purpose MDR-TB case management is to provide patients an appropriated treatment for free of charge with good adherence under reliable network and trained for MDR-TB treatment and care. This network composes of trained program management team from national level, physicians, nurses, and community health volunteers who providing the best care to patients including ongoing psychological and sociological supports and applying TB infection control in their community.

10.3.3. Monitoring of Treatment Response

MDR-TB treatment follow-up aims to monitor their treatment response and provide on time management, especially related side effect management and MDR-TB regimen adjustment.

10.3.3.1. Clinical Follow-up

The physician monitors patients closely on a monthly basis during the entire treatment for adherence, side effects, clinical improvement or treatment failure.

10.3.3.2. Clinical Monitoring

The most important objective evidence of improvement is conversion of smear and culture to negative. Performing smear and culture are recommended to do on monthly basis for patient intensive phase during injection or patients likely to fail their treatment; and on quarterly basis for the rest of treatment duration. The second line DST is recommended if the patient is persistently culture positive at month 4 or if the patient reconverts to culture positive at any point in the treatment.

10.3.3.3. Monitoring of Side Effects

Physicians or healthcare worker provide proper explanation to patients on adverse effects that may appear while receiving second line anti-TB drugs. A mobile team should be established for early side effect management composing of physician and nurse with emergency tool kit available.

Table 2: Activities for Monitoring Treatment Response

MONITORING EVALUATION	RECOMMENDED FREQUENCY
Evaluation by clinician	At baseline, at least monthly until conversion, then every 2–3 months, and when clinical indicated.
Treatment adherence and tolerance	Daily at every DOT encounter by the DOT provider
Sputum smear and Culture	Monitoring smears and culture monthly throughout treatment. (Note: programmes with limited resources may choose to do monthly smears and cultures until conversion and then monthly smears with every other month cultures)
Weight	At baseline, then monthly
Drug susceptibility testing	At baseline in programmes doing individualized treatment testing (DST) or in programmes doing standardized treatments that need to confirm MDR-TB. For patients who remain culture positive, it is not necessary to repeat DST within less than 3 months of treatment.
Chest X-ray	At baseline, and then every six months.
Serum creatinine	At baseline, then monthly while receiving an injectable drug. Every 1–3 weeks in HIV-infected patients, diabetics, and other high-risk patients.
Serum potassium	Monthly while receiving an injectable agent. Every 1–3 weeks in HIV-infected patients, diabetics and other high-risk patients
Thyroid stimulating hormone (TSH)	Every 6 months if receiving ethionamide/prothionamide and/or PAS; and monitor monthly for signs/symptoms of hypothyroidism. TSH is sufficient for screening for hypothyroidism; it is not necessary to measure hormone thyroid levels
Liver serum enzymes	Periodic monitoring (every 1–3 months) in patients receiving pyrazinamide for extended periods or for patients at risk for or with symptoms of hepatitis. For HIV-infected patients, do monthly monitoring
HIV screening	At baseline, and repeat if clinically indicated
Pregnancy tests	At baseline for women of childbearing age, and repeat if indicated
Haemoglobin and white blood count	If on linezolid, monitor weekly at first, then monthly or as needed based on symptoms; there is little clinical experience with prolonged use. For HIV-positive patients on an ART regimen that includes AZT, monitor monthly initially and then as needed based on symptoms
Lipase	Indicated for work up of abdominal pain to rule out pancreatitis in patients on linezolid, D4T, ddI, ddc.
Lactic acidosis	Indicated for work up of lactic acidosis in patients on linezolid or ART
Serum glucose	If receiving gatifloxacin, monitor glucose frequently (weekly) and educate patient on signs and symptoms of hypoglycaemia and hyperglycaemia

10.4. Treatment of DR-TB in Special Conditions

At the same time, CENAT also develop strategies to manage and treat other forms of drug resistant TB such as mono DR-TB, Poly DR-TB, and MDR-TB in special conditions including MDR-TB in HIV infected patient, MDR-TB in diabetes mellitus patient, MDR-TB in pregnancy or breast feeding, MDR-TB in severe liver diseases or renal failure patients, MDR-TB in psychosis or addicted patient, or MDR-TB in children.ect. Providing MDR-TB treatment to these group should be consulted with MDR-TB Management Committee (MMC) in each MDR-TB treatment site. The detail information of treating DR-TB in special conditions refers to national guideline of programatic management for drug resistant TB or contact technical bureau of CENAT.

10.5. MDR-TB Shorter Treatment Regimen

Based on the WHO's recommendation in 2016, CENAT is considering for the possibility starting MDR-TB shorter treatment regimen for all RR/MDR-TB patients and preparing the relevant policy and guideline. CENAT is going to start this regimen in mid-2017. The duration of this regimen is only 9 to 12 months. The regimen composes of:

4-6 Km-Mfx-Pto-Cfz-Z-H_{high-dose}-E / 5 Mfx-Cfz-Z-E

The intensive phase composes of Km, Mfx, Pto, Cfz, Z, H in high dose, and E for 4 to 6 months and continuation phase remains only Mfx, Cfz, Z, and E for more 5 months. The detail information of the MDR-TB shorter treatment regimen will be issued soon before CENAT applying this new regimen.

11. TB/HIV

11.1. Introduction

Worldwide, tuberculosis (TB) is the leading cause of death among HIV-infected persons. HIV is the most potent risk factor for the development of active TB. HIV-infected persons with TB are far more likely to die before completing their 6-month TB treatment course than TB patients who are not co-infected with HIV. While HIV is known to increase susceptibility to new infection with *Mycobacterium tuberculosis*, it also fuels the TB epidemic by increasing susceptibility to reactivation of recent and latent *M. tuberculosis* infection and increasing the risk of recurrent TB. While an HIV-negative person infected with *M. tuberculosis* has a 5 to 10% **lifetime** risk of developing active TB, an HIV-positive person who is co-infected with *M. tuberculosis* has a 10% **annual** risk of developing active TB.

TB is the most common AIDS-related illness in Cambodia as it is observed in up to 20- 30% of People Living with HIV (PLHIV) enrolled in OI/ART cohorts. In addition, the mortality rate among TB-HIV co-infected patients is high within two months of TB treatment among TB HIV co-infected patients with CD4 cell counts less than 200 not yet receiving anti-retroviral therapy (ART) (CENAT, August 2005).

Since 2000, the National Centre for Tuberculosis and Leprosy Control (CENAT) and the National Centre for HIV/AIDS, Dermatology and STD (NCHADS) together with partners have been working to define the key activities necessary to reduce the impact of TB-HIV co-infection and develop the Framework for TB/HIV in Cambodia as well as Standard Operating Procedure (SOP) for prompt HIV testing of TB patients.

TB/HIV collaborative activities were piloted in Banteay Meanchey, Battambang, Phnom Penh and Preah Sihanouk provinces with the support of Cambodia US-CDC, FHI, JICA and WHO respectively. Experiencing and learning from the pilot activities, the expansion plan of the collaboration has been made from year to year; and by 2010, all Operational Districts are implemented TB/HIV collaborative activities. The referral of registered TB patients for HIV testing and referral of PLHIV for TB screening are gradually increased from 30% at start of the project up to above 80% in 2015.

11.2. TB/HIV Collaborative Activities

11.2.1 HIV Testing among TB Patients

Patients co-infected with both TB and HIV have a high risk of death. All registered TB patients are counseled and then referred for HIV testing at VCCT or onsite testing voluntary principles and confidentiality. The activity is carried out in order to know HIV status and to provide treatment and care of TB/HIV in timely manner.

HIV testing among registered TB patients can be done as below:

- Refer TB patient to VCCT for HIV testing
- On-site HIV testing at health centers

TB healthcare staff have to ask if TB patients were tested for HIV and result of HIV testing. For those who are known HIV status, TB health staff have to know whether the patients received CPT and ART or not. For the co-infected TB/HIV patients not received CPT and ART yet, the health staff have to refer the patients to OI/ART service to receive the co-management since they are eligible for TB and HIV treatment regardless of CD4 count.

For registered TB patients who have tested for HIV yet, health staff have to provide counseling and encourage them for HIV testing.

TB health care workers have a responsibility to record HIV test results into TB patient book and refer co-infected TB/HIV patients to OI/ART for HIV management.

11.2.2. TB screening among PLHIV (Three I's Strategy)

TB is most commonly found at initial HIV diagnosis, during the period prior to ART initiation, and shortly after ART initiation.

HIV-infected patients should be screened for TB at the OI/ART clinic during their initial visit, prior to initiating ART, and at every follow-up visit thereafter.

Voluntary Confidential Counseling and Testing (VCCT) counselors and Home-Based Care (HBC) staff can also screen PLHIV for symptoms suggestive of TB and refer to the OI/ART clinics for further diagnosis if the symptom screening is positive (at least one symptom present). While TB symptom screening of PLHIV will be performed at OI/ ART clinics, VCCT, and HBC, those suspected of having TB will have the diagnostic workup performed by the TB care clinicians who will be co-located in the OI/ART site or at TB service.

The three following key elements of the Three I's strategy are as below:

- Intensified TB case finding (ICF) among PLHIV and their household contacts,
- Isoniazid Preventive Therapy (IPT) for PLHIV unlikely to have active TB,
- Improved TB infection control (IC) measures at health facilities, Continuum of Care (CoC) and home-base care (HBC) settings.

1. Intensified TB Case Finding among PLHIV

Activities of Intensified TB case finding among PLHIV are described along with an algorithm in next page. The algorithm is based on a strong evidence for use in identifying TB suspected cases and diagnosing TB among PLHIV enrolled in pre-ART register and ART register.

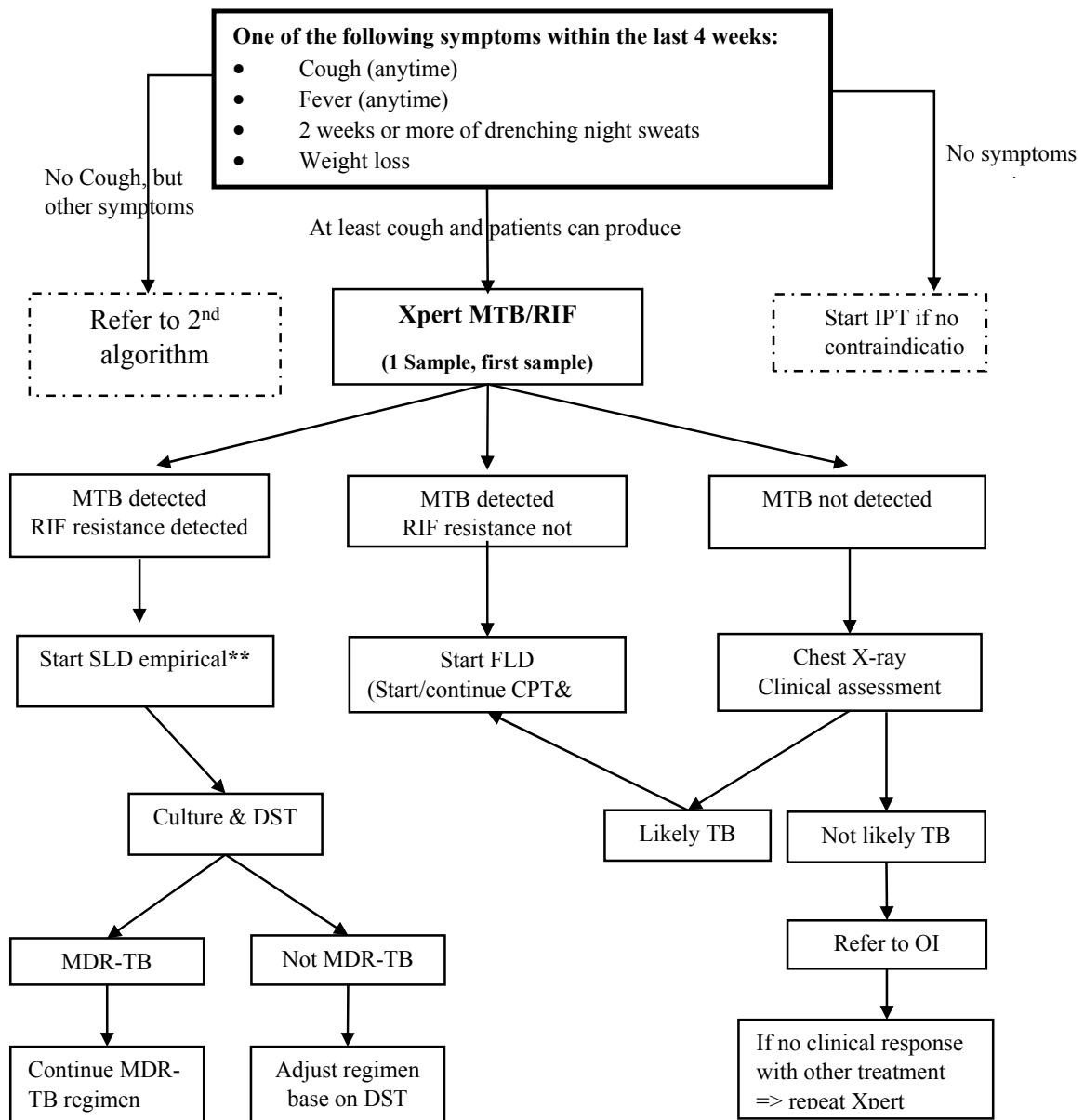
All PLHIV are asked about the following 4 symptoms for TB screening:

- **Fever**, anytime of any duration, more than 2 weeks
- **Cough**, anytime of any duration, more than 2 weeks
- **Drenching night sweats**, more than 2 weeks
- **Weight loss**, more than 2 weeks

If patients have none of the four symptoms, they are considered unlikely to have active TB (without the need of further examination) and are eligible for IPT if they have no contraindications.

If patients have one of the four symptoms, they are considered to have active TB and then they are requested for TB diagnostic workup by Xpert and Chest X-ray. If patients are able to produce sputum, collect one sputum for Xpert testing (see the algorithm below).

Algorithm for Diagnosing TB among PLHIV, Xpert Available



* Contraindications to IPT in case of active hepatitis, heavy alcohol consumption, and peripheral neuropathy

** SLD regimen with CAT-4 for MDR TB patient (Km/Cm, Lfx/Mfx, Eto, Cs/PAS, Z) and regimen adjust will be done after DST result.

2. Isoniazid Preventive Therapy

PLHIV with none of the four TB screening symptoms are considered unlikely to have active TB and are eligible for IPT, which should be started as soon as possible. IPT should be given to such individuals irrespective of the degree of immunosuppression, and also to those on ART, those who have previously been treated for TB and no history of receiving IPT before. However, IPT should not be started in case of the following contraindications:

- AST or ALT > 3 times the Upper Limit of Normal (ULN), or a single elevation > 250, or lower elevations of AST or ALT with symptoms (nausea, vomiting, abdominal pain, anorexia or jaundice). Patients with evidence of active liver disease with baseline ALT or AST > 3x ULN should not be started on IPT until their enzymes have dropped well below this level. A positive hepatitis serology is not itself a contraindication to IPT, but warrants monthly monitoring of liver function tests (LFTs) until it is clear that the drug is well tolerated.
- Active alcohol abuse.
- Past history of severe adverse side effects to Isoniazid.

IPT will be prescribed by the OI/ ART clinician:

- For at least a 6 month duration.
- Pyridoxine is given to prevent peripheral neuropathy with the dose of 50mg daily.
- The patient should be provided a one month INH supply at the OI/ART clinic visit with the dose of 5 mg/ kg (standard adult dose of **300 mg***) once daily.
- The patient should be closely monitored.

3. TB Infection Control

The main important activities are as below:

- Minimize referrals of PLHIV to TB clinics/ward. Instead, provide TB symptom screening or appropriate referrals.
- Organize patient flow (make sure that coughing patients do not wait in the waiting room with other patients but at an outside space dedicated to them).
- Diagnose and treat TB as early as possible through TB Intensified TB Case Finding.
- Ensure good ventilation within all rooms in which Presumptive TBs and TB patients could be present.
- Assign an OI/ART nurse counselor to be in charge of detecting and separating coughing presumptive TB and known sputum smear-positive pulmonary TB patients from PLHIV.
- Provide TB patients with masks (or if not available encourage using Kramar or moto mask) to be used when coughing.

12. CHILDHOOD TB

12.1. Introduction

TB in children is a priority of the National TB Control Program (NTP). In Cambodia, proportion of childhood TB cases among all TB cases is higher than other countries, which accounted for up to 20% or more. At global level, the proportion of childhood TB cases among all TB cases is between 13% and 15%.

In general, making TB diagnosis for children is very difficult even for pediatric expert physician. In principle, making diagnosis depending on careful evaluation on evidence of previous history, clinical exam and relevant further evaluation such as tuberculin skin test, chest x-ray and smear microscopic examination.

For TB in children, although bacteriological confirmed cannot be performed all time, this should be performed if possible. For instance, sputum microscopic examination should be performed for presumptive pulmonary TB children who can produce sputum. Decision of treating children should be taking into account and that if the decision of treatment is made the completion of course of treatment should be provided.

12.2. Approach to Diagnose TB in Children

- asking history including history of staying with TB patient and symptoms
- conducting clinical investigation, including evaluation of physical development
- perform tuberculin skin test
- finding evidence of bacteriologically confirmed, if applicable
- doing further relevant evaluation on pulmonary and extra pulmonary
- conduct HIV testing

Please read detail information in National Guidelines for Diagnosis and Treatment of TB in Children (CENAT, 2008) and diagnostic algorithm for TB in children in the next pages.

Role of health center staff in childhood TB activities:

1. Identify presumptive TB in children:

Health center (HC) staff has to screen for presumptive TB in children when:

- Parents or care givers bring children for general consultation (OPD)
- Volunteers or community based organization staff refer close contact children who live with TB patient especially with pulmonary TB patient with smear positive, from community to HC after HC staff copied TB patient's names and gave it to them.
- In case of children have any symptoms suggestive of TB, those children have to be referred to referral hospital for diagnosis.

2. Treatment: when HC staff receive child TB after being diagnosed at the referral hospital and referred back to HC. HC staff register child TB on treatment for those who being referred from diagnostic center or treated another health condition.

3. Provision of INH Preventive Therapy (IPT): HC staff asks for history and do carefully clinical investigation to evaluate children under 5 years old who are close contact with smear positive TB patient to ensure that they are healthy and register them on IPT.

4. Treatment Follow-up and Side Effects: HC staff monitors patients taking TB drugs and its side effects on a regular basis.

5. Health Education: HC staff has to provide health education to community and parents on TB and main symptoms of TB for children.

Seven symptoms suggestive of TB for children that HC staff has to screen for:

1. Persistent chronic cough more than two weeks, dyspnea, chest pain (especially after taking antibiotic but no improvement).
 2. Children living with smear positive TB patient newly diagnosed.
 3. Lymph node (neck, armpit, groin) more than 2 cm diameter, firm, painless, mobile and/or squishy (abscess)
 4. Chronic pain in joint/bone, spiral mass, joint swelling with/without paralysis of lower two limbs or all limbs.
 5. Fever ($>38^{\circ}\text{C}$) and/or night sweat for more than 14 days
 6. Weight loss or failure to thrive (after providing supplementary food according to nutrition program for six weeks, there is no any gain weight).
 7. Fatigue, loss of playing, reduction of daily activities
- Read detail information in diagnostic algorithm 1 to 4 TB in children in section 12.

Highly presumptive TB in children who are at risk of infection:

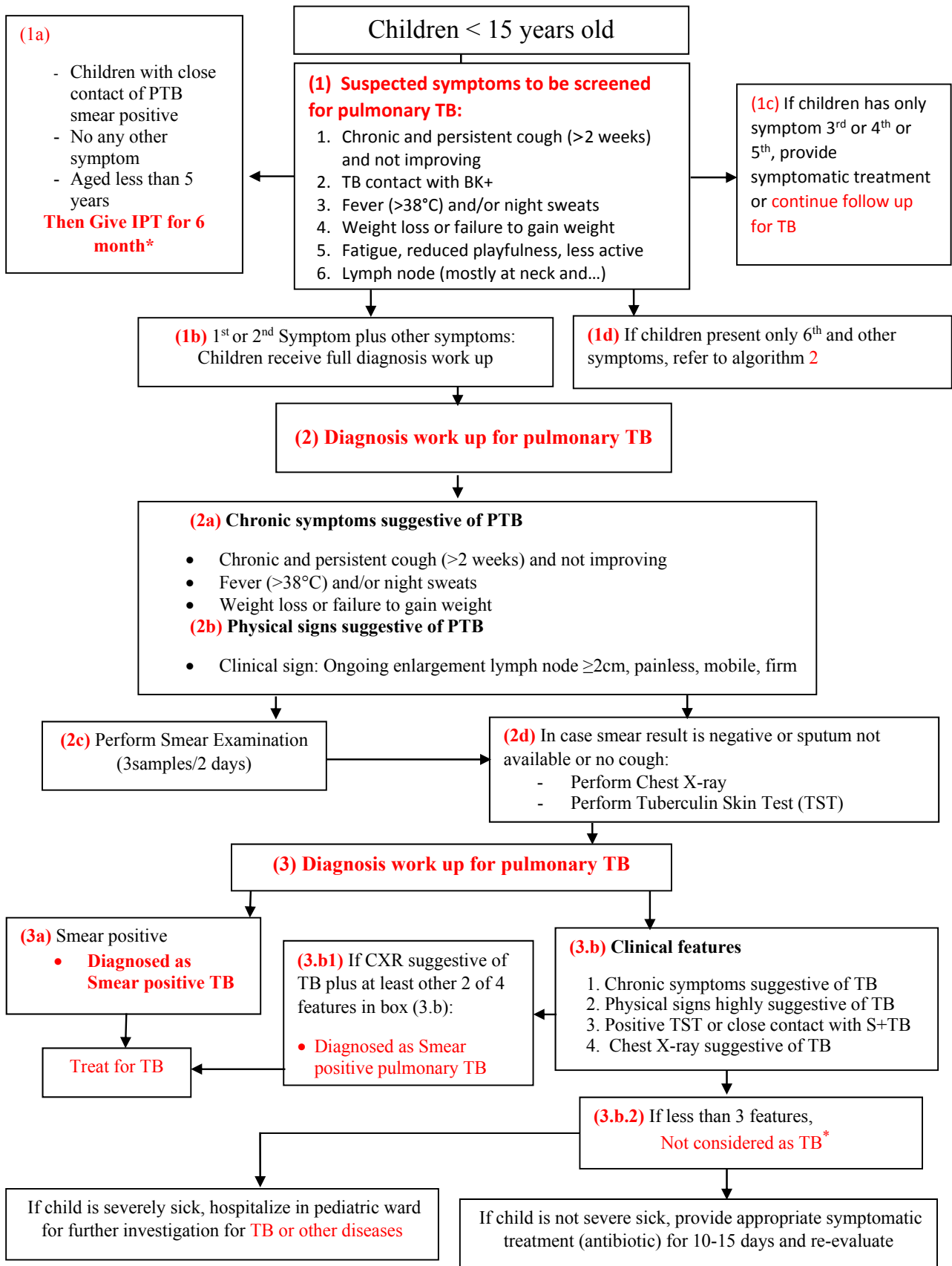
- a. Living with smear positive TB patient recently diagnosed, in a family
- b. Aged less than 5 years
- c. HIV infected
- d. Severe malnutrition

All presumptive TB patients must be recorded telephone number and proper address, and refer to referral hospital for TB diagnosis:

INH Preventive Therapy (IPT) must be registered on treatment at HC level:

- a. Only children history and clinical exam conducted by HC staff. Recommendation in IPT for children who referred from community must be made.
- b. Criterion of IPT provision for children is that children living with smear positive TB patient (children under 5 years old and healthy).
- c. Based on IPT card, HC staff can identify criteria of healthy children; children must not have any of the above seven symptoms.
- d. If decide to provide IPT, HC staff need:
 - Before provision of IPT, education and clear messages on benefit of IPT to parents or caregivers must be provided. IPT can prevent children from development of TB disease.
 - Provide INH once a month with a dose of 10mg/kg/day for six months
 - Regular monitoring of taking INH and its side effects
 - During the IPT, if there is any symptoms suggestive of TB appeared, must immediately refer children to RH for further TB evaluation. If children found to have active TB, IPT must be stopped.

Child TB Algorithm 1: for Pulmonary TB in Children



* If children is a contact with BK⁺ TB patient and CXR shadow suggestive of TB, try to find other evidences including regular follow-up every 3 months through telephone or bringing children for screening.

Child TB Algorithm 1:

(1) Children aged less than 15 shall be screened for suspected pulmonary TB as follows:

1. Persistent cough for more than 2 weeks and not improving
2. TB contact with smear positive patients
3. Fever ($>38^{\circ}\text{C}$) and/or night sweats
4. Weight loss or failure to gain weight (check yellow card)
5. Fatigue, reduced playfulness, less active
6. Lymph node (mostly at neck....)

(1a) If children are TB contact with smear positive patient, and do not have any other symptom (healthy) and aged less than 5 years, *then those children shall be given IPT (dose: 10mg/kg/day for 6 months).*

* For children aged 5 to 14 years old, they are not given IPT except children living HIV/AIDS.

(1b) If children have symptom 1 plus other symptoms (Ex. Symptom 1 plus 2 and 3) or symptom 2 plus other symptoms (Ex. Symptom 2 plus 3 and 4), children shall be received full diagnosis work up carefully.

(1c) If children have only symptom 3, 4, 5, should provide appropriate symptomatic treatment or continue follow up for TB.

(1d) If children have symptom 6 in combination with other symptoms, refer to algorithm 2

(2) In order to diagnose pulmonary TB, TB physicians shall find out:

(2a) Chronic symptoms suggestive of TB:

- Persistent cough for more than 2 weeks and not improving
- Fever ($>38^{\circ}\text{C}$) and/or night sweats
- Weight loss or failure to gain weight

(2b) Physical signs highly suggestive of TB:

- Ongoing enlargement lymph node $\geq 2\text{cm}$, painless, mobile, firm, mostly at neck and rarely at axillaries and other place

(2c) If children can produce sputum (3 samples/2 days), smear microscopy shall be performed. If the specimen is not enough, contact HC and community to follow up for additional specimens

(2d) If sputum examination result is negative or sputum is not available or no cough, TB physicians shall:

- Perform Tuberculin Skin Test (TST)
- Perform chest X-ray

(3) Diagnosis pulmonary TB for children is based on below for features:

(3a) If sputum examination result is **positive**, (NTP definition), those children shall be diagnosed as *smear positive TB or bacteriologically confirmed TB* and registered for TB treatment

(3b) If sputum examination is negative or not able to get sputum or no coughing, diagnosis is based on the four points below:

- Chronic symptoms suggestive of TB
- Physical signs highly suggestive of TB
- Positive TST or close contact with S+TB
- Chest X-ray suggestive of TB
 - Opacity± excavate
 - Infiltrate,
 - Hilar lymphadenopathies, etc...

(3.b.1) If children have chest x-ray suggestive of TB plus at least 2 other features, then children shall be diagnosed as *Smear Negative TB or clinically diagnosed TB*

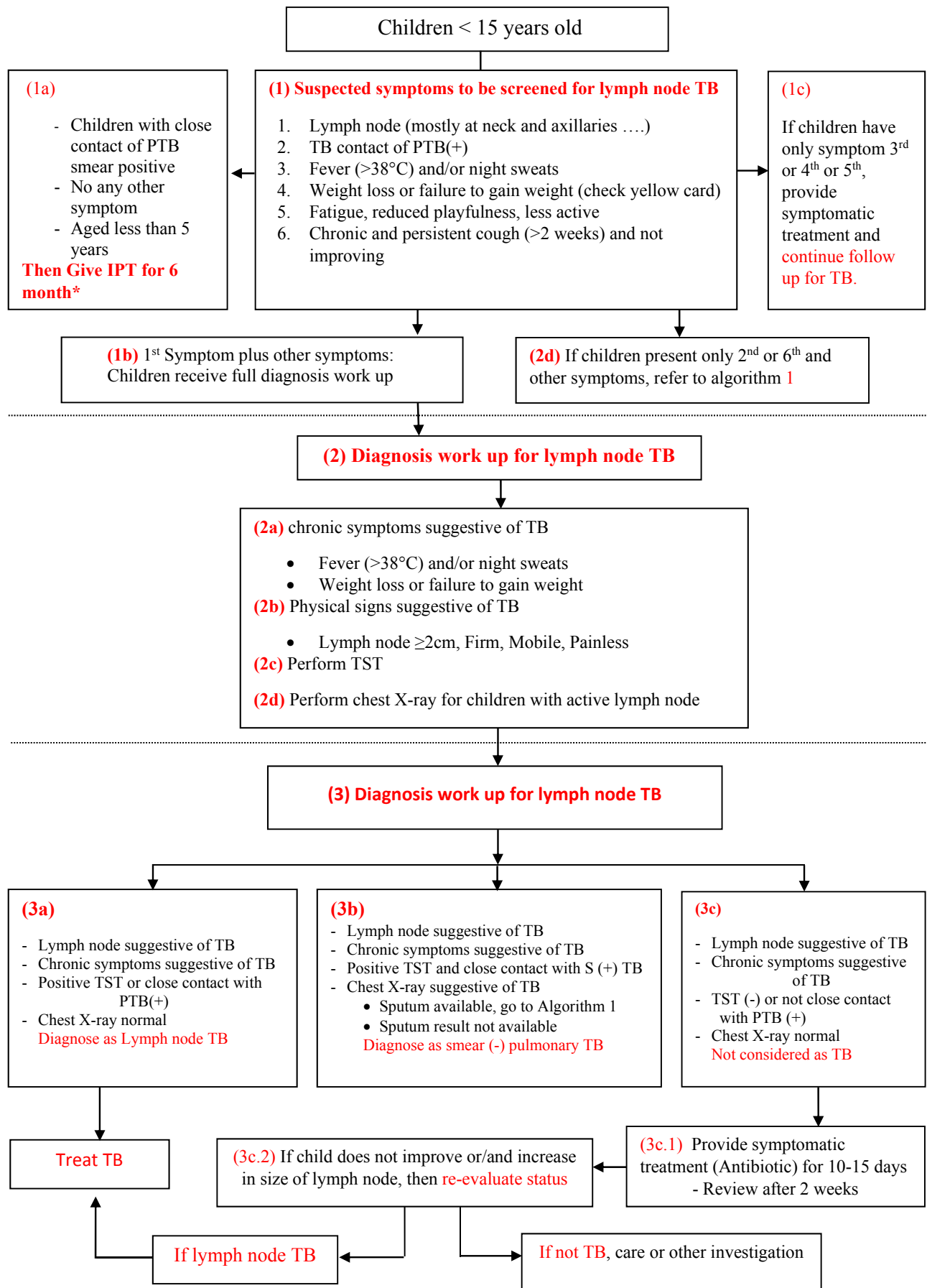
(3.b.2) If children have less than 3 features, *they are not considered as TB**; and:

- If they are severely sick, then they shall be referred to pediatric ward in hospital for further investigation for TB or other diseases
- If they are not severely sick, then they shall be provided with symptomatic treatment (antibiotic) for 10-15 days and re-evaluate.

* If children is a contact with BK⁺ TB patient and CXR shadow suggestive of TB, try to find other evidences including regular follow-up every 3 months through telephone or bringing children for screening.

Note: for detailed information, please refer to the National Guidelines for Diagnosis and Treatment of TB in Children, 2008.

Child TB Algorithm 2: for TB Adenitis in Children



Child TB Algorithm 2:

(1) Children aged less than 15 shall be screened for suspected lymph node TB as follows:

1. Lymph node (mostly at neck, axillaries)
2. TB contact
3. Fever ($>38^{\circ}\text{C}$) and/or night sweats
4. Weight loss or failure to gain weight (check yellow card)
5. Fatigue, reduced playfulness, less active
6. Chronic and persistent cough (>2 weeks) and not

(1a) If children are TB contact with smear positive patient, and do not have any other symptom (healthy) and aged less than 5 years, then **those children shall be given IPT (dose: 10mg/kg/day for 6 months)**.

***** For children aged 5 to 14 years old, they are not given IPT except children living HIV/AIDS.

(1b) If children have symptom 1 plus other symptoms (Ex. Symptom 1 plus 2 and 3 or symptom 1 plus 3 and 4), children shall be received full diagnosis work up carefully.

(1c) If children have only symptom 3, 4 or 5, should provide appropriate symptomatic treatment or continue follow up for TB.

(1d) If children have symptom 2 or 6 in combination with other symptoms, refer to algorithm 1

(2) In order to diagnose lymph node TB, TB physicians shall find out:

(2a) Chronic symptoms suggestive of TB:

- Fever ($>38^{\circ}\text{C}$) and/or night sweats
- Weight loss or failure to gain weight

(2b) Physical signs highly suggestive of TB:

******Lymph node assessment suggestive of TB

- Ongoing enlargement lymph node located at neck mostly, rarely at axillaries and other place
- Lymph node size: $\geq 2\text{cm}$, painless, mobile, firm, or fistula formation
- If available and possible, perform lymph node biopsy or fine needle aspiration

(2c) Perform TST

(2d) Perform chest X-ray for active lymph node

(3) In order to diagnose for TB, if there is a presence of lymph node suggestive of TB plus at least 2 others of 4 features, children shall be diagnosed as **lymph node TB or PTB**

(3a) If there is a presence of lymph node suggestive of TB, plus any chronic symptom suggestive of TB plus positive TST or TB contact with smear(+) patient and plus chest X-ray normality, then he/she shall be diagnosed as **lymph node TB**

(3b) If there is a presence of lymph node suggestive of TB, plus any chronic symptom suggestive of TB, plus positive TST or TB contact with smear + patient, plus chest x-ray suggestive of TB and:

- In case children have sputum, then go to childhood TB algorithm 1
- In case children do not have sputum, then he/she shall be diagnosed as **smear negative TB**

(3c) If there is a presence of lymph node suggestive of TB, plus any chronic symptom suggestive of TB, plus TST (-) or not close TB contact and plus chest x-ray normality, then:

(3c.1) children shall be give symptomatic treatment (antibiotic) for 10-15 days

(3c.2) if children are not improved or lymph node is still ongoing enlargement, then re-evaluate for TB:

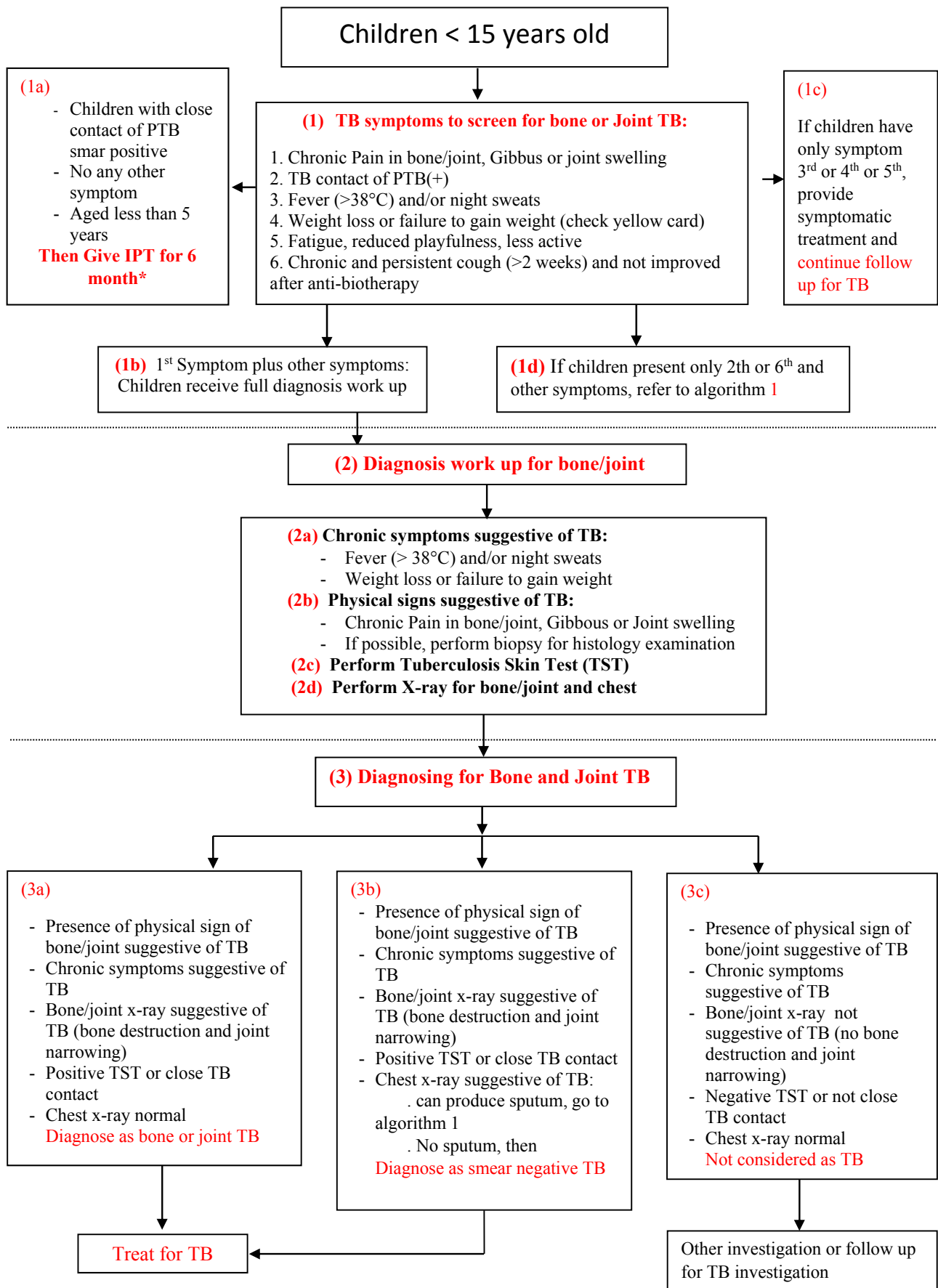
- If **lymph node TB** found, **treat TB**
- If not TB, need other care or investigation

Remarks:

- HIV+ children and severe malnutrition: TST positive ≥ 5 mm
- Other children and BCG vaccination: TST positive ≥ 10 mm

Note: for detailed information, please refer to the National Guidelines for Diagnosis and Treatment of TB in Children, 2008.

Child TB Algorithm 3: for Bone and Joint TB in Children



Child TB Algorithm 3:

(1) Children aged less than 15 shall be screened for suspected lymph node TB as follows:

1. Chronic Pain in bone/joint, Gibbus or joint swelling
2. TB contact especially with pulmonary TB smear positive
3. Fever ($>38^{\circ}\text{C}$) and/or night sweats
4. Weight loss or failure to gain weight
5. Fatigue, reduced playfulness, less active
6. Chronic and persistent cough (>2 weeks) and not improved after anti-biotherapy

(1a) If children are TB contact with smear positive patient, and do not have any other symptom (healthy) and aged less than 5 years, then **those children shall be given IPT (dose: 10mg/kg/day for 6 months)**.

* For children aged 5 to 14 years old, they are not given IPT except children living HIV/AIDS.

(1b) If children have symptom 1 plus other symptoms (Ex. Symptom 1 plus 2 and 3 or symptom 1 plus 3 and 4), children shall be received full diagnosis work up carefully.

(1c) If children have only symptom 3, 4 or 5, should provide appropriate symptomatic treatment or continue follow up for TB.

(1d) If children have symptom 2 or 6 in combination with other symptoms, refer to algorithm 1

(2) In order to diagnose bone/joint TB, TB physicians shall find out:

(2a) Chronic symptoms suggestive of TB:

- Fever ($>38^{\circ}\text{C}$) and/or night sweats
- Weight loss or failure to gain weight

(2b) Physical signs highly suggestive of TB:

- Chronic pain in bone/joint, Gibbus or joint swelling with or without neurological problem (paraplegia)
- If possible, perform biopsy for histological examination

(2c) Perform TST

(2d) Perform chest X-ray and bone/joint x-ray.

(3) Presence of bone/joint X-ray image suggestive of TB plus other 2 of 4 features, diagnosed as bone/joint or PTB

(3a) Presence of physical sign of bone/joint suggestive of TB

- Chronic symptoms suggestive of TB
- Bone/joint x-ray suggestive of TB (bone destruction and joint narrowing)
- Positive TST or close TB contact
- Chest x-ray normal

Then, child shall be diagnosed as **bone or joint TB**

(3b) Presence of physical sign of bone/joint suggestive of TB

- Chronic symptoms suggestive of TB
- Bone/joint x-ray suggestive of TB (bone destruction and joint narrowing)
- Positive TST or close TB contact
- Chest x-ray suggestive of TB and:

. If child can produce sputum, **go to algorithm 1**

. If child cannot produce sputum, then child shall be diagnosed as **smear negative TB**

(3c) Presence of physical sign of bone/joint suggestive of TB

- Chronic symptoms suggestive of TB
- Bone/joint X-ray not suggestive of TB (no bone destruction and joint narrowing)
- Negative TST or not close contact with TB patient
- Chest X-ray normal

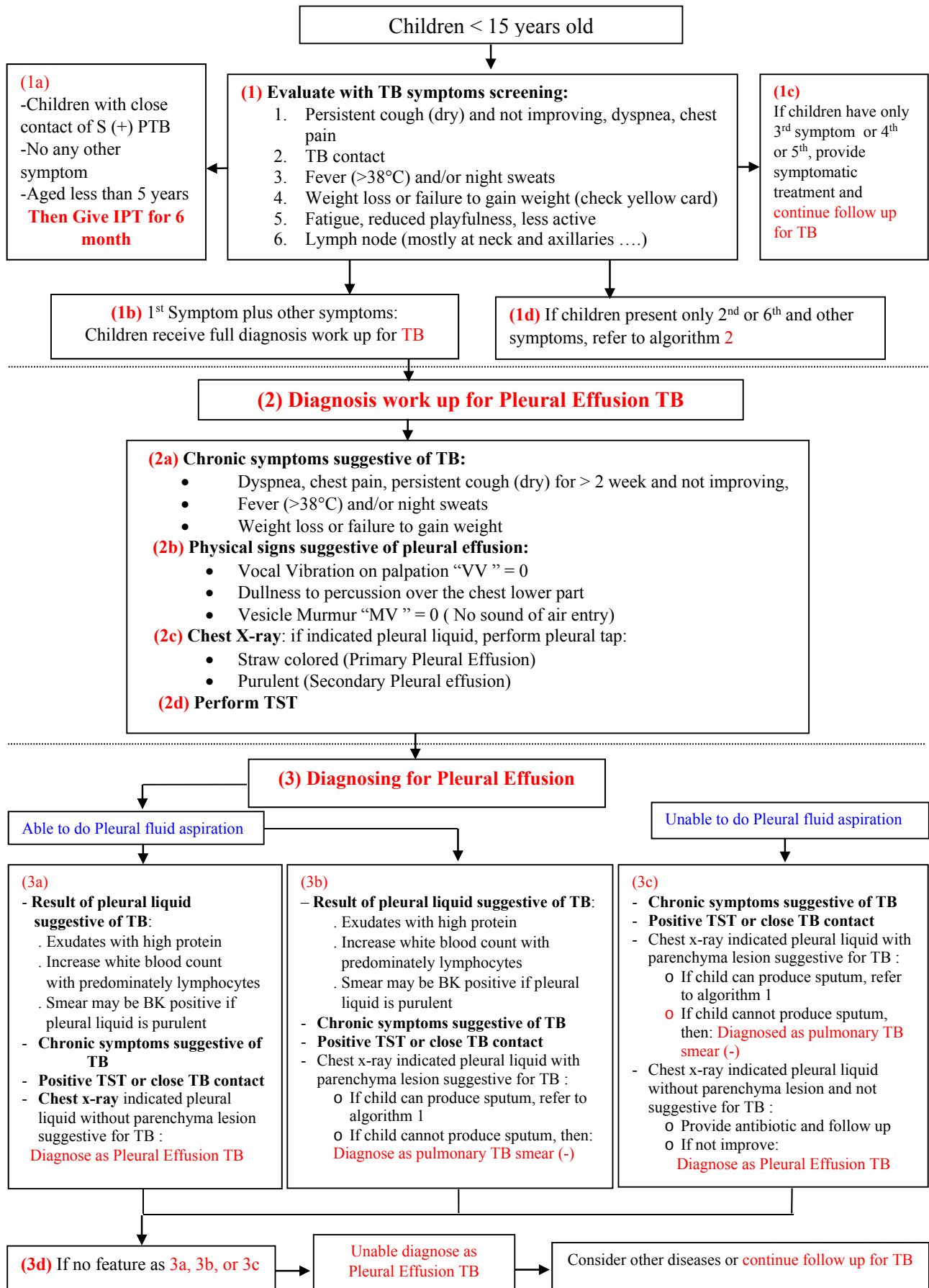
Then, child shall not be considered as TB, other investigation should be done.

Remarks: - for bone/joint TB and if patient has BK+ pulmonary TB, then patient is diagnosed as BK+ pulmonary TB and record result as cured if patient has appropriate follow up but need to continue treatment until 12 months (2RHEZ/10RH)

- record as BK+ PTB in white treatment card but in the note in the remarks as BK+ PTB and bone/joint TB

Note: for detailed information, please refer to the National Guidelines for Diagnosis and Treatment of TB in Children, 2008.

Child TB Algorithm 4: for Pleural Effusion TB in Children



Child TB Algorithm 4:

(1) Children aged less than 15 shall be screened for suspected Pleural Effusion TB as follows:

1. Persistent cough (dry) and not improving, dyspnea, chest pain
2. TB contact especially with patient pulmonary TB smear positive
3. Fever ($>38^{\circ}\text{C}$) and/or night sweats
4. Weight loss or failure to gain weight
5. Fatigue, reduced playfulness, less active
6. Lymph node (mostly at neck and axillaries)

(1a) If children are TB contact with smear positive patient, and do not have any other symptom (healthy) and aged less than 5 years, then *those children shall be given IPT (dose: 10mg/kg/day for 6 months)*.

* For children aged 5 to 14 years old, they are not given IPT except children living HIV/AIDS.

(1b) If children have symptom 1 plus other symptoms (Ex. Symptom 1 plus 2 and 3 or symptom 1 plus 3 and 4), children shall be received full diagnosis work up carefully.

(1c) If children have only symptom 3, 4 or 5, should provide appropriate symptomatic treatment or continue follow up for TB.

(1d) If children have symptom 2 or 6 in combination with other symptoms, refer to algorithm 2

(2) In order to diagnose pleural effusion TB, TB physicians shall find out:

(2a) Chronic symptoms suggestive of TB:

- Persistent cough (dry) and not improving, dyspnea, chest pain
- Fever ($>38^{\circ}\text{C}$) and/or night sweats
- Weight loss or failure to gain weight

(2b) Physical signs highly suggestive of TB:

- VV=0
- Reduced breath sounds (MV=0)
- Dullness on percussion

(2c) Perform chest x-ray: If there is pleural liquid

- Perform pleural tap:
 - o Straw colored: Primary Pleural Effusion and
 - o Purulent: Secondary Pleural effusion
- Perform biopsy for histology if possible

(2d) Perform TST

(3) Presence of pleural liquid or parenchyma suggestive of TB plus other 2 of 4 features, diagnose as **pleural effusion or PTB**

(3a)

- There shall be a presence of 3 below features if result of pleural liquid suggestive of TB:
 - Exudates with high protein
 - White blood count increase
Predominately lymphocytes
 - Smear positive if pleural liquid is purulent (Secondary pleural effusion)
- Chronic symptoms suggestive of TB
- Positive TST or close TB contact
- Chest X-ray indicated pleural liquid without parenchyma attack suggestive of TB, then:
Child shall be diagnosed as *Pleural Effusion TB*

(3b)

- There shall be a presence of 3 below features if result of pleural liquid suggestive of TB:
 - Exudates with high protein
 - White blood count increase
Predominately lymphocytes
 - Smear positive if pleural liquid is purulent (Secondary pleural effusion)
- Chronic symptoms suggestive of TB
- Positive TST or close TB contact
- Chest x-ray indicated pleural liquid with parenchyma attack suggestive of TB, then:
 - o If child can produce sputum, go to algorithm 1
 - o If child cannot produce sputum, then:
Diagnose as *smear negative TB*

(3c) If unable to do pleural fluid aspiration, consider below:

- Chronic symptoms suggestive of TB
- Positive TST or close TB contact
- Chest x-ray indicated pleural liquid with parenchyma attack suggestive of TB, then:
 - o If child can produce sputum, go to algorithm 1
 - o If child cannot produce sputum, then:
Diagnose as *smear negative TB*
- Chest X-ray indicated pleural liquid without parenchyma attack suggestive of TB, then:
 - o Give common antibiotics and follow up
 - o If not improved with anti-biotherapy, diagnose as *Pleural Effusion TB; then treat for TB.*

(3d) If result does not fit with any of the above features 3a or 3b or 3c, not diagnose as TB; but consider other diseases or continue follow up for TB.

Note: for detailed information, please refer to the National Guidelines for Diagnosis and Treatment of TB in Children, 2008.

12.3. Treatment of TB in Children

1. Treatment of TB in children, HC staff receive children with TB for treatment when:
 - a. Children referred back to HC after being diagnosed at RH
 - b. Children referred from other diagnostic health facilities
2. Before put children on treatment, health education about benefit of treatment must be provided to parents or care givers in order that they will provide TB drugs to children regularly and until completion of treatment.
3. Treatment regimens for TB in children of national TB program:
 - Treatment category-1a: new case: pulmonary smear positive TB, pulmonary smear negative (severe form), extra pulmonary TB (severe form), TB-HIV (2RHZE/4RH).

Category-1a: 2RHZE/4RH (Children)

Use Pediatric Drugs for Children Weighted ≤ 20 Kg					
Weight Bands (Kg)	Intensive Phase: 2 months			Continuation Phase: 4 months	
	RHZ 60/30/150	RH 60/60	E 100	RH 60/30	RH 60/60
5-7	1 tab	1 tab	1 tab	1 tab	1 tab
8-14	2 tab	1 tab	2 tab	2 tab	1 tab
15-20	3 tab	2 tab	3 tab	3 tab	2 tab
Use Adult Drugs and RH (60/60) for Children Weighted 21-30 Kg					
Weight Bands (kg)	Intensive Phase: 2 months		Continuation Phase: 4 months		
	RHZE 150/75/400/275	RH 60/60	RH 150/75	RH 60/60	
21-30	2 tab	2 tab	2 tab	2 tab	
Dosage / kg / day	R= 15 mg (10-20), H= 10 mg (10-15), Z= 35 mg (30-40), E= 29 mg (15-25)				

- Treatment category-1b: new case: TB meningitis and bone/joint TB (2RHZE/10RH).

Category-1b: 2RHZE/10RH (Children)

Use Pediatric Drugs for Children Weighted ≤ 20 Kg					
Weight Bands (Kg)	Intensive Phase: 2 months			Continuation Phase: 4 months	
	RHZ 60/30/150	RH 60/60	E 100	RH 60/30	RH 60/60
5-7	1 tab	1 tab	1 tab	1 tab	1 tab
8-14	2 tab	1 tab	2 tab	2 tab	1 tab
15-20	3 tab	2 tab	3 tab	3 tab	2 tab
Use Adult Drugs and RH (60/60) for Children Weighted 21-30 Kg					
Weight Bands (kg)	Intensive Phase: 2 months		Continuation Phase 4months		
	RHZE 150/75/400/275	RH 60/60	RH 150/75	RH 60/60	
21-30	2 tab	2 tab	2 tab	2 tab	
Dosage / kg / day	R= 15 mg (10-20), H= 10 mg (10-15), Z= 35 mg (30-40), E= 29 mg (15-25)				

- Treatment category-2: retreated case: Relapse, Failure, RAD and Others (2RHZES/1RHZE/5RHE).

Category-2: 2RHZES/1RHZE/5RHE (Children)

Use Pediatric Drugs for Children Weighted ≤ 20 Kg										
Weight Bands (kg)	Intensive Phase: 3 months							Continuation Phase: 5 months		
	Intensive Phase (2m)				Intensive Phase (M3)					
	RHZ 60/30/150	RH 60/60	E 100	S 1 g	RHZ 60/30/150	RH 60/60	E 100	RH 60/30	RH 60/30	E 100
5-7	1 tab	1 tab	1 tab	100 mg	1 tab	1 tab	1 tab	1 tab	1 tab	1 tab
8-14	2 tab	1 tab	2 tab	200 mg	2 tab	1 tab	2 tab	2 tab	1 tab	2 tab
15-20	3 tab	3 tab	3 tab	300 mg	3 tab	2 tab	3 tab	3 tab	2 tab	3 tab
Use Adult Drugs for Children Weighted 21-30 Kg										
Weight Bands (kg)	Intensive Phase: 3 months						Continuation Phase: 5 months			
	Intensive Phase (2m)			Intensive Phase (M3)						
	RHZE (150/75/400/275)	RH 60/60	S 1 g	RHZE (150/75/400/275)	RH 60/60		RH 60/30	RH 60/30	E 100	
21-30	2 tab	2 tab	100 mg	2 tab	2 tab		2 tab	2 tab	1 tab	
Dosage / kg / day: R= 15 mg (10-20), H= 10 mg (10-15), Z= 35 mg (30-40), E= 29 mg (15-25)										

- Treatment category-3: new case: pulmonary smear negative (non-severe form), extra pulmonary (non-severe form) (2RHZ/4RH)

Category-3: 2RHZ/4RH

Use Pediatric Drugs for Children Weighted ≤ 20 Kg				
Weight Bands (Kg)	Intensive Phase: 2 months		Continuation Phase: 4 months	
	RHZ 60/30/150	RH 60/60	RH 60/30	RH 60/60
5-7	1 tab	1 tab	1 tab	1 tab
8-14	2 tab	1 tab	2 tab	1 tab
15-20	3 tab	2 tab	3 tab	2 tab
Use Adult Drugs and RH (60/60) for Children Weighted 21-30 Kg				
Weight Bands (kg)	Intensive Phase: 2 months		Continuation Phase: 4 months	
	RHZE 150/75/400/275	RH 60/60	RH 150/75	RH 60/60
21-30	2 tab	2 tab	2 tab	2 tab
Dosage / kg / day	R= 15 mg (10-20), H= 10 mg (10-15), Z= 35 mg (30-40), E= 29 mg (15-25)			

Remarks: The above treatment regimens for childhood TB will soon be revised. Then NTP will organize training for relevant health workers for implementation arrangement.

Monitoring treatment of TB in children: HC staff has to do:

- Smear follow up at the end of month 2, beginning of month 5 and mid-last month of the treatment, if children are pulmonary smear positive TB
- Clinical monitoring for pulmonary smear negative and extra pulmonary TB
- Regular monitoring the treatment compliance
- Monitoring children weight
- Monitoring side effects

Health education is introduced in general part as the same as adult.

13. TB AMONG HIGH RISK GROUP

13.1. TB in Elderly

Among the general population, people aged 55 years old or older is about 7% but prevalence of TB among those age groups (434/100,000 population) is around 3 times higher than general population (118/100,000 population) according to second national TB prevalence survey 2011.

During 10 years from 2001 to 2010, although National TB control Program has increased TB case detection up to 41,628 in 2010, currently the number of old people access to TB service is still limited. Therefore, to improve TB case detection and treatment among elderly population, all elderly aged from 55 years old whether they present TB symptom or not, need to come to check for TB disease at least once a year. Moreover, NTP are finding resources and moving passive and active case findings among elderly people aged 55 years old and over.

13.2. TB in Prison

TB control in prison is one among the other priorities of NTP. TB prevalence among prisoners is higher than general people. Now we have 26 prisoners implementing TB control in Prison. NTP will strength cooperation with involved partners and mobilized resources in order to comprehensively carry out all activities and principles mentioned in standard operating procedure for TB in prison. TB in prison activities need to emphasized either routine or active case findings.

13.3. TB in Migrants

This is also one of priority tasks of NTP. NTP is cooperating with IOM to screen TB among migrant people. Moreover, we are doing mobilization resources to identify problem as well as TB control in migrant people including TB screening among migrant people at the border between Cambodia and neighboring countries.

13.4. TB and Diabetes

The number of patients with diabetes is expected to increase twice between 2000 and 2030 because of prolonged life expectancy in urban and the change of life style in foods and nutrition.

In early 2013, NTP and URC have conducted a study and found that the prevalence of pre-DM and DM among people living in Siem Reap was 25% and 7% respectively. And the prevalence of pre-DM and DM among people living in Prey Veng was 22% and 5% respectively. But TB prevalence among diabetes patients was 4.43% in Siem Reap, this number indicated that it was 6 times higher than TB prevalence in general people. Therefore, diabetes screening among TB patients and TB screening among diabetes is very important and is a new priority of TB program.

All diabetic patients (all new cases or follow-up case/old cases) need to be screened to find presumptive TB with symptom for at least 2 weeks (Cough, fever, night sweet and weight lost with unknown reason). When doing diabetic consultation, if there is symptom as mentioned above, TB diagnosis should be considered. If patients have any symptoms above before appointment date, patients need to come back early to diabetes clinics. For patients who diagnosed as extra pulmonary TB is also possible to have others symptoms such as lymph node and joint swelling.

Patients with diabetes are at high risk of TB disease. Now the prevalence of TB among diabetic patients is higher than in general population. According to WHO and IUATLD, the diabetes screening should do for all TB patients during TB identified or anytime during TB treatment.

Primary examination for diabetes among TB patients can be done by using portable glucometer with glucose strip at TB ward at health center or during community outreach activities by trained physician, nurse laboratory technician, peer education or community volunteer DOT watcher.

Pilot project on primary examination for diabetes are operating in Siem Reap, Soth Nikum ODs in Siem Reap province and Pearaing OD in Prey Veng province. This pilot project is planning to expand to others diabetes clinics and related TB services.

13.5. TB and Tobacco Smoking

Tobacco is a risk factor to harm TB disease. Every year number of people who got disease and died from respiratory disease are increasing, these due to some risk factors like habit to smoke cigarettes. A study research in Cambodia has showed that the prevalence of smoker in Cambodia is highest in the southeast region: 54% among men and 6% among women and 14% of TB patients are smoker.

In order to get improvement or treatment outcome successfully, TB smoker patients need to quit smoking. To get patients quit smoking unless health staff at every health center got the training on strategy to quit smoking and continuous education and provide counseling to patients. Health staff should receive training of trainer on strategy to quit smoking, counseling courses and those course are including in the following topics; harmful chemical in tobacco smoke, passive smoking health effect, treatment on tobacco addicted and summary strategy to help patients to quit smoking, we have 5 A: Ask to know who are smoker and Advice to quit smoking, Assess to determine the level to quit smoking, Assist to help patients to quit and to Arrange methods to quit smoking. After the training, health staff who got the skill need to continues educating to all TB patients who come to treat TB and help them to quit smoking.

14. TB PREVENTION

14.1. Introduction

Like the prevention of active TB development, prevention of transmission plays an important role in provision of safety of health care environment for general people especially children and health staff who are working in peripheral health facility and community. There are essential methods for prevention as follows: 1) early active TB case detection and treatment; 2) detection and IPT provision for PLHIV; 3) BCG vaccination; and 4) TB infection control.

14.2. Isoniazid Preventive Therapy

The purpose of the Isoniazid Preventive Therapy (IPT) is to stop the development from TB infection to active TB disease. For IPT, the National TB Control Program (NTP) focuses on two target groups: 1) under 5 year-old children with good health who live with smear positive TB patient and 2) PLHIV who do not have active TB.

14.2.1. Under 5 Year-old Children

Children who are eligible for IPT must meet 3 criteria: under 5 year old, living with smear positive TB patient and having good health. Healthy Children should not have any of the below symptoms:

- Chronic cough $\geq$ 2 weeks
- Chronic fever $\geq$ 2 weeks
- Weight loss or failure to thrive
- Less daily active
- Visible lymph node at neck and armpit
- Gibbous

IPT drug:

- Dose: 10 mg/kg/1day
- Treatment period: 6 months

14.2.2. People Living with HIV/AIDS

The study reveals that the provision of IPT for PLHIV who have TB infection can reduce the chance of TB active development (TB disease). The counseling and IPT management for PLHIV should be performed at OI/ART service:

- Client eligible for IPT is the PLHIV who do not have any TB symptom suggestive of active TB
- IPT management should be started as soon as possible, but if PLHIV are banned from using contraindication drug, the IPT should be cancelled
- The treatment period is 6 months and drug dose is 5 mg/kg/day for adult and 5 mg/kg/day for children under 15 years old
- Vitamin B6 of 50 mg/kg/day should be given along with IPT

14.3. BCG Vaccination (Bacillus Calmette-Guerin)

BCG vaccine is prepared from a strain of the attenuated (virulence-reduced) live bovine tuberculosis bacillus, *Mycobacterium bovis*, that has lost its ability to cause disease in humans. In the countries like Cambodia where the TB prevalence is high, BCG vaccination is recommended for new

born babies (less than 1 month) to prevent the infection of severe TB case especially TB meningitis and military TB. In Cambodia, BCG immunization is under the management of national immunization program of the ministry of health. The main source of TB infection is from TB active patient especially smear-positive TB patient.

14.4. TB Infection Control

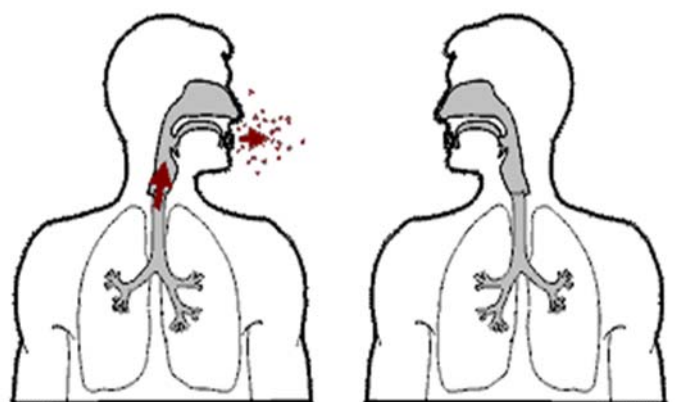
The National Standard Operating Procedure (SOP) of TB infection aims at targeting to health care facilities as well as congregated setting such as prison and factory was developed in 2011. The chapter for TB infection control in the MDR-TB guideline was updated. In Cambodia the infection control through air was integrated into the national policy for infection control and in the technical guideline as well as in guideline for OI/ART. However, more than 75% of health staff have not been trained for TB infection control and their understanding of TB infection control is limited. The ventilation in most of the hospitals is good, but the implementation is not proper (door, or window or working place is not properly prepared). The separation of coughing patients in the OPD service and hospitals is not well prepared and the education materials for infection control are limited in health centers.

14.4.1. Purpose of TB Infection Control

TB infection control is aimed at providing people with prevention of TB infection from TB patients who speak, cough, sneeze (picture 1) to health staff, guests and other patients who stay in hospital or health center as well as in congregate setting such as prison and factory.

- The TB infection precaution consists of three ways:
 - 1) Contact precaution
 - 2) Droplet precaution
 - 3) Airborne precaution
- The TB infection control program should be based on three levels of control measures and include:
 - 1) Administrative measures
 - 2) Environmental controls
 - 3) Use of respiratory protective equipment

Picture 1: TB infection from contact TB



TB patient

TB Contact

14.4.2. Risk Factors for Health Staff

- Delay in diagnosis and treatment for TB
- TB treatment is not properly implemented based on national guideline
- Making mistake in sputum collection and preparation causes high risk for biological safety
- There are no proper cough triage in health care setting
- There is no proper ventilation in health care setting
- Lack of personal protection equipment such as surgical mask for consultation
- Do not implement TB infection measures correctly
- Health staff have HIV infection

14.4.3. Measurement for TB Transmission

There are 4 measures to minimize TB infection in healthcare setting:

1. Managerial TB infection control

In every hospital managerial TB infection control committee (TBIC) must be set up. This committee is a part of the hospital prevention control committee with at least 5 members from hospital committee, TB ward, OPD, laboratory and OI/ART clinic. The managerial TB infection control committee has the following tasks:

- Check and strengthen local health services and implement TB infection control including administrative measure, environmental measure and personal protection measure.
- Recheck the ventilation system to make sure that it is correctly installed naturally and mechanically.
- Follow up, evaluate and make plan for the implementation of the three TB infection control measures (administrative measure, environmental measure and personal protection measure).

2. Administrative measure

Administrative measure is the main measure and easy to implement among the three measures for TB infection control at local health setting. Administrative measure includes:

- Establish proper cough triage for both OPD and IPD as well as other services in hospital
- Separate TB patients, especially smear positive TB or Presumptive TBs from other patients
- Instruct patients to cover their mouth by Krama or mask when they speak, cough sneeze to prevention TB infection to neighbors
- Start first consultation for Presumptive TBs (esp. cough) to minimize time for attending in hospital
- Conduct active case finding among health staff for every two years. If health staff have HIV infection, they should be given ART or IPT if they do not have active TB

3. Environment measure

- Natural ventilation: Set up proper natural and mechanical ventilation
- Ultraviolet: Ultrasound kills bacteria
- Building with good ventilation: shall have plan to renovate building or build new one if necessary

4. Personal protection equipment measure

- Who wear mark: Everyone who contact with TB patient such as health staff, TB patient's relatives who come to visit patient must wear mark to prevent TB infection especially at high risk areas such as laboratory, MDR-TB ward and smear positive TB ward.

- Type of mask:
 - √ Health staff, accompanied or relatives should wear N95 mask
 - √ TB patients or Presumptive TBs (cough) should wear surgical mask
- Time of mask wearing:
 - √ New smear positive TB patients should wear mask at least two weeks
 - √ Old smear positive TB patients should wear mask at least two months
 - √ MDR-TB patients should wear mask at least 4 to 6 months

14.4.4. Scope of TB Infection Control Strategy

The strategy of TB infection control is implemented at the health facilities as well as congregated areas such as prison and factory. However, the implementation of this strategy at those areas is limited due to the lack of supportive resource. In spite of this, hospitals should strengthen administrative measure at least like setting up of cough triage, proper separation for smear positive TB and MDR-TB patients at OPD and IPD as well as emergency ward and other services. For environment measure, hospital should prepare a good quality of ventilation system. The hospitals should prepare a good plan of securing the supply and storage of personal protection equipment.

15. ADVOCACY, COMMUNICATION, AND SOCIAL MOBILIZATION

Advocacy, communication, and social mobilization (ACSM) or called Information, Education, and Communication Advocacy (IEC Advocacy) is an important part of TB control program. It is an effective strategy in controlling TB, especially in:

- Improving TB case detection and treatment adherence
- Combating stigma and discrimination
- Empowering community and people affected by TB
- Mobilizing political commitment and resource for TB control

The key activities of ACSM should be done in difference ways such as through:

- Conducting workshops and meetings
- Different types of human resource mobilizations
- Different types of providing information and communication such as public broadcasting (Radio and Television), News, Magazines, social media. etc.
- Individual education is a strategy of providing information from one person to another such as health care worker educates patient during providing medical treatment and community outreach (community education).
- All activities lead to change the behavior of people and health care workers
- Mobilizing the supports from community, volunteers, religious, and social media broadcasting.
- Conducting World TB Days nationwide of all levels

These activities should be done at all levels including country level and global level. All relevant projects and partners of TB control program have to apply ACSM strategies in order to achieve their goals under coordinating and facilitating from national TB control program.

16. KEY ACTIVITIES OF NATIONAL TB PROGRAM

16.1. Public-Private Mix in Fighting against TB

16.1.1. Introduction

The private sector is the first place where most people who seek health care and treatment including TB. CENAT, in 2005 with the collaboration of partners had launched a Public Private Mix (PPM) strategy as a key approach to improve access to TB services through various types of private healthcare providers who also from pharmacies and clinics.

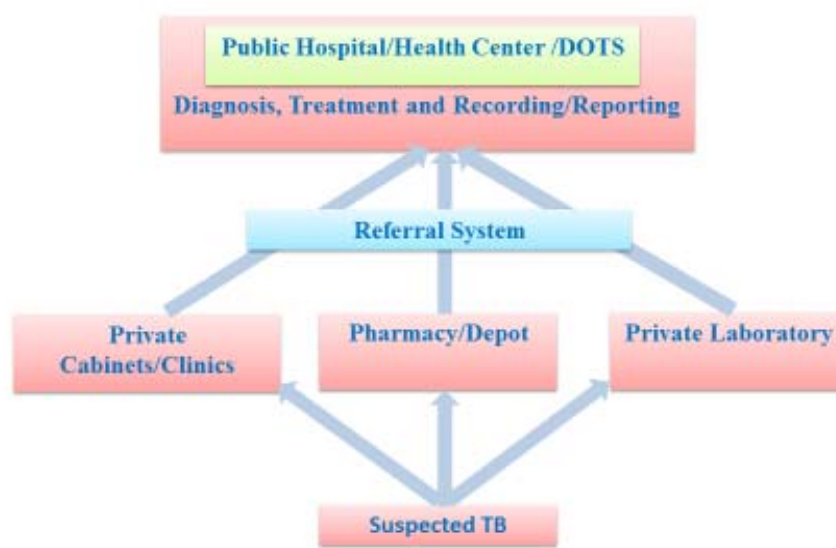
Since 2005 NTP had expanded the activity of Public Private Mix in implementing DOTS strategy (PPM-DOTS) to cover 35 ODs in 2007 and increased to 37 ODs countrywide. Consequently, between 2011 and 2013, the number of ODs coverage had decreased to 36 ODs and continue to decline to 27 ODs in 2014. Due to the lack of resources, the Public Private Mix DOTS had been discontinued since 2015 up to present. The first phase of PPM-DOTS is to urge private sectors to refer presumptive TBs to public sector for diagnosis and treatment. As a result, the private sectors had contributed 6% of all TB cases detected in all ODs coverage. The second phase is yet not to be launched, will engage of private health care providers in comprehensive TB case management including diagnosis and treatment.

Ministry of Health, in 2011 and 2012, issued the Prokas to prohibit the importation and sale of anti-TB drugs and serological tests that resulted remarkable successful. Private laboratories and other private sectors have been encouraged to follow NTP Policy and Guidelines.

16.1.2. Objectives

- Mobilize resources of public and private providers to fight TB through establishing referral system using DOTS strategy aiming to reduce morbidity and mortality rates to reach NTP goal
- Increase TB case detection and to prevent the emergence of MDR-TB

Referral System



16.1.3. Evaluation

Hospitals, both public and private were found during the 2012 Program Review, did not screen routinely for TB symptoms. Even worse, the referrals of Presumptive TBs to hospitals, or the referrals among departments in hospitals, and feedback back to clinics were often lost. The volume of Presumptive TBs referrals from private clinics or sectors is remaining low. Some clinics diagnosed and managed TB cases not following NTP guidelines.

The program review indicated that many private laboratories diagnosed TB cases from private providers, but not reporting them to CENAT. The total number of private providers involved in PPM-DOTS has been falling down so as for the number of cases identified by this activity.

16.1.4. Next Plan

PPM-DOTS will be expanded to cover all provinces and ODs. Urban and peri-urban areas, which have much higher concentration of private providers and large hospitals, continue to require special PPM approaches, separate from C-DOTS. Harmonization of C-DOTS activities and Private Public Mix (PPM) approaches aimed at engaging all care providers, including pharmacists is needed at grassroots in rural areas, to ensure early referral of persons with suspected TB.

The successful bans on importation and sale of anti-TB drugs and serological tests in private pharmacies will be maintained, re-enforced and monitored. Private laboratories and care facilities will be encouraged to follow NTP policy guidance.

16.2. Community DOTS

16.2.1. Introduction

As stated in its five-year strategic plan for TB control 2001-2005 with the purpose of expanding the DOTS strategy to all health centers by 2005, NTP has reached its target whereby 98% of health centers covered with DOTS services by the end of 2004. At the beginning of the implementation of the DOTS strategy at health center, NTP focused mainly on Ambulatory DOT, i.e. TB patients have to come to health center to take anti-TB drugs every morning under direct observation of health staff. According the review made after a certain period of implementation, it was observed that not all diagnosed patients could receive TB treatment. This was due to a number of reasons such as elderly people or patients living far away from health centers. To address this issue, new priority was considered in order to ease the access to proper TB diagnosis and treatment. To respond to the necessary needs mentioned above, through collaboration between NTP and NGO partners, pilot implementation of Community DOTS has been started in some ODs in Svay Rieng, Pursat, Kampot, and Kampong Chhnang provinces since 2002.

16.2.2. Definition of C-DOTS

Community DOTS is the involvement of community to supplement to TB control activities at community level. Community may be at least involved with Community DOTS in 4 main activities: (1) participating in TB case findings; (2) Support of TB treatment in the community (DOT observer); (3) TB prevention by providing health education to the community to promote TB awareness and provision of Isoniazid Preventive Therapy [IPT] to children under five years old; and (4) Participation in dealing with certain social problems which prevent community people from accessing to TB services.

16.2.3. Objectives and Activities

Overall objective of community DOTS implementation is to help improve access of general population including high risk group to quality DOTS services as well as to receive proper TB diagnosis and treatment on time.

In order to achieve this overall objective of C-DOTS, some activities below should be conducted:

- Provide training for village health support group (VHSG) on basic TB information
- Coordination meeting at health center level between health center staff, VHSGs/Community DOT Watchers (CDWs) etc.
- Provide health education to the community about TB
- Routine TB case finding and semi-active TB case finding at pagoda, mosque, Church, village focusing on high risk group or key affected population
- Contact investigation of the people living in close contact with bacteriologically confirmed pulmonary TB patient
- Provide TB treatment in the community under direct observation (DOT) from VHSG/trained volunteer (DOT-watcher)
- Provide Isoniazid Preventive Therapy (IPT) to healthy children aged less than five years old living in close contact with bacteriologically confirmed pulmonary TB

16.2.4. Main Activities

16.2.4.1. Raising TB Awareness in the Community

Training and refresher trainings for village health support group (VHSGs), DOT-watchers, and regular meetings is an essential approach to promote TB awareness of DOT-watchers in order to ensure that they have full capacity in assisting in early TB case findings and treatment. In order to properly monitor and evaluate community DOTS, it is necessary to regularly conduct bi-monthly or quarterly meeting.

16.2.4.2. Routine and Semi-active TB Case Findings at Pagoda and Village

Community volunteers will routinely help identify presumptive TB (TB suspect) through their daily activities. In addition to routine TB case findings, semi-active TB case findings which focuses on elderly people at pagoda, other religious temples, and remote villages are implemented as supplementary activity to assure that high risk group get access to quality TB services as the others i.e. access to proper TB diagnosis and treatment on time.

16.2.4.3. TB Contact Investigation

The investigation for TB disease among people living in close contact with bacteriologically confirmed pulmonary TB patient has two main purposes: 1) to find missing TB cases; and 2) to rule out active TB disease in order to provide IPT for healthy children aged less than five years old living in close contact with bacteriologically confirmed pulmonary TB patient. The target groups for the contact investigation are people living in the same household with bacteriologically confirmed pulmonary TB patient (family members or relatives), neighbors, and those who are in close contact with BK+ confirmed pulmonary TB patient.

16.2.4.4. TB treatment Support through DOT Approach

If someone is found to have TB disease, he or she will receive TB treatment with DOT approach. In addition to hospitalization or ambulatory DOT, TB patient may also take anti-TB drugs in the community under the direct observation from a community member. In this case, community DOT-watcher has to observe TB patient taking anti-TB drugs every morning and ensures that TB patient swallow anti-TB drugs regularly at the right time and with the appropriate prescribed dosage. DOT-watcher or TB patient needs to withdraw anti-TB drugs from health center every week or every two weeks in intensive phase of the treatment, according to their agreement with health staff.

In continuation phase, TB patient or DOT-watcher has to withdraw anti-TB drugs from health center at least twice per month. Health center staff may also bring anti-TB drugs for DOT-watchers or TB patients if possible. Besides observing patient swallowing anti-TB drugs, community DOT-watcher should encourage TB patient to adhere to TB treatment until it is complete.

**MINISTRY OF HEALTH
National Center for Tuberculosis and Leprosy**

**KINGDOM OF CAMBODIA
National Religion King**

Guidance for Case Finding of Multi-Drug Resistant Tuberculosis

A. Introduction

Tuberculosis (TB) is still a major public health problem to Cambodia. Therefore, the existence of Multi-Drug Resistant TB (MDR-TB) would become an additional burden to National TB Control Program (NTP). In Cambodia, the first case of MDR-TB was found and enrolled for treatment in 2006. WHO expert on MDR-TB has estimated that, in Cambodia, approximate 400 newly MDR-TB cases occur every year.

Furthermore, the morbidity and mortality of HIV patients are mainly due to TB and to diagnose TB among PLHIV is really difficult, especially to diagnose by smear microscope. Anyway, sputum testing by Xpert is a new technical method to diagnose TB, through detecting TB bacilli in a short time and also detecting Rifampicin Resistant strain. WHO has recommend to use this method (Xpert) for diagnosing TB among the target populations such as presumptive MDR-TB patients, presumptive TB among PLHIV, and other high risk groups for TB (prisoners, diabetes, elderly, ...etc.)

Three key effective strategies to fight against MDR-TB are to:

- 1) Timely identify and treat drug susceptible TB patients and effectively, i.e. treat them with CAT1 or CAT2 or CAT3 until they are cured;
- 2) Timely identify and treat Multi-Drug Resistance TB;
- 3) Prevent transmission of MDR-TB to other (contacts or those who live nearby the patients included neighbors and health care staff).

This guidance has been developed through revising “**Guidance for Case Finding of Multi-Drug Resistant Tuberculosis**” No. 737 and issued on September 25, 2012 base on the new updated guidelines of WHO including revised case definition and Xpert MTB/RIF for the diagnosis of pulmonary and extra-pulmonary TB, policy update, and consulting with concerned partners of the National Center for TB and Leprosy Control (CENAT). It is aimed at providing with three options of strategies for referring MDR-TB suspected patients or sputum specimen as well as providing information related to some rates and procedures for spending and referral, which include how to provide support to patients and staff who handle the job. This document went into effect from April 01, 2014.

B. Criteria for MDR-TB Suspect

Currently, priority criteria for MDR-TB suspect in Cambodia are the following:

1. Pulmonary TB with previously treated TB for more than 1 month:
 - Failure: A patient who had treated for TB (the last TB treatment) and confirmed as failure (defined as sputum smear positive at month 5 or later during treatment).
 - Relapse: A patient who had treated for TB and was confirmed as cured or completed. Currently the patient was diagnosed as TB.
 - Loss to follow up : A patient who had treated for TB and the last treatment had been taken at least one month and confirmed as interrupted his/her treatment for the 2 consecutive months. Currently, patient is re-diagnosed as TB again.
 - Others: A patient who has been treated for TB for more than one month and the treatment outcome was not known or did not record his/her treatment outcome. Currently, patient is re-diagnosed as TB again.
2. Smear positive pulmonary TB cases, whose sputum smear examination remains positive at month 3 during treatment.

Non converter at month 2 can be referred for diagnosing MDR-TB in ODs or provinces where non converter at month 3 was referred for testing MDR-TB more than 80%.
3. New pulmonary TB cases or symptomatic TB who are close contact with MDR-TB patients.
4. Pulmonary TB smear positive cases are co-infected with HIV.

Persons, who are working for TB, have to refer patient or sputum samples of above MDR-TB suspects to Xpert centers by any options mentioned in section C. Presumptive TB, who was previously treated for TB, should be tested for TB or MDR-TB by Xpert (smear microscopy was not required).

C. Procedure for Sputum/Specimen Handling or Referral

To facilitate the process for identifying MDR-TB cases and provide financial support to MDR-TB related activities that included specimen collection and transport, the national program endorses 3 options for operation:

1) Option 1:

- When a MDR-TB suspect was identified as mentioned in section “B” above, community workers included C-DOT watchers or villages health support group (VHSG), staff of health center or former district hospital (FDH) will refer the MDR suspect to referral hospital (RH) or CENAT using tetra-cycle motor or motor-bike taxi, horse card etc. In case of referred patient taking risk for infection such as by taxi or bus, a patient is encouraged to cover his/her mouth with mask at all the times. The patient will be offered/reimbursed for their local transport and accommodation while staying in the referral hospital or CENAT for specimen collection (3 sputa) and diagnosis performance (please see annotation 1).
- When MDR-TB suspect patients arrive at RH or CENAT, the patient’s samples were collected to perform by Xpert MTB/RIF. For RHs that Xpert MTB/RIF is not available, TB supervisor or TB lab technician of the RH have to collect 3 sputum specimens within 2 days

and place the sputum cups in cold box and transport to culture center at Kampong Cham (KPC) or Battambang (BTB) RHs or CENAT for performing Xpert MTB/RIF or culture (please read section “D” concerning provinces for which to culture sites where they need to refer specimens). The person who in charge of packaging and transport the samples will receive financial support for the completion of 3 sputa (see annotation 2).

- The referral hospital also has to collect samples from MDR-TB suspect patients from their own hospital for Xpert MTB/RIF or transport to culture center at KPC-PRH or BTB-PRH or CENAT (please read annex 1 for more detail).
- Specimen referral should be carried out using practical means which included paid vehicle and taxi. Payment will be covered depending on actual road condition and distance (see annotation 3). Duration of transport and the assurance of sputum condition and result returned from culture sites BTB, KPC & CENAT to the referring facilities will have to be clearly indicated.
- Lab technician at any RHs who has Xpert machine, and BTB PRH, KPC-PRH, and CENAT have to perform all of these relevant activities including effective feedback on lab result. They also get some compensation (see annotation 4). *It is an option, since patient is observed and correctly followed up by the health staff.*

2) Option 2:

- When a MDR-TB suspect was identified as mentioned in section “B” page 2, a provincial or OD TB supervisor or a person being in charge of TB control activities or lab technician of RH, will go and collect 3 cups of sputum within two days, pack them up in cool box as directed by NTP guidance (please read section E.1 for detail) and transfer to provincial RH for Xpert MTB/RIF. For RHs that Xpert MTB/RIF is not available, the specimens will be sent to the culture centers of Kg Cham, BTB, or CENAT for Xpert testing and culturing. The person who perform these tasks will be paid when quality of sputum have been maintained, accepted by lab technician (see annotation 5).
- The referral hospital also has to collect samples from MDR-TB suspect patient from their own hospital to test MDR-TB by Xpert or refer to Kg Cham, BTB, and CENAT cultures.
- Specimen referral and financial support for this Option 2 is similar to those in Option 1 (see annotation 3 and read annex 2 for more details).

3) Option 3:

- When a MDR TB suspect was identified as mentioned in section “B” page 2, community workers included C-DOT watchers or VHSG or staff of HC/FDH, will collect 3 cups of sputum from patient within two days, put them in cool box as directed by NTP guidance (please read section E.1 for detail) and transport to the provincial referral hospital for Xpert MTB/RIF. The person who perform these tasks will get paid when quality of sputum have been maintained and accepted by lab technician. For the government staff, mission and travel authorization letters are required and used for financial reimbursement/liquidation (see annotation 6).

- In case, RHs does not have Xpert MTB/RIF, the specimens will be sent to culture center at KPC or BTB PRHs or CENAT for performing Xpert MTB/RIF or culture.
- The referral hospital also have to collect samples from MDR-TB suspect patients from their own hospital for Xpert MTB/RIF or transport to culture center at KPC-PRH or BTB-PRH or CENAT.
- Specimen referral and financial support is the same as those in Option 1 (see annotation 3 and read annex 3 for more detail).

D. Provincial Clusters for Specimen Referral to Culture Centers

- Provinces that sputum specimens will be transported to culture at Kampong Cham RH lab: Kampong Cham (KPC), Kampong Thom (KTH), Kratie (KTE), Stung Treng (STR), Rattana Kiri (RTK), Mondul Kiri (MDK) and Preah Vihea (PVH).
- Provinces that sputum specimens will be transported to culture at BTB RH lab: BTB (Battambang), Pailin (PLN), Pursat (PST), Siem Reap (SRP), Banteay Meanchey (BMC) and Oudor Meanchey (OMC).
- Provinces that sputum specimens will be transported to culture lab at CENAT in Phnom Penh: Kandal (KDL), Kampong Speu (KPS), Kampong Chhnang (KCH), Prey Veng (PVG), Svay Rieng (SVG), Preah Sihanouk (PSH), Koh Kong (KHK), Takeo (TKO), Kampot (KPT), Kep (KEP) and Municipality Phnom Penh (PHN).

E. Other Key Tasks for TB Staff

- 1) Sputum packaging (specimen sample): sputum (specimen sample) must be packed in three layers:
 - ✓ Sputum collected from a patient will have to put in a container with screw cap and enclosed completely.
 - ✓ 3 sputum cups collected from a patient will need to put into holes of waterproof cap rack-type holder with/surrounded by absorbent cushioning material; then close up the holder firmly.
 - ✓ Next, put the holder/small box into a cool box (bigger box) with dry ice or wet ice maintaining temperature of box (big box) with cool temperature from 2-8 degree Celsius. Close the big box firmly with a safe lock or its deadlock.
 - ✓ After outer packaging is completely done, transport the box (big box) to the culture centers: BTB or KPC or CENAT in Phnom Penh no later than 5 days (within 5 day time).
- 2) Regarding provinces that need to refer MDR-TB suspect patients to CENAT culture center in Phnom Penh (regarding Option 1) in order to help patient and to facilitate case finding process, we may not need to refer patients to RH, but refer them directly to CENAT.
- 3) With the support from NTP, Provincial TB supervisor will (see annotation 7) have to:
 - ✓ Track down sputum transportation
 - ✓ Follow up sputum result return from laboratory to its specimen referring facility

- ✓ Provide feedback to OD TB supervisor for MDR-TB case management
 - ✓ Release payment to all relevant activities as described in this documentation (such as cash advance, reimbursement/ liquidation, voucher clearance with CENAT/NTP and so forth)
 - ✓ Coordinate all relevant MDR-TB control activities in the province, for example, how to identify the target, monitoring and support for OD, HC, etc.
 - ✓ Whilst conducting supervision and meeting on TB, case finding and treatment of MDR-TB should always be reviewed
- 4) In some circumstances, TB staff at central level (CENAT) would visit patient directly and collect sputum, for which special financial support would be paid.
 - 5) Fill up “MDR-TB suspect referral form and lab request form” and attached with sputum or patients any time/while referring the MDR-TB suspect or sputum.
 - 6) Referred specimen for DST is to transport only positive samples within a highly safe pack from BTB & KPC culture centers to CENAT via renting/paid car, arranged by PHD-TB Supervisor. Duration and sputum condition assurance should be clearly indicated.
 - 7) Follow seriously all infection control measure, which includes entire process of specimen collection, packaging, referral and culture. Involved working staff should be trained and instructed appropriately.
 - 8) TB staff of HC & RH who diagnose and treat TB patients will curiously examine/take history of outpatient department attendants, monitor the patients while caring for them as well as reviewing the register to find out possible MDR-TB cases based on criteria described in section “B”; and then organize and coordinate for sputum specimen or patient’s referral according to option as stated in section “C”.
 - 9) To facilitate of these activities and communication, CENAT, BTB PRH, and KPC PRH has to assign someone to be responsible for receiving the referred patients or samples. Each province (BTB, KPC, or CENAT) has to maintain the update data, telephone number of referring centers, and other relevant focal persons.

Annotations:

1. USD 15 will be paid to a patient for their local travel and accommodation while staying at RH/CENAT for 3-sputum collection (with quality).
2. USD 4 will be paid to the person who packs and communicates for referring 3 sputum specimens. 1 USD will be paid to the person who packs and communicates for referring 1 sputum specimen for Xpert testing among PLHIV.
3. Sputum transportation fee (bus/taxi) will be covered depending on distance and road condition according to the policy and tariff/rates of individual funding agency.
4. Lab technician will be supported as the following:
 - 1 USD per case for performing Xpert MTB/RIF and transport other 2 remaining samples to culture center. The sputum samples transportation from Xpert center to culture center will be done 2 times per week by bus/taxi/carrier agency. 1 USD will be also paid for performing Xpert among Presumptive TB/PLHIV.

- 1 USD per case for performing culture.
 - 2 USD per case for performing DST.
5. A person who undertakes/performs this job (PHD TB supervisor, OD TB Supervisor, RH staff) will receive the support for: Non-overnight stay: a per-diem US\$5 per day and actual travel expense.
 6. A person who undertakes/performs this job (staff of HC or former district hospital or RH/OD) will receive the support for:
 - Sputum collection: a per-diem US\$5 per day and actual local travel cost.
 - Specimen referral to RH: For non-overnight stay: a per-diem US\$5 per day and actual travel expense and US\$15 per an overnight stay with actual travel expenditure.
 7. OD TB supervisor is responsible for collecting and reviewing all expense documents to get clearance from provincial TB supervisor in the first week of next coming month. Provincial TB supervisor is responsible for collecting and reviewing all relevant documents to get clearance from CENAT in the second week of next coming month. These activities will be paid accordingly based on relevant donor's policy. Provincial TB supervisor will be paid US\$10 per month per province for the communication and facilitation.
 8. Xpert/culture/DST centers will be supported US\$10 per month for communication and hot line feedback to the referred local facility and PMDT focal person/manager.

In addition, based on the experience from previous guidance issued more than two year ago, we found some gaps related to communication as well as inflexible implementation which lead to MDR-TB suspect referral not so active. Since guidance/guideline is generally not 100% perfect, the implementers have to use it in a flexible way. Please contact with NTP for any additional explanation, clarification or deal together with the issues that may occur.

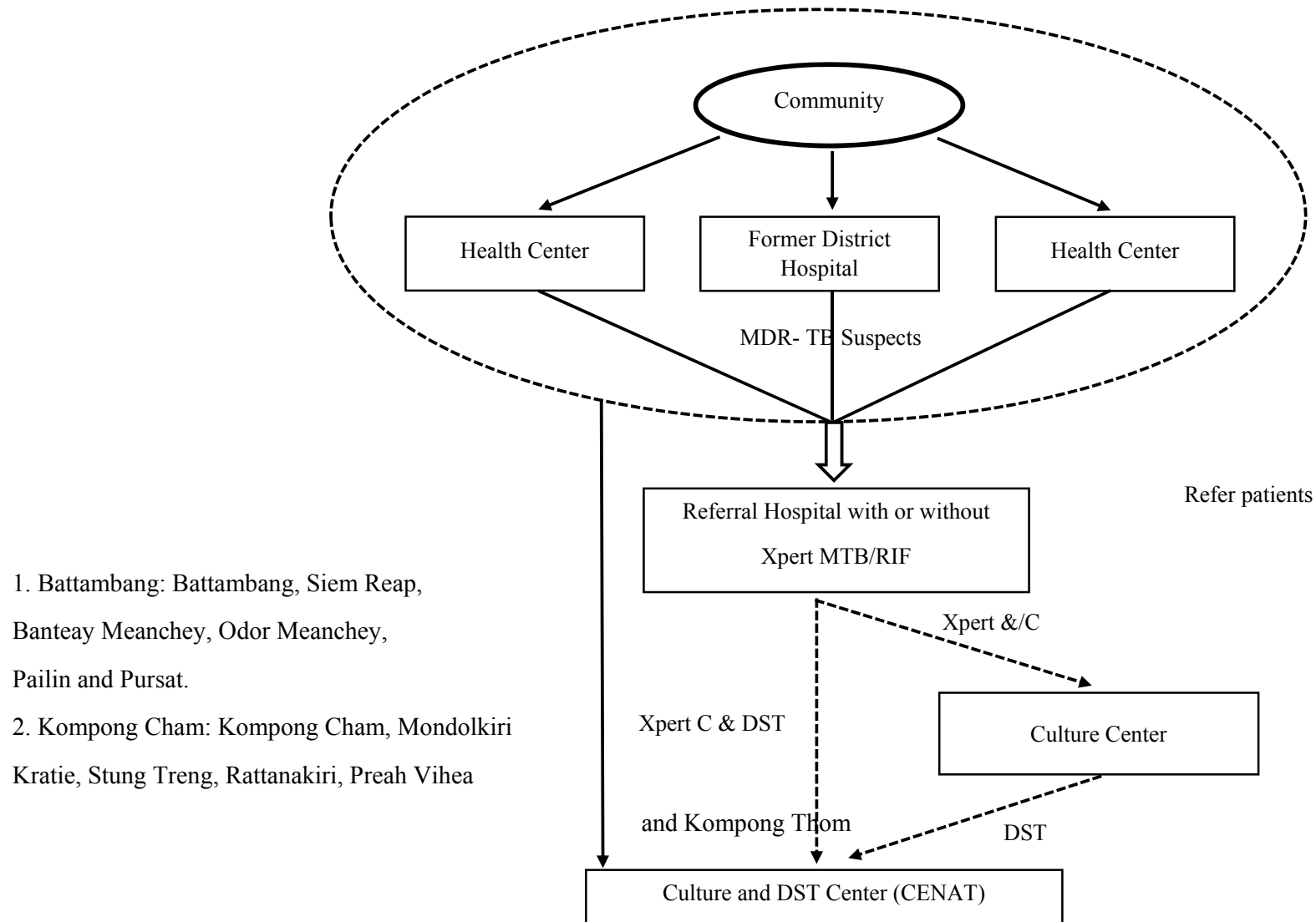
Phnom Penh, March 31, 2014
 Director of CENAT
 Dr. Mao Tan Eang's signature and stamp
 plus CENAT stamp

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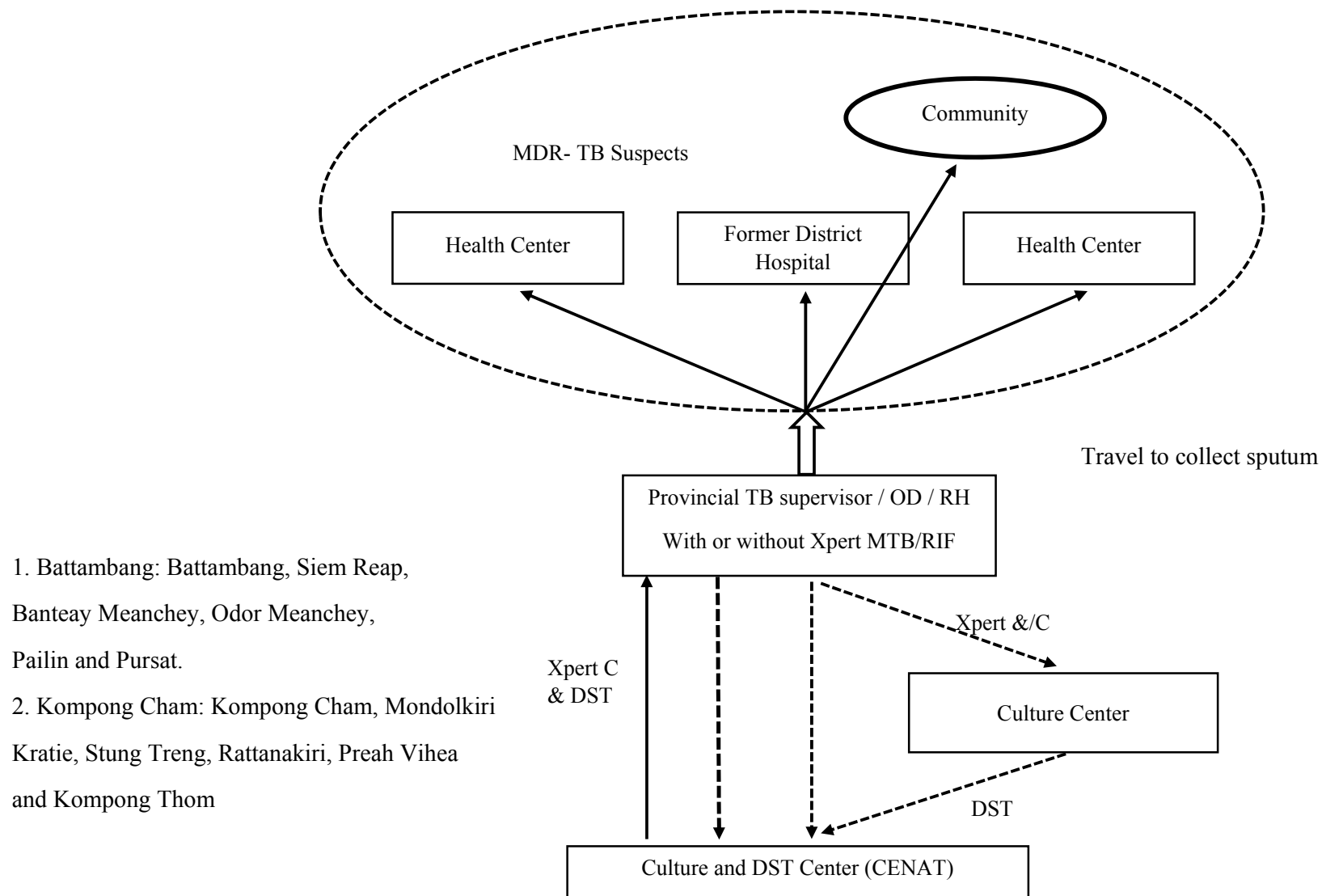
- Provincial/Municipal Health Departments
- Operational districts
- Provincial/OD TB programs
- Relevant partners (NGOs)

“For information and implementation”

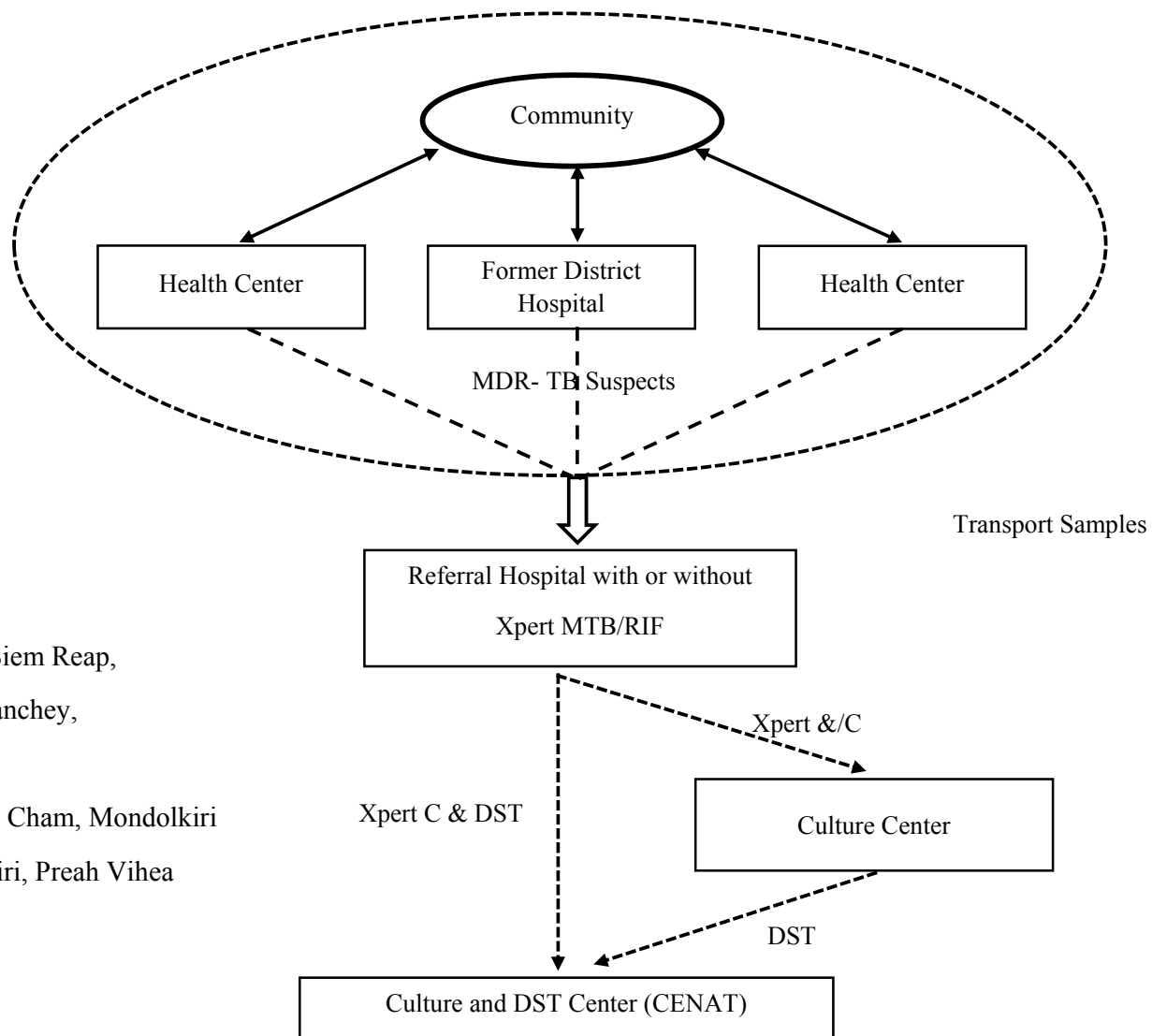
Appendix 1: Option 1: Graph of MDR-TB Suspect (Refer Patients)



Appendix 2: Option 2: Graph of MDR-TB Suspect (Travel to Collect Sputum)



Appendix 3: Option 3: Graph of MDR-TB Suspect (Transport Samples)



1. Battambang: Battambang, Siem Reap, Banteay Meanchey, Odor Meanchey, Pailin and Pursat.
2. Kompong Cham: Kompong Cham, Mondolkiri Kratie, Stung Treng, Rattanakiri, Preah Vihea and Kompong Thom

TB Activity Report for Quarter.....Year.....

1. TB Case Detection

1.1. Distribution by Category and Type of Disease

Case Category (1)	Type of Disease												Total		Total (16)
	New Case	Retreated Case								Unknown History					
		Relapse		Failure		Retreated After Defaulting (RAD)		Others							
	PTB (2)	EP (3)	PTB (4)	EP (5)	PTB (6)	EP (7)	PTB (8)	EP (9)	PTB (10)	EP (11)	PTB (12)	EP (13)	PTB (14)	EP (15)	
BK(+)															
BK(-) / Clinic															
Total															

1.2. Distribution by Age and Sex

1.2.1. New Case

1.2.1.1. Bacteriologically Confirmed Pulmonary TB - New Case

Distribution by Age and Sex of Bacteriologically Confirmed Pulmonary TB – New Case (Column 2)																Total	
0-4 years		5-14 years		15-24 years		25-34 years		35-44 years		45-54 years		55-64 years		≥65 years			
Male	Female	Male	Female	Male	Female	Male	Female	Male	Female	Male	Female	Male	Female	Male	Female	Male	Female

1.2.1.2. Smear Negative/Clinical Pulmonary TB - New Case

Distribution by Age and Sex of Smear Negative/Clinical Pulmonary TB – New Case (Column 2)																Total	
0-4 years		5-14 years		15-24 years		25-34 years		35-44 years		45-54 years		55-64 years		≥65 years			
Male	Female	Male	Female	Male	Female	Male	Female	Male	Female	Male	Female	Male	Female	Male	Female	Male	Female

1.2.1.3. Bacteriologically Confirmed, Smear Negative/Clinical Extra-Pulmonary TB - New Case

Distribution by Age and Sex of Bacteriologically Confirmed, Smear Negative/Clinical Extra-Pulmonary TB – New Case (Column 3)																Total	
0-4 years		5-14 years		15-24 years		25-34 years		35-44 years		45-54 years		55-64 years		≥65 years			
Male	Female	Male	Female	Male	Female	Male	Female	Male	Female	Male	Female	Male	Female	Male	Female	Male	Female

1.2.2. Relapse Case

1.2.2.1. Bacteriologically Confirmed Pulmonary TB - Relapse Case

Distribution by Age and Sex of Bacteriologically Confirmed TB – Relapse Case (Column 4)																Total	
0-4 years		5-14 years		15-24 years		25-34 years		35-44 years		45-54 years		55-64 years		≥65 years			
Male	Female	Male	Female	Male	Female	Male	Female	Male	Female	Male	Female	Male	Female	Male	Female	Male	Female

1.2.2.2. Smear Negative/Clinical Pulmonary TB - Relapse Case

Distribution by Age and Sex of Smear Negative/Clinical Pulmonary TB - Relapse Case (Column 4)																Total	
0-4 years		5-14 years		15-24 years		25-34 years		35-44 years		45-54 years		55-64 years		≥65 years			
Male	Female	Male	Female	Male	Female	Male	Female	Male	Female	Male	Female	Male	Female	Male	Female	Male	Female

1.2.2.3. Bacteriologically Confirmed, Smear Negative/Clinical Extra-Pulmonary TB - Relapse Case

Distribution by Age and Sex of Bacteriologically Confirmed, Smear Negative/Clinical Extra-Pulmonary TB - Relapse Case (Column 5)																Total	
0-4 years		5-14 years		15-24 years		25-34 years		35-44 years		45-54 years		55-64 years		≥65 years			
Male	Female	Male	Female	Male	Female	Male	Female	Male	Female	Male	Female	Male	Female	Male	Female	Male	Female

1.3. Isoniazid Preventive Therapy (IPT) for Children in Quarter..... Year.....

Age	Male	Female	Total
Under 5 years (HIV Negative)			
Under 15 years (HIV Positive)			

1.4. Distribution by Source of Referral for Patients Registered in Quarter..... Year.....

Activities	TB Suspect case Referred from	TB Suspect case Received from	TB Suspect case Examined	Number of TB Patients				Total
				Pulmonary TB		Extra-Pulmonary TB		
				BK(+)	BK(-)/Clinic	BK(+)	BK(-)/Clinic	
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Self-referral								-
Community								-
Private Sector								-
Prison								-
Factory								-
HIV/AIDS Service								-
Other								-
Total				-	-	-	-	-

1.5. Distribution by Type of DOT for Patient Registered in Quarter..... Year.....

Total Number of Patients Registered	Type of DOT					Total
	Hospitalization	Ambulatory	Home Delivery	Community	Non-DOT	

2. Result of Sputum Follow-up at the end of Intensive Phase for Patient Registered in Quarter..... Year.....**2.1. Bacteriologically Confirmed Pulmonary TB - New Case (CAT 1)**

Month		Result of Sputum Follow-up						Total
		BK (+) --> BK (-)	BK (+) --> BK (+)	Not Done	Died	Default	Transfer Out	
C2 (month 2)	Microscopy							
	Culture							
C3 (month 3)	Microscopy							
	Culture							

2.2. Bacteriologically Confirmed Pulmonary TB - Retreated Case (CAT 2)

Month		Result of Sputum Follow-up						Total
		BK (+) --> BK (-)	BK (+) --> BK (+)	Not Done	Died	Default	Transfer Out	
C3 (Month 3)	Microscopy							
	Culture							

3. Result of Sputum Follow-up at Month 5 for Bacteriologically Confirmed Pulmonary TB Registered in Quarter.....Year.....

Month		Result of Sputum Follow-up						Total
		BK (+) --> BK (-)	BK (+) --> BK (+)	Not Done	Died	Default	Transfer Out	
CAT 1	Microscopy							
	Culture							
CAT 2	Microscopy							
	Culture							

4. TB/HIV Collaborative Activities in Quarter..... Year.....

4.1. TB Detection among People Living with HIV/AIDS

1. Number of People with HIV+ Registered at VCCT (Voluntary Confidential Counseling and Testing)					
2. Number of People with HIV+ Registered at VCCT Referred for TB Detection at OI Service					
3. Number of People with HIV+ Arrived at OI/ART for TB Detection					
4. Number of TB Patient Detected among People with HIV+	Pulmonary TB		Extra-Pulmonary TB		Total=
	BK(+)=	BK(-)=	BK(+)=	BK(-)=	
5. Number of People with HIV+ Received Isoniazid Preventive Therapy (at least one dose of INH)					

4.2. HIV Testing among TB Patients

HIV+ HIV Testing among TB Patients										
1. Number of All TB patients registered in TB Patient Book (including people with HIV+ detected TB)										
2. Number of TB patients registered in TB Patient Book excluding people with HIV+ known before this TB diagnosis										
3. Number of TB patients registered in TB Patient Book known HIV status before TB treatment										
4. Number of TB patients registered in TB Patient Book referred for HIV test during TB treatment										
5. Number of TB patients registered in TB Patient Book Arrived at VCCT for HIV testing during TB treatment										
6. Number of TB patients tested for HIV during TB treatment and have result					HIV(+)= 				HIV- =	Total=
					Pulmonary TB		Extra-Pulmonary TB			
					BK(+)=	BK(-)=	BK(+)=	BK(-)=		

4.3. Care for TB patients with HIV+

1. Number of TB patients with HIV+ registered for treatment and received Cotrimoxazole Preventive Therapy (at least one first dose)	
2. Number of TB patients registered for treatment whose HIV+ result known before TB treatment and received Cotrimoxazole Preventive Therapy (at least one first dose)	
3. Number of TB patients with HIV+ registered for treatment received ART during TB treatment	
4. Number of TB patients registered for treatment whose HIV+ result known before TB treatment and received ART during TB treatment	

5. Drug Resistant TB Activities in Quarter..... Year.....

5.1. Identifying Suspect Cases of Drug Resistant TB (reported in Quarter..... Year.....)

	Retreated Pulmonary TB Case				Sputum Positive		Close Contact to Drug Resistant TB	People with HIV+ and TB BK+	Total
	Failure	Relapse	RAD	Others	Month 2	Month 3			
Drug Resistant TB Suspect									
Drug Resistant TB Suspect Tested									
Drug Resistant TB Suspect remaining from the previous quarters tested									

5.2. Detecting Multi-Drug Resistant TB Cases (reported in Quarter..... Year.....)

Type of MDR-TB	Total	Xpert Test Result		DST Test Result			MDR-TB Cases Tested for Susceptibility to Injectable Second-line Quinolone
		Not Resistant to R (T)	Resistant to R (RR)	Susceptible to R & H	Resistant to R or H (with or without resistant to E or S)	Resistant to R & H (MDR)	
Retreated Pulmonary TB Patients							
Failure							
Relapse							
Defaulter							
Other							
Sputum still Positive at Month 2							
Sputum still Positive at Month 3							
Close Contact with Drug Resistant TB							
Bacteriologically Confirmed TB Patients with HIV Positive (BK+ HIV+ TB Patients)							
Total							

5.3. TB Cases Treated with Second-line Drugs (reported in Quarter..... Year.....) for OD with MDR-TB Treatment Site only

Type of Patient	Patient Detected	Patient Treated	Female	< 15 Years	HIV+	ART
MDR-TB						
Xpert						
DST						
Poly-drug Resistant TB (PDR-TB)						
Mono-drug Resistant TB (Mono-DR-TB)						
Extremely Drug Resistant TB (XDR-TB)						
Number of MDR-TB Cases known for the delay of treatment initiation	Time length between the laboratory result (Xpert/DST) issued and treatment initiation date (in day)					
	Average		Minimum		Maximum	

5.4. Interim Treatment Result (reported in Quarter..... Yea) for OD with MDR-TB Treatment Site only

RR-TB/MDR-TB	Number of Registered Cases	Culture Result at Month 6			Discontinued Treatment		Non-MDR-TB
		Positive	Negative	Not Done	Death	Defaulter	
MDR-TB							
Xpert							
DST							
XDR-TB							

5.5. Treatment Outcome (reported in Quarter..... Year.....) for OD with MDR-TB Treatment Site only

Type of Patient	Cured	Complete	Failure	Died	Default	Not Evaluated	Total
MDR-TB							
Xpert							
DST							
PDR-TB							
Mono DR-TB							
XDR-TB							

Reporting Schedule

Activity Report	Quarter 1, 2016	Quarter 2, 2016	Quarter 3, 2016	Quarter 4, 2016
Identifying Suspect Case of Drug-resistant TB	Quarter 1, 2016	Quarter 2, 2016	Quarter 3, 2016	Quarter 4, 2016
Detecting Multi-Drug Resistant TB Cases	Quarter 3, 2015	Quarter 4, 2015	Quarter 1, 2016	Quarter 2, 2016
TB Cases Treated with Second-line Drugs	Quarter 1, 2016	Quarter 2, 2016	Quarter 3, 2016	Quarter 4, 2016
Interim Treatment Result	Quarter 1, 2015	Quarter 2, 2015	Quarter 3, 2015	Quarter 4, 2015
Treatment Outcome	Quarter 4, 2013	Quarter 1, 2014	Quarter 2, 2014	Quarter 3, 2014

6. Treatment Outcome for TB Patients Registered in the past 12 months in Quarter..... Year.....

Note: For OD with functioning TB program, please send the page on the right side of TB Register attached with the above report to CENAT

6.1. TB in Adults (≥ 15 years)

Number of Cases Registered in the Quarter mentioned above	Treatment Regimen	Treatment Outcome						Total
		Cured	Complete	Failure	Died	Default	Not Evaluated	
		(1)	(2)	(3)	(4)	(5)	(6)	
	CAT 1-3							
	New Case							
	BK(+) Pulmonary TB							
	BK(-)/clinical Pulmonary TB							
	BK(+) EP TB							
	BK(-)/clinical EP TB							
	Unknown History							
	BK(+) Pulmonary TB							
	BK(-)/clinical Pulmonary TB							
	BK(+) EP TB							
	BK(-)/clinical EP TB							
	CAT 2							
	Relapse							
	BK(+) Pulmonary TB							
	BK(-)/clinical Pulmonary TB							
	BK(+) EP TB							
	BK(-)/clinical EP TB							
	Failure							
	BK(+) Pulmonary TB							
	BK(-)/clinical Pulmonary TB							
	BK(+) EP TB							
	BK(-)/clinical EP TB							
	Default							
	BK(+) Pulmonary TB							
	BK(-)/clinical Pulmonary TB							
	BK(+) EP TB							
	BK(-)/clinical EP TB							
	Others							
	BK(+) Pulmonary TB							
	BK(-)/clinical Pulmonary TB							
	BK(+) EP TB							
	BK(-)/clinical EP TB							

6.2. TB in Children (< 15 years)

Number of Cases Registered in the Quarter mentioned above	Treatment Regimen	Treatment Outcome						Total
		Cured	Complete	Failure	Died	Default	Not Evaluated	
		(1)	(2)	(3)	(4)	(5)	(6)	
	CAT 1-3							
	New Case							
	BK(+) Pulmonary TB							
	BK(-)/clinical Pulmonary TB							
	BK(+) EP TB							
	BK(-)/clinical EP TB							
	Unknown History							
	BK(+) Pulmonary TB							
	BK(-)/clinical Pulmonary TB							
	BK(+) EP TB							
	BK(-)/clinical EP TB							
	CAT 2							
	Relapse							
	BK(+) Pulmonary TB							
	BK(-)/clinical Pulmonary TB							
	BK(+) EP TB							
	BK(-)/clinical EP TB							
	Failure							
	BK(+) Pulmonary TB							
	BK(-)/clinical Pulmonary TB							
	BK(+) EP TB							
	BK(-)/clinical EP TB							
	Default							
	BK(+) Pulmonary TB							
	BK(-)/clinical Pulmonary TB							
	BK(+) EP TB							
	BK(-)/clinical EP TB							
	Others							
	BK(+) Pulmonary TB							
	BK(-)/clinical Pulmonary TB							
	BK(+) EP TB							
	BK(-)/clinical EP TB							

6.3. Treatment Outcome for BK(+) Pulmonary TB Patient – New Case under Community DOT registered in the past 12 months in Quarter.....Year.....

Number of BK+ Pulmonary TB – New Case registered in the Quarter mentioned above	Treatment Regimen	Treatment Outcome						Total
		Cured	Complete	Failure	Died	Default	Not Evaluated	
		(1)	(2)	(3)	(4)	(5)	(6)	
	CAT 1							
	BK(+) Pulmonary TB							

6.4. Treatment Outcome for IPT Registered in the past 12 months in Quarter..... Year.....

Total	Complete	Developed TB	Default	Died	Other

Calendar

Activity Report	C ₂	C ₅	Treatment Outcome
Quarter 1, 2016	Quarter 4, 2015	Quarter 2, 2015	Quarter 1, 2015
Quarter 2, 2016	Quarter 1, 2016	Quarter 3, 2015	Quarter 2, 2015
Quarter 3, 2016	Quarter 2, 2016	Quarter 4, 2015	Quarter 3, 2015
Quarter 4, 2016	Quarter 3, 2016	Quarter 1, 2016	Quarter 4, 2015

7. Drugs Requested for Use in Quarter..... Year.....

Drugs Requested for use in Quarantine Treatment										
TB Drug	Adult TB						Grand Total (G) = C + F	Buffer Stock (H) = G x 2	Current Stock Remaining (I)	Request (J) = H - I
	CAT1 & CAT3			CAT2						
	No. of Cases (A)	Quantity of Drugs per Case (B)	Total (C) = A x B	No. of Cases (D)	Quantity of Drugs per Case (E)	Total (F) = D x E				
RHZE 150/75/400/275 mg		180			270					
RH 150/75 mg		360			450					
E 400 mg					450					
S 1 g		60			60					
H 100 mg		540								
B6 10 mg		180			90					
TB drug	Childhood TB						Grand Total (G) = C + F	Buffer Stock (H) = G x 2	Current Stock Remaining (I)	Request (J) = H - I
	CAT1 & CAT3			CAT2						
	No. of Cases (A)	Quantity of Drugs per Case (B)	Total (C) = A x B	No. of Cases (D)	Quantity of Drugs per Case (E)	Total (F) = D x E				
RHZE (21-30 Kg) 150/75/400/275 mg		120			180					
RHZ 60/30/150 mg		180			270					
RH 150/75 mg (21-30 Kg)		240			300					
RH 60/60 mg		360			480					
RH 60/60 mg (>21-30 Kg)		360			480					
RH 60/30 mg		360			450					
E 400 mg (21-30 Kg)					150					
E 100 mg		180			720					
S 1 g					60					
H 100 mg		60								
B6 10 mg		90			90					

8. Drugs Received from CMS in the last quarter: day.....moth.....Year.....

Drug Name	Remaining Stock (A)	Expiry Date (B)	Quantity Received (C)	Expiry Date (D)	Total Quantity of Drugs (E)	Other (F)
RHZE 150/75/400/275 mg						
RH 150/75 mg						
Z 400 mg						
E 400 mg						
E 100 mg						
S 1 g						
RHZ 60/30/150 mg						
RH 60/30 mg						
RH 60/60 mg						
H 100 mg						
Vit B6 (50 mg)						
Vit B6 (10 mg)						

9. Laboratory

Number of TB Suspects Submitted Sputum for TB Diagnosis		Number of TB Suspects with Result Positive (count only those who has at least one slide positive)	
--	--	--	--

Aspect of the Collected Sputum:

Number of Salivary Sputum	Number of Pusy Sputum	Number of Bloody Sputum

Smear:

Detection						
	Number of BAAR	1+	2+	3+	Negative	Total
First Slide						
Second Slide						
Third Slide						
Total						

Follow-up						
	Number of BAAR	1+	2+	3+	Negative	Total
C2, CAT1						
C3, CAT1						
C3, CAT2						
C4						
C5, CAT1						
C5, CAT2						
C6, CAT1						
C8, CAT2						
Others						
Total						

Xpert:

Type of Patient	Relapse			Failure			Return After Default (RAD)			Others			Number of Retreated Patients			Number of Patients with Sm+ in Month 2/3			Number of Close Contacts			Number of Patients whose Smear became Positive			Others			Total
HIV/AIDS Status	+	-	?	+	-	?	+	-	?	+	-	?	+	-	?	+	-	?	+	-	?	+	-	?	+	-	?	
Number of Patients																												

Number of Tests Done by Type of Patient

RR																											
T																											
TI																											
N																											
I																											
Total																											

Error Code:

5011	2008	2127	5006	5007					

Request for Laboratory Materials:

Item	Quantity per Quarter (A)	Quantity per Quarter with Buffer Stock (B)	Stock at Reporting Time (C)	Quantity Request (D= B-C)	Other
Fuchsine (ml)	Total slide..... x 4 ml:ml	A.....x2 :.....ml	ml	ml	
Acid-Alcohol (ml)	Total slide..... x 8 ml:ml	A.....x2 :.....ml	ml	ml	
Methylene Blue (ml)	Total slide..... x 4 ml:ml	A.....x2 :.....ml	ml	ml	
Alcohol (ml)	Total slide..... x 4 ml:ml	A.....x2 :.....ml	ml	ml	
Immersion Oil (ml)	Total slide..... x 0.2 ml:ml	A.....x2 :.....ml	ml	ml	
Xylene (ml)	Total slide..... x 2 ml:ml	A.....x2 :.....ml	ml	ml	
Glass Slide (piece)	Number of slides used per quarter: pieces	A.....x2 :.....pieces	pieces	pieces	
Sputum Cup(piece)	Number of slides used per quarter: pieces	A.....x2 :.....pieces	pieces	pieces	
Filter Paper (sheet)	60 sheets x Lab:sheets	60 sheets x 2:sheets	sheets	sheets	

☞ If other materials need to be requested or broken, please specify below

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10. Inventory of IEC Materials

Item	Brief Description	Producer-Sponsor	Source	Incoming Quantity	Distribution

☞ **Note:** Please send this quarterly report to CENAT regularly before the 15th day of the first month of next quarter.

Director of Provincial Health Department

Provincial TB Supervisor

OD Chief

Reporter

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